

Autocrats Remake the State: Evidence from Francoist Spain*

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Abstract

In autocracies, personnel decisions often prioritize ideological alignment over merit. We investigate how such ideology-based personnel selection shapes the quality of the public workforce and its implications for state capacity. We examine this in the context of a large-scale political purge and reorganization of primary school teachers carried out by the Francoist dictatorship following the Spanish Civil War (1936-1939). We construct a novel dataset linking newly digitized purge archives and administrative records that track teacher careers, and document that more competent teachers were disproportionately dismissed and that average experience declined by about 2.6 years (14.4%). Local shortages created by the purge were partly offset by regime efforts to reassign teachers across municipalities, yet coordination frictions in this process left persistent staffing deficits that spread to areas less affected by the purge. Our findings show how ideology-based personnel reforms can erode public workforce expertise and weaken the state's capacity to deliver essential services, providing a mechanism through which autocratic consolidation entails sizable capacity costs.

JEL codes: D73, H41, H83, J45, N44

Keywords: ideology-based selection; state capacity; political purge; loyalty-competence trade-off; autocratic consolidation

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1 Introduction

Public officials are central to state effectiveness and, in turn, to economic development (Finan, Olken, and Pande, 2017). Yet political incentives frequently distort their selection. In democracies, ideological alignment shapes turnover among political appointees but not among career civil servants (Spenkuch, Teso, and Xu, 2023). In contrast, autocracies tend to extend ideological screening deeper into the public workforce, influencing not only political appointees but also the selection of frontline civil servants – such as educators and police officers – responsible for implementing the state’s daily functions (Caplan, 1988).

While ideology-based selection of frontline public workers is a common feature of autocracies, little is known about its implications. Data limitations and the opacity in personnel decisions in such regimes make their study challenging. Yet, the issue is of growing importance: in 2024, 72% of the world’s population lived under autocratic rule, the highest share since 1978 (Nord et al., 2025).

In this paper, we study the relationship between ideology-based personnel policies in autocracies, the quality of the public workforce, and the state’s capacity to deliver essential services. Our setting is a large-scale political purge and subsequent reorganization of primary school teachers carried out by the Francoist dictatorship following the Spanish Civil War (1936–1939), aimed at ensuring ideological conformity. We construct a novel panel tracking teacher careers from 1934 to 1946 by linking newly digitized purge archives and administrative records. We find that more competent teachers were more likely to be dismissed, and the overall experience of the teaching workforce declined. Despite the regime’s efforts to offset local shortages through teacher relocations across municipalities, coordination frictions in this process left persistent staffing gaps even in areas with fewer purge-related losses. Overall, our evidence illustrates how ideology-based screening can erode public-sector expertise and weaken the state’s ability to deliver essential services.

The Francoist purge of primary school teachers provides a unique setting to explore how ideology-based personnel selection shapes workforce quality and local service provision. Although the purge extended across the civil service, the regime placed particular emphasis on education, which it viewed as a source of political dissent and moral decay. In this context, schools were conceived as an instrument to promote national unity and traditional values in the aftermath of the war (Monés i Pujol-Busquets, 1976). We focus on primary school teachers, who formed the backbone of Spain’s education system, given the country’s stage of development and the limited reach of higher schooling. The purge of teachers was legally introduced in 1936, with most sanctions imposed between 1938 and 1941. As a result, approximately 10 percent

of teachers were dismissed and another 15 percent were temporarily suspended or forcibly reassigned to other municipalities. Subsequent rounds of voluntary relocations and new hiring completed the reorganization of the teaching workforce.

Whether ideology-based selection undermines the quality of the public workforce is a priori ambiguous. On one hand, ideology-based screening may reduce workforce quality if more competent teachers are also more likely to express views at odds with the regime. Moreover, to the extent that experience contributes to teaching skills, removing experienced teachers would also reduce overall quality. On the other hand, if teachers' political and social participation against the regime substitutes for their investment in professional skills, removing non-aligned individuals may not lower competence. Beyond quality, large-scale ideological screening also poses practical challenges for staffing: when applied to a frontline workforce such as educators, it can create widespread vacancies that are difficult to fill. While states may attempt to replace non-aligned workers, the extent to which they do so effectively remains also an open question.

Quantifying the effects of ideology-based personnel selection in autocratic regimes is challenging because comprehensive data on individual careers are rarely available. We address this limitation with a large data collection effort, assembling a novel panel that tracks the careers of Spanish primary school teachers from 1934 to 1946, spanning the Second Republic, the Civil War, and the early Francoist years. We digitize three key sources: (i) complete official rosters from 1934 and 1946, including each teacher's locality of employment and experience; (ii) individual purge archival records from fifteen provinces detailing sanction outcomes; and (iii) nationwide hiring lists from 1928 to 1946, recording each entrant's ranking within their entry cohort. We geocode all localities using historical censuses and gazetteer data and link teachers across sources to reconstruct their career histories. The resulting panel covers the universe of over 68,000 teachers, including demographic and experience information, and purge outcomes for over 16,000. This dataset allows us to quantify how ideology-based personnel selection reshaped the teaching workforce and the local provision of education.

We begin by documenting the evolution of the teaching workforce before and after the purge. Using the panel of teacher careers, we show that the number of in-service primary school teachers declined sharply following the purge and did not return to prewar levels until 1946 – seven years after the regime's establishment. We reconstruct the evolution of teacher counts by leveraging the timing of dismissals and new hires in our data. The workforce had grown steadily since the 1920s, with a rapid expansion from 1931 under the Second Republic, before dropping abruptly following the purge. Although hiring resumed soon after, this disruption effectively stalled a decade of potential expansion. By 1946, the system employed roughly the same number of

teachers as ten years earlier. To rule out that this pattern merely reflects an adjustment to lower schooling demand, we estimate the school-age population using historical census data. The resulting teacher-student ratio follows a similar evolution to the number of teachers, indicating that staffing shortfalls persisted well into the 1940s before restoring prewar capacity.

Then, we examine how ideology-based selection shaped workforce quality. Using official hiring lists for the 1928, 1931, and 1933 cohorts, we construct cohort- and gender-specific measures of competence based on teachers' ranks in entry examinations and training performance. We then estimate competence-based differences in dismissal rates, and find that they increase with competence: teachers in the top quintile were about 40 percent more likely to be dismissed than those in the bottom quintile. The result is driven mainly by men, consistent with their greater exposure to sanctions and with historical accounts of higher political participation and visibility during the Second Republic, which was likely correlated with competence. Competence losses are also more pronounced among teachers who were fully trained under the Republic (1933 cohort), but similar patterns across earlier cohorts – trained under the preceding nondemocratic regime – indicate that these results are not driven by ideological bias in the competence measure.

The purge and subsequent reorganization reduced the experience of the teaching workforce. We use the 1946 roster to measure the observed distribution of experience among in-service teachers. Next, we simulate a counterfactual scenario using the 1934 roster, projecting how prewar teachers' careers would have evolved in the absence of the purge. To account for natural attrition, we apply exit rates estimated from non-sanctioned teachers. We find that the average experience fell by about 2.6 years, or 14.4 percent, relative to the no-purge counterfactual, with losses across the entire distribution. While less experienced teachers faced higher dismissal rates, they would have accumulated over a decade of experience by 1946, translating into sizable human capital losses. Alternative counterfactuals assuming lower attrition, plausible in the absence of war, yield experience losses roughly 40 percent larger, suggesting that the baseline estimate is a lower bound.

Next, we explore how ideology-based selection shaped local education provision. We exploit municipal-level variation in purge intensity – measured by the local share of teachers dismissed, suspended, or forced to relocate – to document changes in staffing capacity. To do so, we aggregate our teacher-level panel into a municipality-level panel covering the fifteen provinces for which purge sanction outcomes are collected. The dataset provides measures of municipal-level staffing and workforce composition, including the number of teachers employed, purge intensity, the number of post-purge hires, and the balance of relocations.

Municipalities that were more severely affected by the purge do not exhibit a persistent decline in the number of teachers relative to those less affected. The regime mitigated local shortages through a large relocation process announced in 1941 and fully implemented by 1943. This round of voluntary transfers allowed incumbent teachers to move into posted vacant positions – largely those left by dismissed teachers – and represented the regime’s first major response to staffing gaps. Hiring, slower and administratively more costly than relocations, did not directly target purge-related vacancies.

These relocations, however, created new imbalances. Because geographic transfers were the main mechanism to fill purge-related vacancies, we examine how staffing levels evolved across municipalities with different relocation balances. Our estimates show that municipalities that lost teachers through relocations had lower staffing levels: a one percentage point decrease in the teacher relocation balance is associated with a 0.55 percentage point decline in local staffing in 1946. Although the regime succeeded in closing disparities across municipalities with different purge intensities, the relocation process spread staffing shortfalls to municipalities less affected by the purge.

Several findings suggest that these imbalances reflected coordination and bureaucratic frictions in the relocation and hiring processes rather than deliberate planning. First, the regime determined which vacancies to post for relocation but did not restrict who could apply, allowing teachers to leave already depleted areas. Second, teachers hired while relocations were underway were evenly allocated, whereas subsequent hires were disproportionately assigned to municipalities with larger relocation deficits, indicating timing frictions in hiring. Third, municipalities with negative relocation balances that were not compensated through new hires were not previously overstaffed. Fourth, hiring patterns in provinces with strong regional nationalist movements, such as Catalonia and the Basque Country, are similar to those in other provinces, suggesting that these imbalances did not reflect targeted punishment of disloyal regions.

Finally, we assess the magnitude of allocation frictions in the local distribution of teachers. We do so through a back-of-the-envelope calculation that measures aggregate municipal deviations from the regime policy target of having at least one teacher per 50 students. The analysis shows that a substantial share of in-service teachers – about 10.1% – could have been relocated from municipalities whose staffing exceeded the minimum policy requirement to municipalities that were understaffed. This would have reduced aggregate local staffing deficits by 17.3%. We then compare the 1946 allocation with two benchmarks that differ in how new hires are assigned: one allocates them exclusively to understaffed municipalities, while the other distributes them randomly. We find that the observed distribution lies closer to the random benchmark than to the frictionless one. Overall, these findings highlight how large-scale personnel

reforms under autocracy can generate persistent organizational adjustment costs that weaken local state capacity.

This paper contributes to three different literatures. First, we contribute to the literature on the personnel economics of the state (Finan, Olken, and Pande, 2017). Several papers have analyzed the role in the selection and composition of public sector workers of financial incentives (Dal Bó, Finan, and Rossi, 2013; Deserranno, 2019; Ashraf et al., 2020); corruption (Weaver, 2021); discretionality (Liu and Zhang, 2025); and patronage (Colonnelli, Prem, and Teso, 2020; Brierley, 2021). Closely related to our work, Akhtari, Moreira, and Trucco (2022) show that political discretion in local government personnel decisions undermines public education provision by increasing teacher turnover. Our paper contributes to this literature by documenting how ideology-based screening, a common feature of autocratic governance, shapes the quality of the public workforce.

A related strand of this literature examines the organization of the state and the development of state capacity (Besley and Persson, 2009; Besley et al., 2022; Mastrococco and Teso, 2025). Xu (2018) finds that patronage in bureaucratic appointments under British colonial rule reduced fiscal performance, and Chambru, Henry, and Marx (2024) show how changes in local administrative presence in eighteenth-century France affected public good provision and economic development. Our findings bridge the two by examining how personnel policies relate to the state's capacity to provide local services.

Second, our results add to the literature on the loyalty–competence trade-off in autocratic regimes (Egorov and Sonin, 2011; Zakharov, 2016). Existing work, largely focused on China, examines how bureaucrats are selected into mid- and high-level positions (Shih, Adolph, and Liu, 2012; Jia, Kudamatsu, and Seim, 2015; Fisman et al., 2020; Meng and Paine, 2022; Chen et al., 2023). The evidence suggests that loyalty generally outweighs competence in political selection, though competence is relatively more important in lower ranks (Landry, Lü, and Duan, 2018; Wang, Yao, and Zhang, 2022). Whether similar trade-offs extend to the frontline public workforce remains an open question. Our results show that, in a frontline occupation central to the regime's authoritarian project, loyalty – in the form of ideological conformity – is prioritized at the expense of professional competence.

Third, we contribute to the growing literature on the political and economic legacies of the Spanish Civil War and the Francoist dictatorship. Prior research has documented long-run effects on political participation and preferences (Oto-Peralías, 2015; Albertus, 2023; Martinez-Marquina, 2024; Rodon, 2024; Tur-Prats and Valencia Caicedo, 2025), and on social capital (Muñoz-Blanco, 2024). Related to our analysis, though focused

on the Civil War period, is [La Parra-Pérez \(2020\)](#), who shows that military officers promoted under Republican governments were more likely to remain loyal to the Second Spanish Republic. Our paper differs from the existing literature by focusing on the reorganization of the state and selection of civil servants during Franco’s autocratic consolidation.

Finally, our novel dataset, to the best of our knowledge, offers the first comprehensive data on local education provision in Spain during the first half of the twentieth century and provides a foundation for future research on the historical development of education in Spain.

The remainder of the paper is organized as follows. In the next section, we provide historical background. In Section 3, we describe the data sources and the construction of the teacher-career panel. In Section 4, we present the empirical strategy and main findings on the size, competence, and experience of the teacher workforce. In Section 5, we investigate the role of ideology-based screening on local staffing. Section 6 concludes.

2 Reorganization of Primary Education Under Franco

As provinces fell under rebel forces’ control during the Spanish Civil War, the Francoist regime initiated a far-reaching reorganization of the education system. Ensuring teachers’ ideological conformity was a central aim of the emerging authoritarian regime. This section outlines the prewar structure of the primary school system, the purge of teachers during and after the war, and the subsequent relocation and hiring policies aimed at rebuilding the workforce.

2.1 Primary Education in Spain Before the Civil War

Prior to the Second Republic in 1931, Spain’s primary education system operated under limited administrative capacity, persistent underinvestment, and significant territorial disparities. Although national legislation dating back to the nineteenth century – most notably the 1857 *Ley de Instrucción Pública*, which declared elementary schooling compulsory – established a legal foundation for mass education, implementation remained incomplete. In rural municipalities, where school provision and teacher hiring relied on local public finances, provision was particularly weak. As a result, Spain lagged behind other Western European countries in educational attainment ([Barro and Lee, 2013](#)). According to the 1930 Census, illiteracy among individuals aged 18-40 was 27.8%, with substantial regional and rural-urban gaps. Overall, the pre-Republic education system reflected limited state capacity and unequal access to

primary education.

Between 1931 and 1933, the early Second Republic launched a major expansion of primary education. Reforms prioritized the construction of schools, particularly in rural areas; a large increase in the number of primary school teachers; the promotion of mixed-gender and secular education; and the adoption of modern pedagogical methods. A 1931 hiring round added 4,965 teachers who entered service in 1934, and the 1933 round appointed 11,138 teachers deployed in 1935, before the Civil War halted subsequent rounds. An innovation in the 1933 cohorts, unlike earlier ones, was that pedagogical training was part of the examination process. Though reform implementation slowed after a conservative coalition took office in late 1933 and fiscal pressures from the global economic crisis intensified (Molero Pintado et al., 2001), the primary educational system in 1936 differed markedly from that of 1931.

2.2 The Ideological Purge of the Primary Teacher Workforce

The Civil War's outbreak in 1936 immediately placed the education system under scrutiny. As rebel forces consolidated territorial control, the new Francoist authorities began reorganizing the education system and purging teachers deemed politically or ideologically disloyal. The emerging authoritarian regime regarded Republican educational reforms and the teacher workforce as central to what it viewed as a deterioration of national tradition and religious values. Ensuring teachers' ideological conformity thus became an early state priority. This objective was formalized in one of the regime's first decrees, which established the purge of the teacher workforce:

The fact that, for decades, the teaching profession at all levels has been influenced and almost monopolized—save for increasingly rare exceptions—by ideologies and dissolvent institutions in open opposition to the national spirit and tradition, makes it necessary, in the solemn circumstances we face, to undertake a total and thorough review of Public Instruction personnel. This shall constitute a preliminary step toward the radical and definitive reorganization of education, thereby uprooting those false doctrines and their advocates, which have been the principal factors leading our homeland to its tragic situation.

Decree 66, regulating the purge of teachers. November 8, 1936

The purge extended across the civil service, but education held a central role in the regime's ideological project. Schools were viewed as essential for ideological control and the transmission of regime values.¹ Although the purge affected all levels of the

¹ For a contemporaneous democratic perspective on the role of schools as civic institutions, see (Dewey, 1899; Dewey, 1916).

education system, we focus on primary schooling, which had the broadest reach and influence, particularly given the limited access to secondary and higher education in Spain at the time.²

The purge of primary school teachers proceeded through several administrative steps. First, all in-service teachers were required to submit a petition to a provincial review commission attesting to their suitability to remain in service; non-submission resulted in automatic dismissal, ensuring full screening of the workforce. Second, the commission gathered statements from local authorities and trusted community figures (e.g., the local priest, mayor, and police chief). Third, based on the collected information, the commission issued a preliminary proposal for reinstatement or sanction. Appendix Table C1 summarizes the charges most commonly filed against sanctioned teachers. Teachers facing sanctions could contest the proposal, prompting a second review. Finally, a central commission issued the definitive decision, ratifying or modifying provincial proposals.³ Although decisions could be appealed to a national review board, appeals rarely succeeded and involved long delays. For detailed historical accounts of the purge process, see [Morente Valero \(1997\)](#) and [Díaz Agulló and Fernández Soria \(1999\)](#).

According to our harmonized purge data, 27.3% of primary school teachers were sanctioned, including 9.7% who were permanently dismissed. Additional non-exclusive sanctions included temporary suspensions of six months to five years (7.5%), forced relocations to different municipalities or provinces (10.5%), and bans on promotion to managerial positions, such as headteachers or inspectors (15.3%).⁴ Most sanctions were issued between 1938 and 1941.

2.3 Regime Policies to Relocate and Hire New Teachers

As a consequence of the purge, the regime faced substantial local shortages. To address these staffing gaps, it relied primarily on voluntary relocation contests and, subsequently, new hiring rounds.

The regime's first major response was a large relocation contest, announced in 1941 and implemented in 1943. The Ministry of Education published a list of available vacancies, and non-sanctioned teachers with at least three years of tenure could submit ranked locality preferences. Assignments were determined by stated preferences and

² Historical work on the purge in secondary and higher education includes [Lorenzo Vicente \(2003\)](#) and [González Roldán \(2023\)](#). Purge intensity was highest in tertiary education, where 23.4% of full professors were dismissed. For comparison, [Waldinger \(2010\)](#) documents an 18.3% dismissal rate among mathematics professors in Nazi Germany.

³ Figures C1, C2, and C3 provide examples of purge documentation issued by provincial and central commissions.

⁴ These magnitudes are consistent with those reported in the historical literature ([García Díaz, 2018](#)).

a priority rule based on tenure and loyalty-related credentials (e.g., war service). In total, 12,139 teachers were reassigned, a scale consistent with the regime's objective of backfilling purge-related vacancies. In addition, approximately 10% of teachers were subject to forced relocation sanctions between 1938 and 1941, often facing delays in redeployment.⁵ Together, these voluntary and forced transfers generated substantial mobility. The regime directed teachers toward specific destinations through posted vacancies, but did not restrict origin localities in voluntary relocations, nor did it condition forced relocation sanctions on staffing needs at origin localities.

Hiring rounds were slower and more administratively costly than relocations. Immediately after the war, the regime first held two special openings for former military officers in 1939 and 1941, appointing a total of 2,273 teachers in 1940 and 1942. Two subsequent general hiring rounds were announced in 1941 and 1944, resulting in the appointment of 6,110 and 9,965 teachers, respectively, in 1944 and 1946. Applicants were subject to ideological screening and, as in the relocation contests, priority was given to candidates with demonstrated loyalty. The 1941 hiring round initially aimed to recruit 5,000 teachers, but entry requirements were relaxed in 1942, underscoring persistent staffing shortages created by the purge.⁶

3 A Novel Panel on Teacher Careers

To study how the Francoist regime reshaped the teaching workforce, we construct a panel tracking teachers' careers from 1934 to 1946, spanning the years before, during, and after the purge. We first describe the historical sources digitized for the analysis. Next, we explain how teachers' employment localities are geocoded to municipalities and how individual records are linked across datasets. Finally, we present the resulting panel used in the analysis.

3.1 Collected Data Sources

Teacher rosters. Our starting point for constructing the panel of teacher careers is the teacher rosters. These publications listed all in-service primary school teachers at a given point in time and were issued periodically by the Ministry of Education. For

⁵ Roughly 40% of teachers sanctioned with forced relocation also received temporary suspensions of varying duration, and administrative frictions often slowed redeployment.

⁶ Order of 19 May 1941 announcing competitive examinations for entry into the National Teaching Corps to fill 5,000 positions, *Official State Gazette* (BOE), no. 140 (20 May 1941), p. 3605 ([in Spanish](#)); Order of 11 April 1942 expanding the number of positions to be filled in the competitive examinations for entry into the National Teaching Corps, announced on 19 May 1941, to include all candidates who obtained a score of no less than five points in the third examination. *Official State Gazette* (BOE), no. 107 (17 April 1942), p. 2713 ([in Spanish](#)).

the period relevant to our study, complete rosters were issued in 1934 and 1946, with separate volumes for male and female teachers.⁷

These volumes, available at the Ministry's library, report individual-level information for each teacher, including full name, province and date of birth, years of experience, employment locality, and administrative rank within the public system, which determined pay levels.⁸ Figure C4 shows sample pages from the two rosters.

We digitize the complete teacher roster for 1934 and 1946, yielding 44,217 teachers in 1934 and 55,796 in 1946. Panel A of Table A1 presents descriptive statistics. Women represented 48.7% of the workforce in 1934 and 52.1% in 1946. In 1934, 51.5% of teachers worked in their province of birth, compared to 53.1% in 1946. Average age increased from 37.6 to 40.6 years, and average experience rose from 10.9 to 14.0 years.

Purge records. Data on the purge process comes from individual case files preserved in the *Archivo General de la Administración*. Every teacher was required to undergo the process (see Section 2.2), and records were kept for both sanctioned and non-sanctioned individuals. As a result, the files provide detailed information on sanctions and offer a complete snapshot of the teaching workforce at the onset of the purge.

We digitize and harmonize individual records compiled by historians for fifteen provinces: Albacete, Barcelona, Bizkaia, Cádiz, Castelló, Ciudad Real, Cuenca, Girona, Guadalajara, Huelva, Murcia, Tarragona, Toledo, València, and Valladolid.⁹ These provinces account for approximately 27% of all in-service teachers in 1934. For each record, we collect the teacher's full name, employment locality, sanction imposed, and the date of resolution.

Table A2 reports summary statistics for purge outcomes in the fifteen provinces collected. Overall, 27.3% of primary school teachers were sanctioned, including 9.7% who were permanently dismissed. Additional non-exclusive sanctions included temporary suspensions of six months to five years (7.5%); forced relocations within the province of employment (4.7%), with an average duration of 29.2 months; forced relocations outside the province of employment (5.8%), averaging 52.3 months; and bans on promotion to managerial positions such as headteachers or inspectors (15.3%).

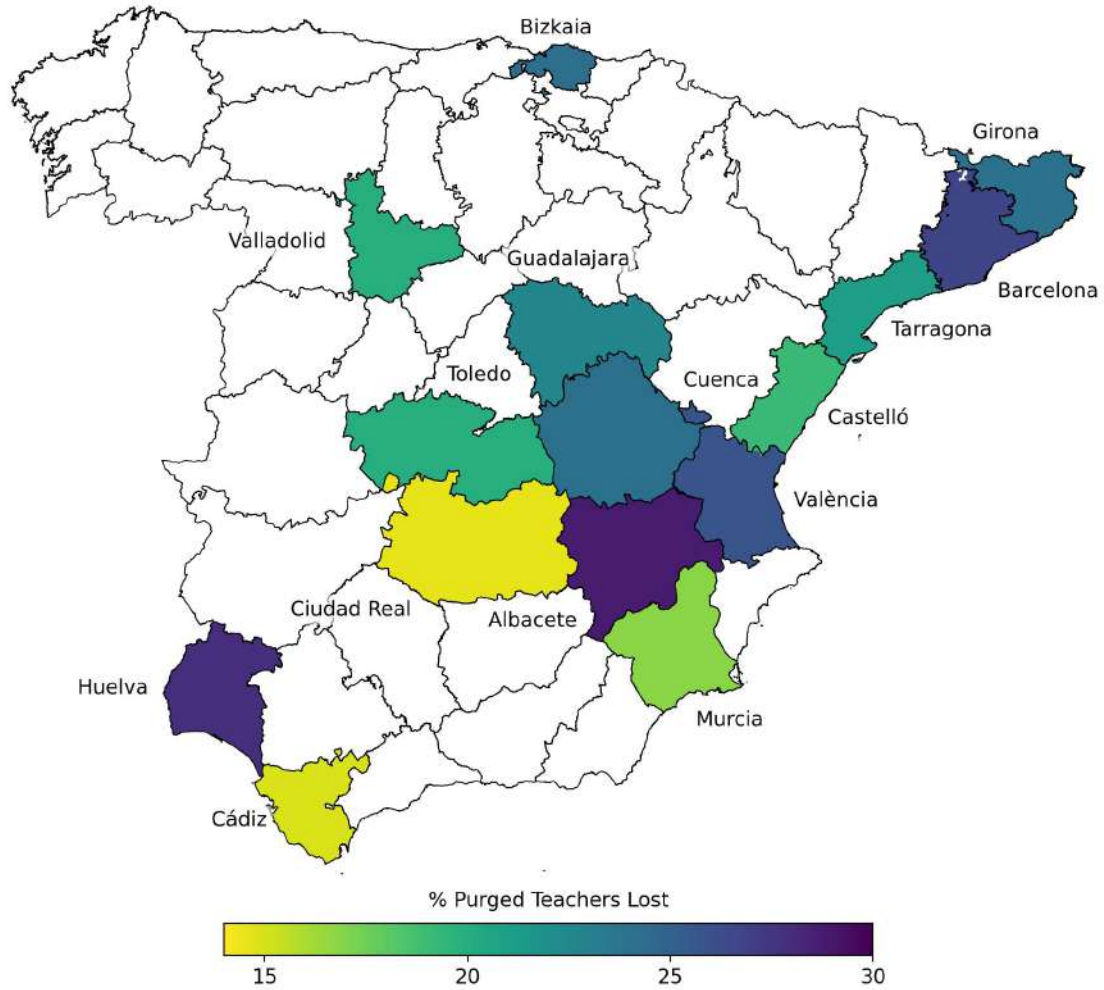
Hiring lists. While the teacher rosters provide comprehensive information on all in-

⁷ The 1934 includes all in-service teachers on December 31, 1933, and all new hires incorporated during 1934. The 1946 roster covers those in-service teachers on December 31, 1945, and all new hires incorporated during 1946.

⁸ Rank was mainly determined by seniority. Among teachers who entered the public education system simultaneously, rank was assigned based on performance in the entry examination or in required training courses. Rank promotions could also be delayed or accelerated for several reasons, such as leaves of absence or sanctions resulting from the purge.

⁹ We have newly collected and digitized archival records for ten additional provinces, which will be incorporated in future versions of this paper.

Figure 1: Share of teachers sanctioned with dismissal, temporary suspension, or forced relocation, by province



Note: The figure shows a map of mainland Spain with all provinces for which we digitize teacher purge records. These provinces include Albacete, Barcelona, Bizkaia, Cádiz, Castelló, Ciudad Real, Cuenca, Girona, Guadalajara, Huelva, Murcia, Tarragona, Toledo, València, and Valladolid. For each of these provinces, we depict the share of teachers that were sanctioned either with dismissal, temporary suspension, or forced relocation.

service teachers, they do not identify the cohort in which each teacher entered the profession. To recover this information, we digitize the official lists of new hires issued by the Ministry of Education, corresponding to the recruitment rounds held between 1928 and 1946.

For the prewar period, we collect data from the last four hiring rounds. First, the 1928 round – the last under the Primo de Rivera dictatorship – incorporated 4,752 new teachers who began service in 1930-1931. Second, the 1931 round, initiated before and completed after the transition to the Second Republic, added 4,965 teachers who entered in 1934. Third, the 1933 round, organized entirely under the Republican government, appointed 11,138 new teachers deployed across the territory in 1935. The final prewar

round, announced in 1935, was interrupted by the outbreak of the Civil War, delaying the appointment of these teachers until the early Francoist period, once the regime screened their ideological conformity through the purge process.

For the Francoist period, we digitize four additional hiring lists. Two correspond to special openings reserved for former military officers, held in 1939 and 1941, which incorporated 1,669 and 604 teachers, respectively. These rounds sought to facilitate the reintegration of demobilized officers and to alleviate teacher shortages, with appointments taking effect in 1940 and 1942. The remaining two openings refer to general recruitment contests announced in 1941 and 1944, which incorporated 6,110 and 9,965 new teachers, respectively, appointed in 1944 and 1946. Although formally open, these contests effectively limited entry to candidates aligned with the regime, as applicants were subject to ideological screening and preference was given to those with proven loyalty, including war-related merits.

Figures C5 illustrate sample pages from the original hiring lists. Each list reports the full name of every newly appointed teacher and their relative ranking within the round, determined by performance in the teacher-training program and the entry examinations. We use this information to construct competence measures for the 1928, 1931, and 1933 rounds, as described in Subsection 4.2.¹⁰

3.2 Assembling the panel of teachers

Geocoding. The data sources used to construct our panel of teacher careers contain detailed information on each teacher's locality of employment. To use this information, we need to assign the corresponding municipality code from the population census. However, two main challenges arise in this process. First, the documentation often refers to small territorial units – such as parishes, hamlets, or neighborhoods – that are smaller than the municipality. In these cases, we need to identify and assign the corresponding municipality. Second, the territorial division of Spain has changed significantly over time, requiring us to reconcile information across census years with different municipal units.

To implement this process, we first assign to each locality listed in the historical sources its corresponding municipality and census code using the closest prior census. For the 1934 roster, we use the 1930 census; for the purge records and the 1946 roster, we use the 1940 census. When the listed locality corresponds to a municipality, we assign the associated census code directly. When the documentation refers instead to territorial units smaller than a municipality, we use historical gazetteers to identify the

¹⁰ In all hiring rounds, male and female teachers were listed separately. Competence measures based on rank are therefore constructed independently by gender.

municipality in which the locality was located.¹¹

Once teachers' employment localities are geocoded to their corresponding municipal codes, we harmonize territorial units across time to account for changes in municipal boundaries. To address this challenge, we rely on the correspondence tables between censuses developed by [Goerlich Gisbert et al. \(2015\)](#) and define *pseudomunicipalities* as the smallest groups of municipalities that remain stable throughout the period under study. In each census, every municipality is assigned to a *pseudomunicipality* that is consistent over time. Municipalities that do not experience any change form a *pseudomunicipality* on their own. This procedure reduces the total number of municipalities from 9,262 in 1930 to 7,933 *pseudomunicipalities* that remain stable over the entire period. For clarity, we refer to these units as "municipalities" throughout the paper. Beyond facilitating the linkage of teacher records and enabling our analysis, this structure also makes the dataset a useful resource for future research, as it can be readily linked to a wide range of contemporary datasets.

Voluntary relocations. To link teachers across the 1934 roster and the purge records, we need information from voluntary relocation contests published in the *Gaceta de Madrid* (Official Gazette of Madrid). Because the purge records do not contain demographic variables for teacher identification, accurate pre-purge employment localities are required to link teachers who relocated after 1934. Voluntary relocations took place through nationwide contests.¹² We digitize data from two prewar rounds, conducted in 1934 and 1935; a third round, announced in 1936, was disrupted by the outbreak of the Civil War.

Relocation contests followed a standardized procedure. The Ministry of Education announced available vacant positions, teachers submitted ranked preferences for reassignment, and placements were made according to these preferences and tenure.

The transfer lists report each reassigned teacher's full name, origin locality, and destination locality. In total, 7,005 and 1,378 teachers were relocated in 1934 and 1935, respectively. Figure [C6](#) reproduces a sample page of the 1934 relocation contest from the original source.

Linking teacher records. To construct our panel of teacher careers, we link individual-level records across the historical sources described above. We follow a three-step procedure that exploits overlapping information across pairs of sources to minimize matching errors.

¹¹ Gazetteers for both 1930 and 1940 are available for our period of study. They provide comprehensive listings of settlements, parishes, and neighborhoods within each municipality, which we use to assign sub-municipal localities to their corresponding administrative units.

¹² Individual transfers outside these contests were uncommon and mostly limited to moves within the same locality.

First, we link the 1934 and 1946 rosters through a sequential procedure that combines exact one-to-one matches on composite identifiers – constructed from teachers’ full name and demographics: sex, month and year of birth, and province of birth – with name-similarity matching within demographic blocks (birth province, birth year, and tenure windows). Conservative similarity thresholds are applied to limit false positives. Borderline matches are reviewed manually, and accepted pairs are deduplicated to ensure a unique one-to-one correspondence between rosters.

Second, we link the 1934 roster to the purge records for the fifteen provinces processed to date, using teachers’ full names and employment localities. When teachers relocated after 1934, we recover teachers’ pre-purge employment municipalities from the voluntary relocation lists. We then apply an analogous matching sequence: exact matching on composite identifiers (name and province of employment), followed by name-similarity matching within province- and municipality-of-employment blocks, and finally without blocking. Conservative thresholds, manual review, and deduplication ensure a one-to-one correspondence.

Third, we link the teacher rosters to the hiring lists to identify recruitment cohorts. The 1934 roster is matched to the 1928, 1931, and 1933 prewar hiring rounds. The 1946 roster is matched to the 1935 round, the 1939 and 1941 military openings, and the 1941 and 1944 hiring rounds. We assign teachers to hiring rounds using the seniority ordering in administrative ranks, as relative positions within entry cohorts are generally preserved. For prewar entry cohorts, we also use within-round rankings to construct our competence measures.

The match rate between the 1934 and 1946 rosters, for teachers already hired by 1934, is 99.3%, due to the detailed identifying information available. The match rate between the 1934 roster and the purge records is 65.3%, reflecting more limited information in the purge records, but it remains within the range reported in comparable historical linking exercises.

Panel B of Table [A1](#) presents descriptive statistics for the subsample of teachers with linked purged records, which represent approximately 25% of the 1934 roster. Compared with the full sample in Panel A, it closely mirrors it in average gender, age, and experience. The 1946 column in Panel B shows the subset of linked teachers who remained employed in 1946. Table [A2](#) summarizes purge outcomes for linked teachers: 30.9% were sanctioned and 10.7% permanently dismissed, compared to 27.3% and 9.7% in the full set of teachers in the fifteen provinces with purge records. The similarity across samples indicates that the matching procedure does not introduce systematic bias.

3.3 Our final dataset of teachers

The final dataset consists of two complementary panels of teachers. The first links the 1934 and 1946 rosters, producing a career panel for the universe of primary-school teachers that spans the years before and after the purge. For each teacher, we observe the municipality of employment, demographics, experience, and hiring round. For teachers in the 1928, 1931, and 1933 entry cohorts, we also obtain a relative measure of competence (see Section 4.2). This panel allows us to document changes in workforce size, experience, gender composition, and geographic allocation across municipalities, and tracks entry and exit over the period. Because it covers the full public teaching force, it provides the baseline structure for studying how the regime rebuilt and reorganized the education system following the purge.

The second panel augments the roster-based panel with purge outcomes for individuals working in one of the fifteen provinces where individual purge records have been collected. The resulting linked purge panel identifies teachers who were permanently dismissed, temporarily suspended, or forcibly relocated, and allows comparisons between sanctioned and non-sanctioned teachers within the same entry cohort. It also allows for aggregation to the municipality level to study how local purge intensity and replacement policies shaped municipal staffing capacity in Section 5.2. Together, the two panels provide comprehensive evidence on how ideology-based screening reshaped both the composition and geographic distribution of frontline public servants.

4 Ideology-Based Selection and the Workforce Size and Quality

This section examines how ideology-based selection in the Francoist purge shaped the size, competence, and experience of the teaching workforce. First, we document that the number of in-service teachers returned to prewar levels only seven years after the purge, while teacher–student ratios remained persistently lower. Second, we provide evidence that the purge disproportionately dismissed more competent teachers, thereby lowering the overall quality of the workforce. Finally, we show that the purge was associated with a sizable decline in teacher experience.

4.1 How the Purge Disrupted the Workforce Expansion

Reconstructing the teaching series. We reconstruct the series of in-service and trained teachers from 1925 to 1946 to assess how the Francoist purge altered the expansion of the teaching workforce. The period covers the years before and during the Second

Republic, the Civil War, and the first seven years of Franco's rule.

For the prewar period (1925–1936), we use teachers in our panel who appear in the 1934 national roster. We leverage the recruitment dates we linked from historical hiring lists to this roster. Using these combined data, we recover the stock of in-service teachers in 1925 and sequentially add new entrants from each hiring round through 1935. We also include those who had completed training by 1936 and were about to enter service when the conflict began. Finally, we adjust the series applying a natural yearly exit rate of 1.05%, which we estimate from attrition patterns in our teacher panel.

For the postwar period (1940–1946), we use the 1946 roster of teachers, which already reflects purge-related dismissals and separations. Applying the same procedure, we trace the rebuilding of the workforce under the new regime by estimating the number of in-service teachers in 1940 and adding successive cohorts hired during the early Francoist years. Again, we adjust the series assuming a 1.05% yearly exit rate. This reconstruction yields a continuous series of teacher counts that closely tracks the evolution of the staffing capacity before, during, and after the purge.

Evolution of the teaching workforce. Figure 2 depicts the evolution of the number of in-service and trained teachers over the 1925–1946 period.¹³ The number of teachers was already growing in the years preceding the Second Republic, but the expansion accelerated after 1931 – especially from 1934, when the first cohort fully trained under the Republic entered service. In 1940, however, the series exhibits a sharp decline of about 24% compared to prewar levels, reflecting the combined effects of the purge dismissals and other war-related losses. Subsequent hiring waves initiated soon after helped the system recover, yet by 1946, the total number of teachers remained nearly unchanged from its 1936 level.

While the evolution of the number of teachers suggests that the purge and subsequent reorganization imposed sizable costs on teaching capacity, a potential concern is that the regime was adapting to a lower schooling demand. To address this, we digitize historical census data for 1920, 1930, 1940, and 1950, which provide nationwide population counts by age group. We then interpolate across census years to estimate the school-age population for each year and compute the teacher–student ratio to track changes in teaching capacity.

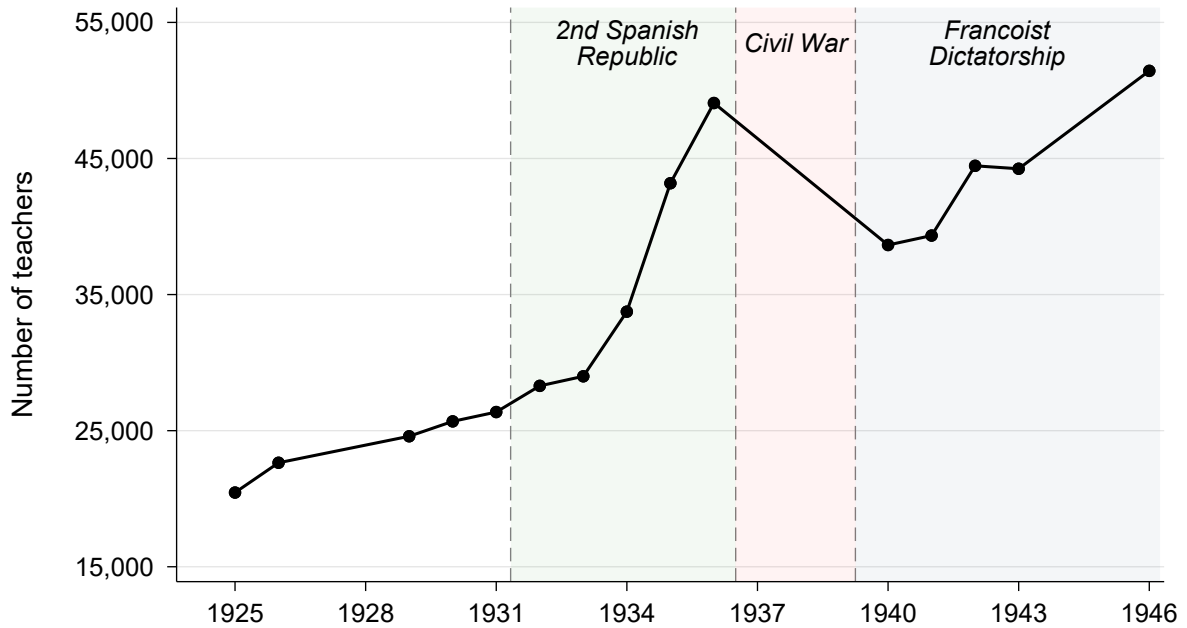
Figure B1 plots the evolution of the teacher-student ratio, expressed as the number of teachers per 50 students.¹⁴ We use this benchmark because both the Second Republic and the Francoist government adopted it as a policy goal.¹⁵ The pattern closely mirrors

¹³ Table B1 provides the corresponding counts for each year.

¹⁴ Table B1 details the evolution of the school-age population and the teacher-student ratio in our series.

¹⁵ This is established in, for the Second Republic, in the Directorate General of Primary Education,

Figure 2: Evolution of the number of in-service and trained teachers in the 1925-1946 period



Note: This figure depicts the evolution of the teacher-student ratio for the 1925-1946 period. The method for reconstructing the series is described in Subsection 4.1 and uses data from our teacher panel and historical census data, including population counts by age group. Table B1 details the evolution of the corresponding teacher counts for each year.

that of the total number of teachers. First, the teacher-student ratio in 1925 stood at only 0.35 teachers per 50 students – well below the stated objective. Then by 1931, it had risen to 0.43, and by the onset of the Civil War, it had almost doubled to 0.77 due to the expansion of the Second Republic. Following the purge, however, the ratio fell back to 0.58 teachers per 50 students, roughly half the policy target. By 1946, it had recovered to 0.84, yet this level was just slightly above to that achieved in 1936. Overall, this evidence confirms that the purge was associated with persistent staffing shortfalls, effectively erasing a decade of potential growth.

Because teacher counts and teacher-student ratios evolve similarly over the period, we use percentage changes in the number of teachers to measure changes in staffing capacity in the remainder of the paper. We additionally report results based on teacher-student ratios as robustness checks when appropriate.

4.2 Ideology-Based Selection and Teacher Competence

Measuring teacher competence. To assess the relationship between ideology-based selection and workforce quality, we begin by constructing a measure of teacher com-

“Circular,” *Gaceta de Madrid* (Official Gazette), no. 265 (22 September 1934): 2542, and for the Francoist government, in the *Law on Primary Education* (17 July 1945), *Boletín Oficial del Estado* (Official State Gazette), no. 199 (18 July 1945), art. 51, among others. See Figure C7 for more details.

petence. We use individual rankings from the official hiring lists, which report male and female candidates separately. Each list ranked candidates according to their performance in the teacher-training program and the entry examinations, including grades from coursework, the traineeship, and final examinations. Figure C8 shows two examples of these official lists, indicating teachers' positions and corresponding scores.

We compute competence for three entry cohorts: 1928, 1931, and 1933. Because the content and grading of the training programs changed over time, the original scores are not directly comparable across cohorts. Therefore, for each cohort, we assign teachers to competence quintiles based on their percentile rank within the list. Our final measure thus captures a teacher's relative competence within their cohort.

Estimating separation rates. We estimate differences in separation rates across competence quintiles using the following regression:

$$\text{Separation}_{i,c} = \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} + \psi_{a(i)} + \psi_{c,g(i)} + \varepsilon_{i,c}. \quad (1)$$

where i denotes the teacher and c the cohort. $\text{Separation}_{i,c}$ is an indicator variable for teachers who were separated from service by 1946. $\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\}$ are indicator variables for competence quintiles, with the bottom quintile omitted as the reference group. $\psi_{a(i)}$ and $\psi_{c,g(i)}$ denote age and cohort-by-gender fixed effects, respectively. The coefficients β_q capture the difference in separation probabilities between competence quintiles and the bottom quintile. Positive and increasing values of β_q for higher competence quintiles would be consistent with a loyalty-competence trade-off in the public workforce.

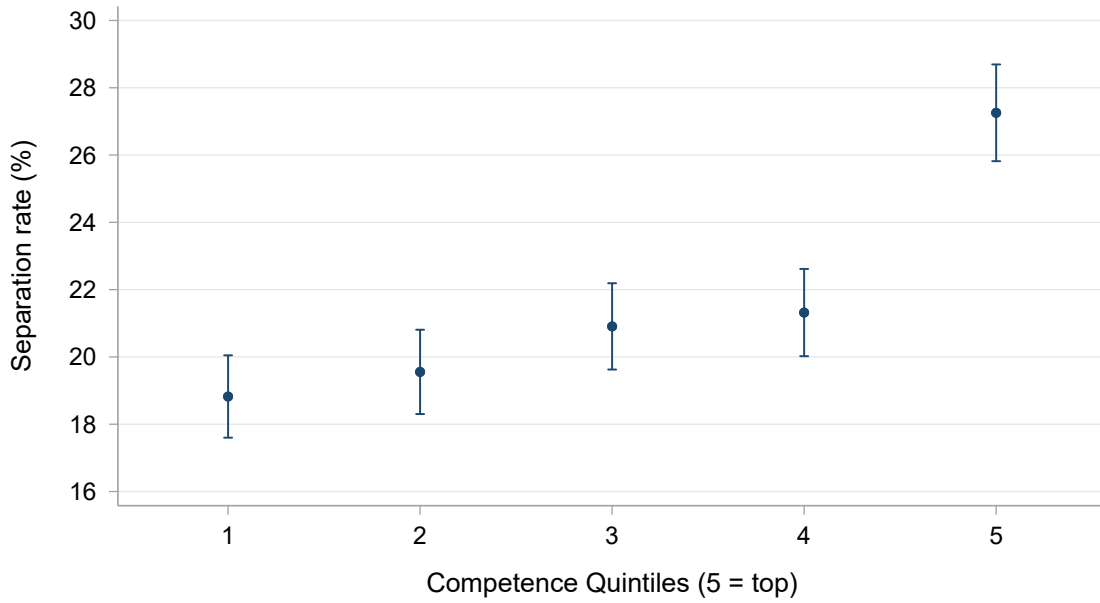
Main results. Figure 3 plots the estimated separation probabilities across competence quintiles from Equation 1. Separation rates increase steadily with teacher competence: teachers in the top quintile faced roughly a 40 percent higher probability of being separated from service than those in the bottom quintile.

Table 1 displays results for different subsamples. Columns (1)–(3) report estimates for the pooled sample of male and female teachers, with Column (3) replicating the results shown in Figure 3. Columns (4)–(5) report estimates for male teachers, whereas Columns (6)–(7) present results for female teachers. Robust standard errors are reported in parentheses.

Table 1 shows that the gradient in separation probabilities across competence levels is mainly driven by male teachers.¹⁶ The coefficients for all quintiles in Columns (4) and

¹⁶ Figure B2 provides a graphical representation of this difference based on the estimates in Columns (5) and (7) of the table.

Figure 3: Teacher separation rates across competence quintiles



Note: The figure presents a non-parametric representation of the results in Column (3) of Table 1. The dependent variable is an indicator variable that equals one for teachers who were separated from service by 1946. The independent variable for binning is a categorical variable for the competence quintile. It is constructed by first partialling out gender-by-cohort fixed effects. Each dot represents the mean separation rate in the residualized data. Blue lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

(5) are statistically significant and increase with competence. The largest difference is observed in the top quintile, where the gap in separation rates between the highest- and lowest-competence group is about 40 percent of the sample mean. Among women, as shown in Columns (6) and (7), only the top-quintile coefficient is statistically significant, and its magnitude is roughly one-third of that for men. Average separation rates also differ markedly by gender: 29.8% for men and 12.5% for women. The larger scale among men is consistent with their greater exposure to sanctions during the purge, as described in historical accounts documenting their higher political and social participation and greater public visibility in the years preceding the Civil War. These characteristics were likely correlated with competence and help explain why the gradient emerges only in this subsample.

Robustness. In the previous analysis, we estimate differences in separation rather than sanction rates, even though separations may also include exits related to other causes. We focus on separations for two reasons. First, this measure allows us to estimate patterns in the nationwide sample of teachers hired between 1928 and 1933, rather than in the subset of fifteen provinces for which purge outcomes are digitized. This is important both for representativeness and for statistical power, given that we already work with a limited set of cohorts. Second, the separation measure also captures exits related to wartime disruptions and political violence, which, although often recorded

Table 1: OLS estimates: teacher separation rates across competence quintiles

	All teachers			Male teachers		Female teachers	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Competence Q2	0.731 (0.894)	0.731 (0.894)	0.639 (0.894)	3.067 (1.413)*	2.944 (1.407)*	-1.802 (1.057)	-1.825 (1.063)
Competence Q3	2.093 (0.905)*	2.093 (0.904)*	1.880 (0.905)*	4.841 (1.425)***	4.510 (1.424)**	-0.887 (1.077)	-0.907 (1.081)
Competence Q4	2.499 (0.910)**	2.497 (0.910)**	2.269 (0.914)*	4.820 (1.427)***	4.368 (1.432)**	-0.022 (1.093)	-0.056 (1.099)
Competence Q5	8.439 (0.964)***	8.438 (0.963)***	8.115 (0.969)***	12.216 (1.491)***	11.807 (1.500)***	4.340 (1.188)***	4.202 (1.194)***
Observations	18,225	18,225	18,225	9,482	9,482	8,743	8,743
R ²	0.05	0.05	0.06	0.01	0.02	0.01	0.01
Outcome average	21.49	21.49	21.49	29.79	29.79	12.49	12.49
Gender FE	Yes	No	No	No	No	No	No
Cohort FE	Yes	No	No	Yes	Yes	Yes	Yes
Gender × Cohort FE	No	Yes	Yes	No	No	No	No
Age FE	No	No	Yes	No	Yes	No	Yes

Note: This table displays the estimates of the OLS regression described in Equation 1. Our sample includes 18,225 teachers that entered the profession between 1928 and 1933; 9,482 male teachers and 8,743 female teachers. Columns (1) to (3) show estimates for the full sample of male and female teachers. Columns (4) and (5) show estimates for the subsample of male teachers, whereas columns (6) and (7) show them for the subsample of female teachers. The complete specification estimated in Column (3) is $\text{Separation}_{i,c} = \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} + \psi_{a(i)} + \psi_{c,g(i)} + \varepsilon_{i,c}$, where i denotes the teacher and c the cohort. $\text{Separation}_{i,c}$ is an indicator variable taking value one for teachers who were separated from service following the purge. $\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\}$ are indicator variables for competence quintiles, with the bottom quintile omitted as the reference group. $\psi_{a(i)}$ and $\psi_{c,g(i)}$ denote age and cohort-by-gender fixed effects, respectively. We report heteroskedasticity-robust standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

among sanctions, may still be underreported in official purge records.

A potential concern is that voluntary exits may have been more common among teachers with higher competence levels. However, such exits are unlikely to represent a substantial fraction of separations at any competence level. Evidence from the labor-market monopsony literature suggests that teachers are among the occupations more exposed to employer concentration, reflecting their limited outside options and low exit rates (Ransom and Sims, 2010; Schubert, Stansbury, and Taska, 2022). These constraints on outside options were likely even tighter in the immediate postwar period, when public-sector employment offered stable pay and few alternatives existed. Nonetheless, we test this possibility by estimating the following regression:

$$\begin{aligned}
\text{Separation}_{i,c} = & \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} \\
& + \sum_{q=2}^5 \gamma_q \left(\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} \times \text{Dismissal}_{i,c} \right) \\
& + \delta \text{Dismissal}_{i,c} + \psi_{a(i)} + \psi_{c,g(i)} + \varepsilon_{i,c}
\end{aligned} \tag{2}$$

where $\text{Dismissal}_{i,c}$ is an indicator variable for teachers who were permanently dismissed in the purge, and all other elements have the same definition as in Equation 1. The coefficients γ_q capture differences in separation rates across competence quintiles not directly explained by dismissal sanctions. Figure B3 shows that these differences are small and statistically insignificant, suggesting that selective voluntary exits are not a concern.

Furthermore, Table B2 reports estimates of Equation 1 for the subsample of teachers from the 1928, 1931, and 1933 cohorts for whom we digitized and linked purge records. We present results for three outcomes: separations, purge dismissals, and any type of purge sanction. The patterns observed are consistent across all three measures and closely mirror those in Table 1. Higher competence is associated with higher separation, dismissal, and sanction rates, although estimates are less precise – particularly for women – because of the smaller sample size. The magnitudes across the three outcomes are also comparable, given the lower average level of purge dismissals relative to separations. Importantly, separation and dismissal rates follow nearly identical trends within this group, supporting the use of separations as a valid proxy for purge dismissals in our nationwide sample. These findings suggest that our earlier results are not driven by higher outside options or selective attrition among more competent teachers.

A plausible alternative interpretation of our results is that the competence measure captures ideological alignment with preceding regimes rather than professional ability. This could occur if higher scores reflected conformity with the norms or expectations of training institutions rather than better performance. To examine this, Table B3 estimates Equation 1 separately by hiring cohort and gender. We find that the pattern of increasing separations with competence holds across all cohorts, including those trained under the earlier nondemocratic regime (1928 and 1931), though it is somewhat steeper for the 1933 cohort, who were fully trained under the Republic. The stronger gradient among the 1933 teachers may reflect some degree of political targeting of those trained under the Second Republic. It could also be partly explained by the greater political and social activism and the visibility of younger teachers, which – as

discussed above for gender differences – was likely correlated with competence and increased their exposure to sanctions. Nonetheless, the correlations observed across cohorts confirm that the relationship between competence and separations cannot be attributed to ideological bias in the competence measure.

4.3 How the Purge Shaped the Distribution of Teacher Experience

The observed distribution of experience. Using our panel of teachers, we estimate the distribution of experience among those in service in 1946 – seven years after the establishment of the regime. For this, we rely on the total experience reported in the 1946 roster. Figure B4 illustrates the resulting distribution, and Table B4 reports the corresponding summary statistics. In 1946, teachers averaged about 15 years of experience, with a median of 12; 25% had fewer than 7.5 years, and 10% had fewer than 2.

Estimation of the counterfactual. To assess how the purge shaped the distribution of experience, we estimate the counterfactual distribution that would have prevailed in 1946 had the purge not occurred. We use teachers listed in the 1934 roster, along with those who had completed training and were ready to be deployed in 1936, as the baseline sample. We take this roster as the starting point because it captures the full teaching force before any large-scale dismissals or war-related disruptions. For each teacher, we then compute potential experience by 1946, assuming continuous employment after 1934.

We impose three assumptions governing retirement, hiring, and non-purge separations. First, teachers are assumed to retire at age 70 (the official retirement age), after which they are replaced in that same year by a new entrant who begins accumulating experience from that point onward. Second, we assume no additional hiring beyond replacement, which yields a workforce size similar to that observed in 1946, as shown in Subsection 4.1. Third, based on our purge-linked data, we estimate a cumulative separation rate of about 10% between 1934 and 1946 among teachers not dismissed in the purge. We apply this same fixed rate randomly across non-retired teachers, assigning the year of exit uniformly over the period and replacing each with a new entrant in that year. This last assumption makes the simulation conservative: the observed 10% separation rate includes war-related exits that would likely have been lower in the absence of conflict, and the uniform timing of attrition delays replacements that likely occurred earlier in practice.

Main Results. Figure 4 compares the observed and counterfactual experience distributions in 1946. Panel (a) shows both distributions, while Panel (b) plots their difference in 2.5-year bins. The results reveal that the purge was associated with a sizable replace-

ment of experienced teachers – those with more than 10 years of tenure – by younger and largely inexperienced ones. The largest shortfall appears among teachers who would have accumulated 10 to 15 years of experience by 1946, but the deficit extends throughout the right tail of the distribution, reflecting the loss of teachers with considerable experience and, arguably, professional expertise. The sharp decline among those who would have reached 10 to 15 years of experience reflects higher dismissal rates among younger teachers in the purge, who would have attained mid-career status by 1946. From the regime’s perspective, dismissing these teachers could be ideologically desirable, given suspicions of their ideological alignment with Republican educational values. Yet many of them had been trained in modern and innovative pedagogical approaches and, by 1946, would have accumulated a decade of professional experience, making their removal a significant loss of teaching and human capital.

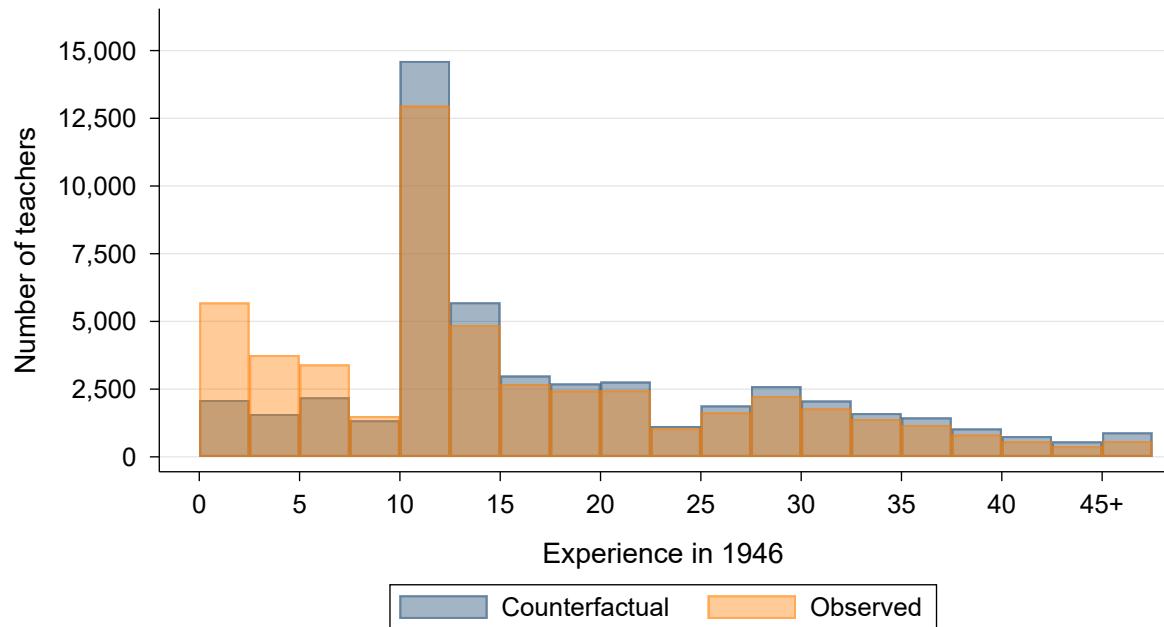
Table B4 reports summary statistics for the counterfactual distribution. Average experience in the counterfactual scenario is 17.8 years—about 2.6 years higher than in the observed distribution. The difference is even larger at the 75th percentile (25.3 versus 21.6 years) and particularly pronounced in the lower tail: the 10th percentile rises from 1.92 to 6 years, indicating that even among the least experienced teachers, the counterfactual workforce would have accumulated substantially more tenure. Overall, our results imply that, in six out of ten cases, a child who is randomly assigned to a teacher in the counterfactual scenario would be taught by a more experienced teacher than in the observed case.¹⁷

Alternative counterfactuals. We assess changes in experience under two alternative counterfactual scenarios. First, we relax the assumption on the natural separation rate. The baseline 10% cumulative rate may overstate normal turnover due to wartime disruptions and adverse postwar conditions. We therefore explore a lower value of 2.5%, approximating a scenario without additional conflict-related losses. Second, we relax the assumption of no new hiring. It is plausible that, had the Second Republic remained in power, teacher recruitment would have continued in the years that followed. We therefore impose a target of one teacher per 50 students – the policy goal of the Second Republic, as discussed in Subsection 4.1 – and assume a linear increase in the number of teachers to reach this objective by 1946.

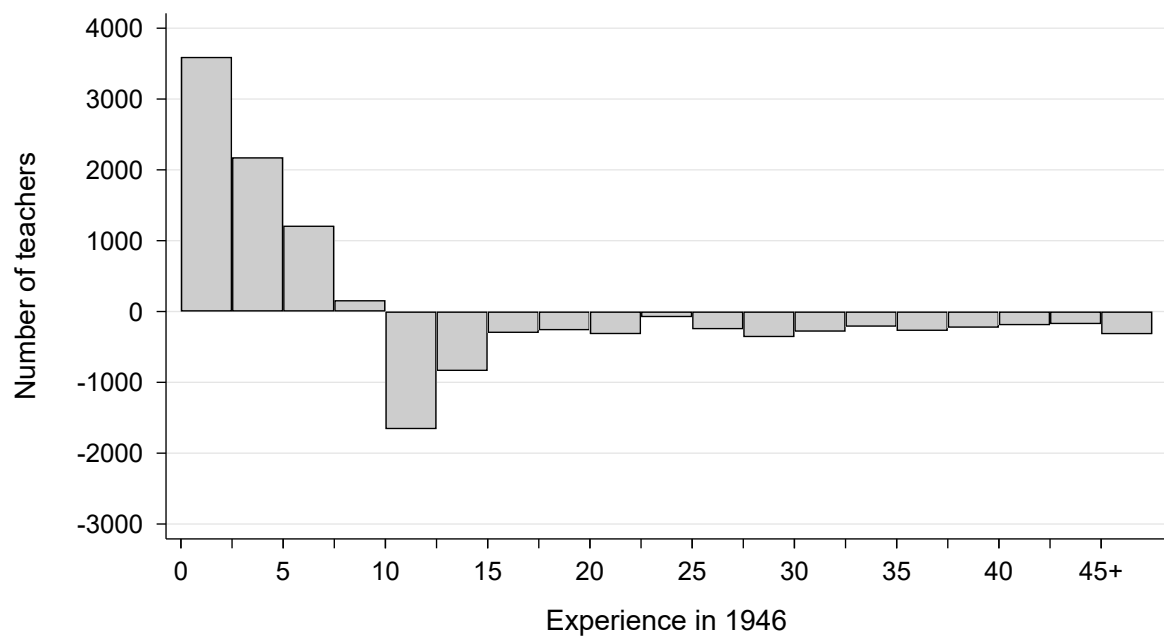
Figure B5 compares the observed and counterfactual experience distributions under the lower attrition assumption. The results show that, in this scenario, experience losses are larger for all experience levels above 10 years. The main differences arise from a

¹⁷ Formally, this corresponds to the probability-of-superiority statistic, defined as $P(\text{Experience}_i^{\text{counterfactual}} \geq \text{Experience}_j^{\text{observed}})$, where i and j denote teachers randomly drawn from the counterfactual and observed distributions, respectively. The measure captures the likelihood that a teacher in the counterfactual distribution has more experience than one in the observed distribution.

Figure 4: Observed and counterfactual distributions of teacher experience in 1946



(a) Distribution of teacher experience in 1946: actual vs no-purge counterfactual



(b) Difference between actual and counterfactual distributions

deeper shortfall among teachers who would have accumulated 10 to 15 years of experience by 1946 and a greater concentration of inexperienced teachers in the observed workforce. As reported in Table B4, the gap in average experience widens to about 3.6 years – roughly 38% larger than in the baseline case – suggesting that the baseline estimates might be a lower bound for the actual magnitude of the purge’s associated experience losses. The change in the distribution is visible across all percentiles but is especially pronounced at the 10th percentile, which increases from 1.9 years in the observed distribution to about 10 years in the counterfactual scenario, confirming the higher prevalence of inexperienced teachers in the 1946 workforce.

Figure B6 displays the observed and counterfactual distributions of teachers under the assumption of additional hiring to achieve the policy goal of a 50:1 student–teacher ratio. In this scenario, the pattern differs markedly from the baseline. First, the number of teachers in the observed distribution is lower across nearly all experience levels, not only among those with more than 10 years of tenure. Second, despite this overall difference, a noticeable shortfall remains among teachers who would have accumulated 5 to 15 years of experience by 1946.

If we focus on the relative distributions, the two profiles appear similar, yet they correspond to very different schooling systems. As shown in Table B4, the counterfactual is equal or lies slightly above the observed distribution across all percentiles, but differences in average experience and percentile values are negligible. However, the teaching workforce is about 18% larger in the counterfactual – rising from 0.84 to 1 teacher per 50 students. Thus, even if the Second Republic had expanded hiring to meet schooling demand and achieve its policy goal, the experience distribution would have been at least as favorable as that observed in 1946, while providing substantially greater staffing capacity.

5 Ideology-Based Selection and the Local Provision of Education

This section analyzes how the Francoist regime’s dismissal, relocation, and hiring policies reshaped the local provision of primary education. We first assemble a municipality-level panel of teacher staffing from the teacher-level data introduced in Section 3.3. We then examine how the regime responded to the staffing gaps created by the purge. Finally, we show that coordination frictions led to persistent local teacher shortages.

5.1 A Municipal-Level Panel of Teacher Staffing

To study how ideology-based selection reshaped local education provision, we construct a municipality-level panel of teacher staffing and purge outcomes. We aggregate our teacher-level panel to the geocoded municipality of employment for the fifteen provinces with digitized purge records (see Section 3.1). The panel covers three points in time: 1934, the onset of the purge (1938–1940), and 1946.¹⁸

For each municipality-year, we compute the number of teachers, gender composition, average experience, distribution across hiring cohorts, and the mean competence of teachers entering in the 1928, 1931, and 1933 rounds. We also construct measures of purge intensity by sanction type (e.g., permanent dismissal, forced relocation, temporary suspension), along with the net balance of relocations after the purge and the number of new hires.

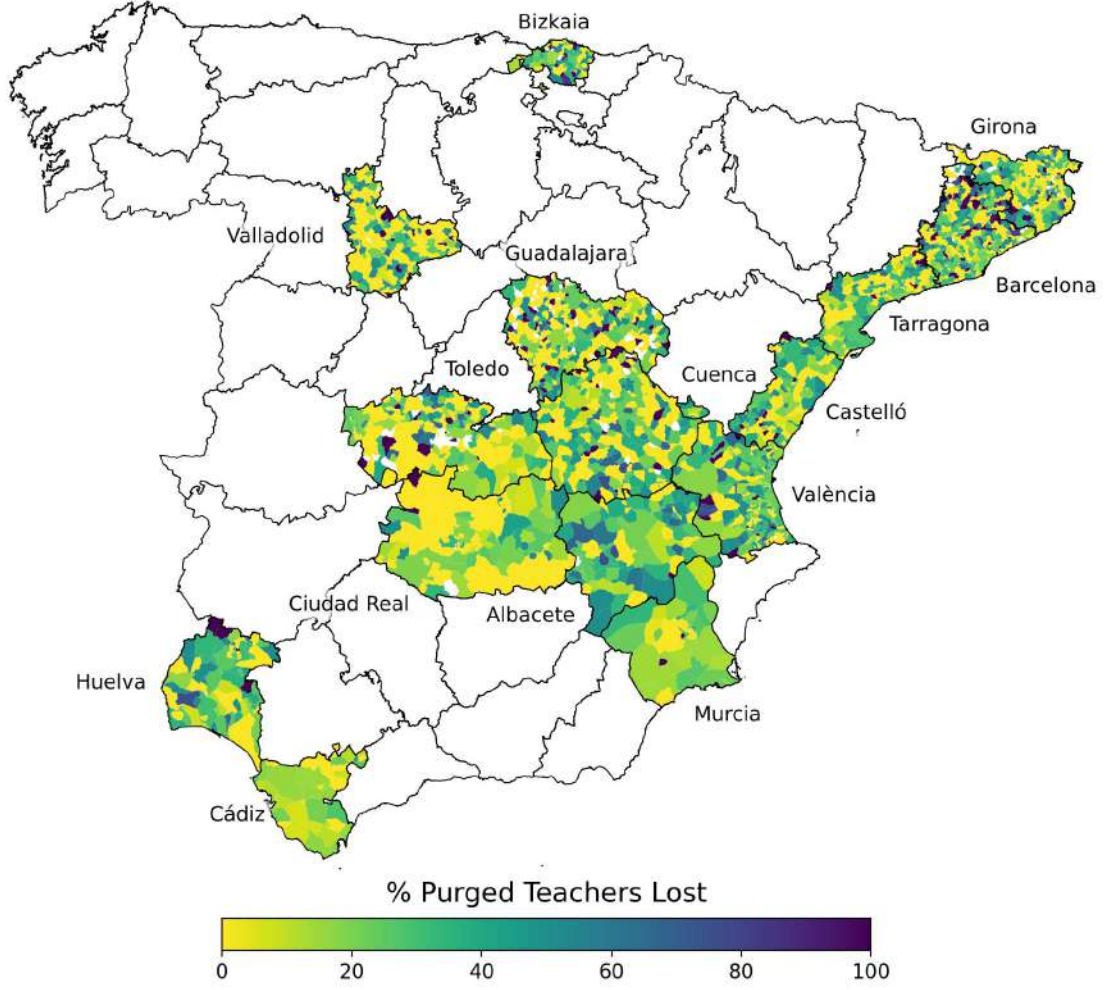
The resulting dataset includes 2,477 municipalities across 15 provinces, representing 31% of all Spanish municipalities and 27% of the teacher workforce in 1934. Table A3 reports summary statistics. The average number of teachers per municipality rose from 3.12 in 1934 to 5.55 before the purge and remained almost unchanged in 1946 at 5.48. On average, 23.6% of teachers were removed from each municipality during the purge. Municipalities in the sample have an average population of about 3,480 inhabitants, and 80% have fewer than 2,105 residents.

Figure 5 presents a map showing the municipal share of teachers lost – through permanent dismissal, temporary suspension, or forced relocation – across the available provinces. The distribution exhibits significant variation not only across provinces, but also within them. Moreover, Figure A1 displays the evolution of local teacher-to-student ratios over time. Average municipal teacher-to-student ratios were 0.65 per 50 students in 1934, 1.17 before the purge, and 1.38 in 1946.¹⁹ Southern municipalities exhibit lower ratios throughout the period, yet their staffing increased at similar rates as in other regions.

¹⁸ The pre-purge year varies across municipalities depending on when the purge process began; for most municipalities, this occurred between 1938 and 1940.

¹⁹ Average municipal ratios exceed the aggregate ratios in Section 4.1 because very small municipalities – with schooling population below 50 students – exhibit large teacher-to-student ratios once they have one teacher assigned.

Figure 5: Municipal Loss of Teachers Through Dismissal, Suspension, or Forced Relocation



Note: The figure presents a map of mainland Spain showing municipalities in the 15 provinces for which teacher purge records were digitized: Albacete, Barcelona, Bizkaia, Cádiz, Castelló, Ciudad Real, Cuenca, Girona, Guadalajara, Huelva, Murcia, Tarragona, Toledo, València, and Valladolid. Shading indicates, at the municipal level, the share of teachers sanctioned through permanent dismissal, temporary suspension, or forced relocation.

5.2 The Regime's Response to Local Shortages Caused by the Purge

Components of the 1946 local staff. To examine the evolution of municipal-level staffing, we first define its components. For 1946, the municipal teacher staff can be expressed as:

$$\begin{aligned} \text{Teachers}_m^{1946} = & \text{Teachers}_m^{\text{Pre-purge}} - \text{Purged Teachers Lost}_m \\ & + \text{New Hires}_m + \Delta \text{Relocations}_m - \text{Retirements}_m \end{aligned} \quad (3)$$

where m denotes the municipality. Teachers_m^{1946} and $\text{Teachers}_m^{\text{Pre-purge}}$ refer to the num-

ber of teachers in 1946 and just before the purge, respectively. $\text{Purged Teachers Lost}_m$ refers to the number of teachers removed from a municipality due to sanctions, and includes teachers lost through dismissals, temporary suspensions, or forced relocation. New Hires_m denotes the number of new teachers appointed by the Francoist regime and assigned to the municipality in the years following the purge. $\Delta \text{Relocations}_m$ represents the net balance of relocations in the post-purge period and includes two main components: incoming teachers assigned to the municipality as a result of forced relocation from another municipality, and both incoming and outgoing teachers who voluntarily relocated.²⁰ Finally, Retirements_m captures the number of teachers who had retired by 1946.

We compute the components of Equation 3 using our municipal-level panel. First, the variables Teachers_m^{1946} , $\text{Teachers}_m^{\text{Pre-purge}}$, and New Hires_m are obtained directly from the data. The first two correspond to the number of teachers employed in each municipality at those two points in time. New Hires_m is measured as the number of teachers in the 1946 local staff who appear in the regime's hiring lists, which we digitized and merged into our teacher panel. Second, with these components in place, we compute $\Delta \text{Relocations}_m$ as the residual change in the local staff not explained by new hires. Finally, we omit Retirements_m , which cannot be directly measured from the available records. This could pose a concern if retirements were systematically correlated with purge intensity across municipalities. However, this is unlikely to affect our results, as teachers at retirement age represented only a small share of the workforce, and any systematic correlation seems implausible.

Main estimating equation. We explore the relationship between municipal-level purge intensity and the size and composition of the 1946 teaching staff by estimating the following equation:

$$Y_m = \beta \text{Purged Teachers Lost (\%)}_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m \quad (4)$$

where m denotes the municipality and $p(m)$ the province. Y_m represents one of three outcomes of interest. The first is the percent change in the number of teachers between the pre-purge level and 1946. The second is the number of new hires appointed between 1940 and 1946 and assigned to the municipality by 1946, expressed as a percentage of the pre-purge teaching staff. The third is the net balance of relocations during the period, expressed as a percentage of the pre-purge staff. Expressing all outcomes relative to the initial (pre-purge) number of teachers allows comparability across municipalities

²⁰ While these teachers relocated voluntarily, the set of positions they could apply for was determined by the regime. That is, teachers could not freely choose any municipality but had to select from a list of designated openings.

of different sizes. Purged Teachers Lost (%)_{*m*} is the share of teachers in municipality *m* removed in the purge due to sanctions, including dismissals, temporary suspensions, or forced relocations. Finally, $\psi_{q(m)}$, $\psi_{n(m)}$, $\psi_{p(m)}$ denote fixed effects for population quintiles, pre-purge staffing levels, and provinces, respectively, accounting for potential differences in purge intensity or in staff evolution across municipalities with different sizes, initial staffing levels, or provincial contexts.

Main results. Table 2 presents the results. Columns (1) and (2) report estimates for the change in the number of teachers, Columns (3) and (4) for the number of new hires, and Columns (5) and (6) for the net balance of relocations. All outcomes are expressed relative to the initial (pre-purge) number of teachers. To address the limited number of clusters, we report wild-cluster bootstrap *p*-values by province, using Webb weights and 100,000 replications, in brackets.

Table 2: OLS: municipal purge intensity, size and composition of the 1946 teaching staff

	Change number of Teachers (%)		New Hires (%)		Net balance of relocations(%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Purged Teachers Lost (%)	-0.120 [0.130]	-0.081 [0.173]	-0.060 [0.321]	-0.017 [0.742]	0.940 [0.000]***	0.936 [0.000]***
Observations	2,388	2,388	2,388	2,388	2,388	2,388
R ²	0.24	0.33	0.12	0.25	0.36	0.38
Outcome average	10.36	10.36	38.37	38.37	-4.38	-4.38
Clusters	15	15	15	15	15	15
Pre-purge staff FE	Yes	Yes	Yes	Yes	Yes	Yes
Population FE	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	No	Yes	No	Yes	No	Yes

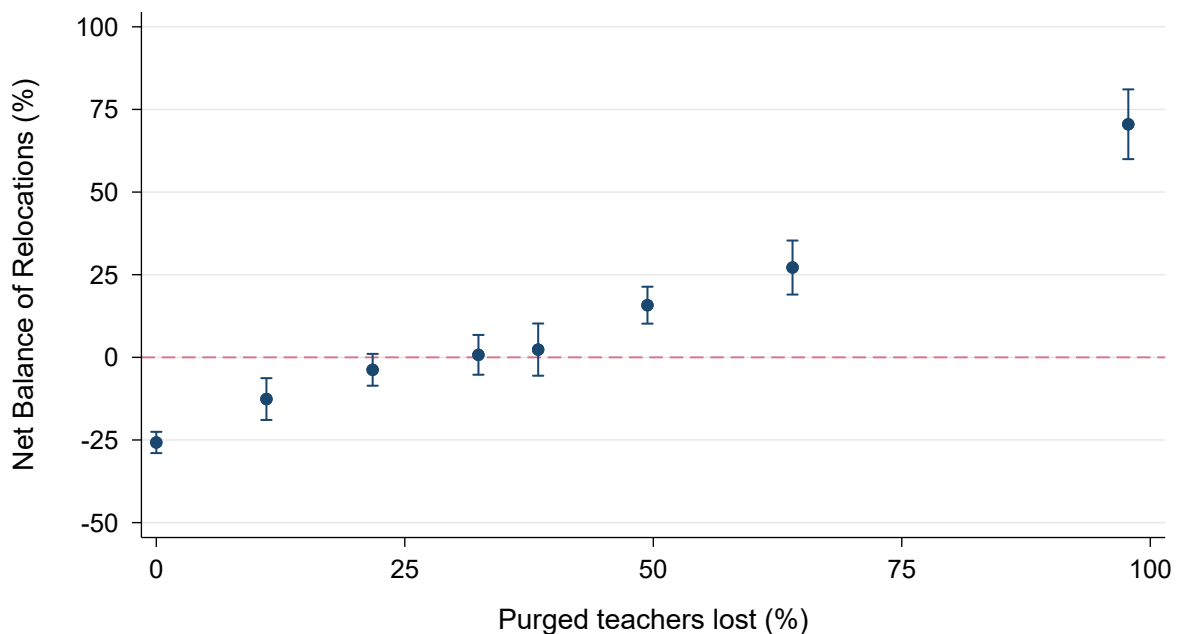
Note: This table displays the estimates of the OLS regression described in Equation 4. Our sample includes 2,388 municipalities in 15 provinces (89 municipalities are excluded as they did not have any teacher assigned at the time of the purge). The complete specification we estimate is $Y_m = \beta \text{Purged Teachers Lost } (\%)_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m$ where *m* denotes the municipality and *p*(*m*) the province. Columns (1) and (2) show estimates using as an outcome the percent change in the number of teachers between the pre-purge level and 1946. Columns (3) and (4) use the number of new hires appointed between 1940 and 1946 and assigned to the municipality in 1946. Columns (5) and (6) use the net balance of relocations during the period. All outcomes are expressed relative to the initial (pre-purge) number of teachers, for comparability. Purged Teachers Lost (%)_{*m*} is the share of teachers removed in the purge due to sanctions, including dismissals, temporary suspensions, or forced relocations. Columns (1), (3) and (5) include fixed effects for population quintiles and pre-purge staffing levels ($\psi_{q(m)}$, $\psi_{n(m)}$), and Columns (2), (4) and (6) province fixed effects ($\psi_{p(m)}$). We report wild-cluster bootstrap *p*-values by province, using Webb weights and 100,000 replications, in square brackets, to account for the low number of clusters. *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.10.

Results reveal that, by 1946, municipalities more affected by the purge experienced changes in total staff similar to those that lost no teachers through sanctions. However, this was not because they received a larger allocation of new hires, but rather because significant inflows of relocated teachers compensated for purge-related losses. Columns (1) and (2) indicate that, although the coefficients are negative and sizable, municipalities with a higher share of sanctioned teachers did not experience a statistically significant decline in total staff by 1946 compared to unaffected ones. This pattern

cannot be explained by the allocation of newly hired teachers: as shown in Columns (3) and (4), the Francoist regime did not disproportionately assign new appointments to municipalities more affected by the purge. Instead, Columns (5) and (6) show that the shortfall was offset through teacher relocations across the territory: for every 1 percentage point increase in the share of teachers lost to the purge, municipalities exhibit a net relocation balance 0.94 percentage points higher than those with no dismissals – relative to their pre-purge staff levels.

Figure 6 illustrates the relationship between purge intensity and the net relocation balance, corresponding to Column (6) of Table 2. The pattern shows that the magnitude of the estimated coefficient reflects two offsetting dynamics: municipalities unaffected by the purge exhibit negative relocation balances, while those more heavily affected show positive ones. On average, municipalities that lost between 20% and 40% of their teachers display a relocation balance close to zero. Thus, the coefficient's value – close to one – should not be interpreted as indicating that all teacher losses were replaced one-for-one through relocations, but rather as the result of transfers from unaffected municipalities compensating for deficits in those more affected by the purge.

Figure 6: Relocation balances across levels of municipal purge intensity



Note: The figure presents a non-parametric representation of the results in Column (6) of Table 2. The dependent variable is the net balance of relocations during the 1940-1946 period, expressed as a percentage of the pre-purge staff. The independent variable for binning is the share of teachers removed from the municipality due to sanctions, including dismissals, temporary suspensions, or forced relocations. It is constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels, and provinces. Each dot represents the mean net balance of relocations in the residualized data. Blue lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

Staff composition. Figure B7 depicts the final composition of the 1946 teaching workforce across municipalities more and less affected by the purge. To this end, we construct

a dummy variable that equals one for municipalities above the 25th percentile of purge-related losses among those with the same pre-purge staff size, and zero otherwise. We then estimate, for each municipality's 1946 staff, the shares of teachers who were stayers (already working there at the time of the purge), relocated, or newly hired. While the share of new hires is similar across both groups, municipalities more affected by the purge display a markedly higher proportion of relocated teachers. This pattern indicates that experience levels do not differ substantially across the two groups, as relocated teachers and stayers had comparable experience.

Robustness. Staffing patterns might vary across provinces. To ensure that our results are not driven by differences in teacher allocation patterns specific to a certain province, we replicate Table 2 by sequentially excluding one province at a time. As shown in Figure B9, the coefficients remain stable, confirming that province-level differences do not drive the results.

In the previous analysis, all measures were expressed as changes in local teacher staff relative to pre-purge levels, allowing comparisons across municipalities of different sizes. However, this approach may conceal potential heterogeneities in how local staff evolved across municipalities of varying sizes. To assess this, we compare the evolution of local staffing between municipalities with the same pre-purge staff size but differing purge intensities. For this purpose, we construct a dummy variable that equals one for municipalities above the 25th percentile of purge-related losses among those with the same pre-purge staff size, and zero otherwise. We then estimate differences in the number of teachers within each size group between high- and low-purge municipalities.

Figure B8 provides a graphical, non-parametric representation of these results. Panel (a) shows that municipalities with high purge intensity experienced an immediate and sizable decline in teacher numbers relative to comparable municipalities with fewer sanctioned teachers – confirming that our measure captures the direct staff losses caused by the purge. Panel (b) indicates that this initial gap was closed by 1946 across all municipal sizes, confirming that the patterns documented in the main results are consistent and not driven by differences in baseline staffing levels.

5.3 The Relocation Policy and Local Staffing Capacity

Results from the previous section show no persistent differences in total staff levels across municipalities with varying purge intensities. Our findings suggest that the regime largely offset purge-related losses through an extensive relocation process that redistributed teachers across municipalities. In this section, we examine this relocation policy in greater detail to understand how it operated and assess its effectiveness in addressing the staffing disruptions caused by the purge.

Main estimating equation. We analyze how municipal relocation balances relate to the size and composition of the 1946 teaching staff by estimating the following specification, analogous to Equation 4:

$$Y_m = \beta \Delta \text{Relocations}_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m \quad (5)$$

where m denotes the municipality and $p(m)$ the province. Y_m represents either the percent change in the number of teachers between the pre-purge level and 1946 or the number of new hires appointed between 1940 and 1946 and assigned to the municipality by 1946, expressed as a percentage of the pre-purge teaching staff. $\Delta \text{Relocations}_m$ is the net balance of relocations during the period, expressed as a percentage of the pre-purge staff. $\psi_{q(m)}$, $\psi_{n(m)}$, and $\psi_{p(m)}$ have the same definition as in Equation 4.

Main results. Table 3 presents our results. Columns (1) and (2) report estimates for the percentage change in teacher supply from pre-purge levels to 1946. Columns (3) and (4) show the share of new hires working in the municipality relative to the pre-purge supply. Again, we report wild-cluster bootstrap p-values by province, using Webb weights and 100,000 replications, between brackets.

Table 3: OLS: relocation balances, size and composition of the 1946 teaching staff

	Change number of Teachers (%)		New Hires (%)	
	(1)	(2)	(3)	(4)
Net balance of relocations (%)	0.556 [0.000]***	0.564 [0.000]***	-0.179 [0.006]**	-0.171 [0.003]**
Observations	2,388	2,388	2,388	2,388
R ²	0.43	0.52	0.15	0.28
Outcome average	10.36	10.36	38.37	38.37
Clusters	15	15	15	15
Pre-purge staff FE	Yes	Yes	Yes	Yes
Population FE	Yes	Yes	Yes	Yes
Province FE	No	Yes	No	Yes

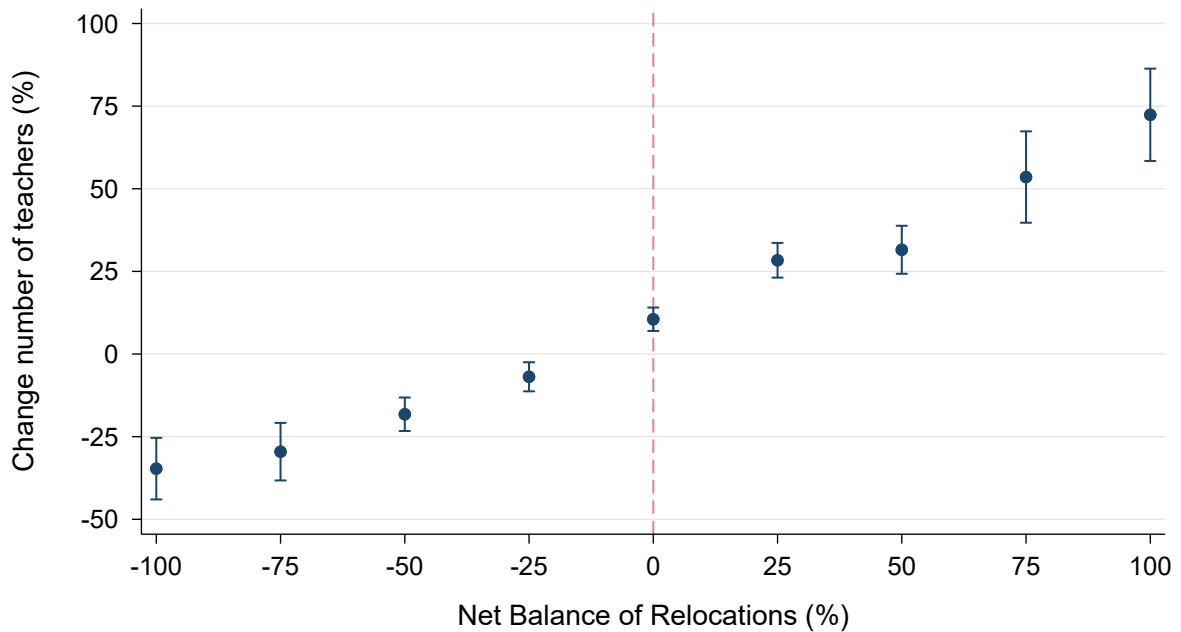
Note: This table displays the estimates of the OLS regression described in Equation 5. Our sample includes 2,388 municipalities in 15 provinces (89 municipalities are excluded as they did not have any teacher assigned at the time of the purge). The complete specification we estimate is $Y_m = \beta \Delta \text{Relocations}_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m$ where m denotes the municipality and $p(m)$ the province. Columns (1) and (2) show estimates using as an outcome the percent change in the number of teachers between the pre-purge level and 1946. Columns (3) and (4) use the number of new hires appointed between 1940 and 1946 and assigned to the municipality in 1946. Both outcomes are expressed relative to the initial (pre-purge) number of teachers, for comparability. $\Delta \text{Relocations}_m$ is the net balance of relocations during the period, expressed as a percentage of the pre-purge staff. Columns (1) and (3) include fixed effects for population quintiles and pre-purge staffing levels ($\psi_{q(m)}$, $\psi_{n(m)}$), and Columns (2) and (4) province fixed effects ($\psi_{p(m)}$). We report wild-cluster bootstrap p -values by province, using Webb weights and 100,000 replications, in square brackets, to account for the low number of clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Our results show that, while purge losses are not correlated with changes in municipal staffing, relocation balances are. Columns (1) and (2) indicate that each additional

percentage point of net relocations is associated with a 0.56 percentage point increase in municipal staff size. Moreover, relocation balances are also significantly correlated with the allocation of new hires: as shown in Columns (3) and (4), for every additional percentage point of net relocations, the share of new hires assigned to the municipality declines by 0.17 percentage points.

Building on these results, we further explore how changes in municipal staff levels relate to differences in relocation balances. Figure 7 replicates the estimates in Column (2) of Table 3 using a non-parametric representation and shows that the negative correlation is explained by staff increases in municipalities with positive relocation balances and declines in those with negative ones. While municipalities with a neutral relocation balance still show modest staff growth, average declines of about 35% are observed in municipalities that lost their entire teaching staff through relocations.

Figure 7: Changes in the 1946 local staff across relocation balances



Note: The figure presents a non-parametric representation of the results in Column (2) of Table 3. The dependent variable is the percent change in the number of teachers between the pre-purge level and 1946. The independent variable for binning is the net balance of relocations during the 1940-1946 period, expressed as a percentage of the pre-purge staff. It is constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels and provinces. Each dot represents the mean net percent change in the number of teachers in the residualized data. Blue lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

Staff composition. Figure B10 presents the composition of the 1946 teaching workforce across municipalities with positive and negative relocation balances. As mentioned in Section 5.2, we calculate the shares of teachers who were stayers (already working there at the time of the purge), relocated, or newly hired. The proportion of stayers is similar across both groups, but municipalities with negative relocation balances show a markedly higher share of new hires – twice as large as in those with positive

balances, reaching nearly half of their total staff. This leads to substantially lower average experience in municipalities that lost teachers through relocations.

Robustness. As we did for the results in Section 5.2, we also replicate Table 3 by sequentially excluding one province at a time. Figure B11 displays the results: again, the coefficients remain unchanged across estimations, ruling out that province-level differences drive the results.

Mechanisms explaining relocation gaps

Our results show that municipalities losing teachers through the relocation process experienced significant declines in their teaching staff by 1946. So far, however, we have not examined the mechanisms explaining this pattern. These imbalances could reflect deliberate planning choices by the regime – such as adjusting staffing levels to local schooling demand or prioritizing certain municipalities for ideological reasons, among others. In what follows, we assess these alternative explanations and show evidence more consistent with a different mechanism: coordination and bureaucratic frictions in the reorganization process.

Design of the relocation. First, we examine the design of the relocation process. The official decree that established it, issued in April 1941, explicitly stated that one of its main objectives was to fill the vacancies left by purge-related dismissals (see Figure C9). Under this process, the regime published a list of available teaching positions open to in-service teachers. Those not subject to disciplinary sanctions or ongoing purge proceedings were allowed to request transfers voluntarily, provided they had three years of tenure. This design meant that, although the regime controlled the set of available destinations, there were no explicit restrictions on which teachers could apply for relocation.

This design raises the question of whether the absence of restrictions on which teachers could relocate may have exacerbated staffing imbalances in the areas most affected by the purge. If teachers from heavily purged municipalities were also able to apply for transfers, the relocation process could have further reduced staff in already affected municipalities.

To explore this possibility, we divide municipalities into four groups according to the share of teachers removed in the purge. Figure B12 presents boxplots showing the 5th–95th percentile range of relocation balances for these four groups. Overall, municipalities with greater purge losses tend to display larger and more frequent positive relocation balances – consistent with inflows to replace dismissed teachers and with the fact that heavily affected areas offered less scope for additional departures. Yet the figure also reveals that, among the most severely purged municipalities, substantial

outflows persisted. This pattern suggests that allowing teachers to transfer freely enabled departures from already strained municipalities, aggravating local staffing shortages.

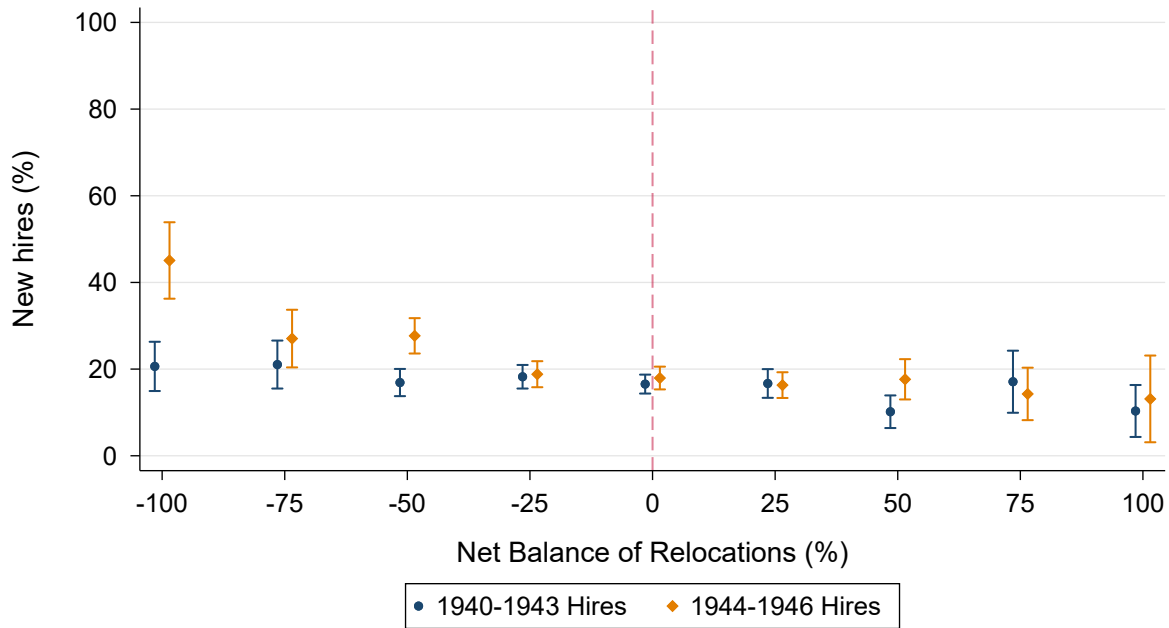
Timing of relocations and hirings. Second, we examine the timing of the relocation process and the regime's hiring rounds. As described in Section 2.3, the relocation contest, announced in 1941, was not implemented until 1943. In the meantime, new lists of available positions were gradually published as additional vacancies emerged. Successive official documents attribute the delay to bureaucratic obstacles that slowed the process and emphasize that promptly filling vacancies was considered in the best interests of both education and the government.

Between 1940 and 1946, the regime also conducted four rounds of teacher recruitment. In three of them, appointments occurred while the relocation process was still underway, and the positions offered for relocation were withheld from new hires. This overlap may have created mismatches in local staffing, as the vacancies created by relocations were not yet known when new hires were appointed. The final recruitment round, announced in 1944 and completed in 1946, took place after the relocation process had concluded and local staffing needs had been identified.

We hypothesize that the overlap between relocation and hiring rounds generated frictions in the allocation of new teachers, leading to an underallocation of hires to municipalities that lost more staff through relocations. If this mechanism holds, we should observe that hires from the three early recruitment rounds were distributed more evenly across relocation balances, whereas those from the final round – conducted once relocations had been completed – were disproportionately assigned to municipalities with larger negative balances. We test this prediction using a non-parametric representation of the distribution of new hires across municipalities with different relocation balances, distinguishing between early and late recruitment rounds.

Figure 8 displays the relationship between relocation balances and the share of new hires for early (1940–1943) and late (1944–1946) recruitment rounds. For early hires, the allocation remains relatively flat across the distribution of relocation balances. Only those with very large positive balances – above 50% – show a modest decline in new hires. In contrast, the distribution of late hires presents a distinct pattern: while allocations are similar for municipalities with relocation balances above –25%, they rise sharply among those with large negative balances. This pattern is consistent with an adjustment in the allocation of new hires once the relocation process had concluded, when local staffing deficits became fully visible – supporting the presence of timing frictions in the reorganization process.

Figure 8: Share of new hires across relocation balances, by recruitment rounds



Note: The figure presents a non-parametric representation of the distribution of new hires across municipalities with different relocation balances. The dependent variable for the blue set is the number of new hires appointed between 1940 and 1943 and assigned to the municipality by 1946, expressed as a percentage of the pre-purge teaching staff. The orange set is analogous to the 1944 to 1946 hires. The independent variable for binning is the net balance of relocations during the 1940-1946 period, expressed as a percentage of the pre-purge staff. It is constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels, and provinces. Each dot represents the mean number of new hires, expressed as a percentage of the pre-purge teaching staff, in the residualized data. Blue and orange lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

Adjusting staffing levels to local schooling demand. The previous results suggest that the declines in local staff associated with negative relocation balances stem from coordination frictions in the post-purge reorganization of the workforce. However, alternative explanations remain. One possibility is that these municipalities were initially overstaffed relative to their schooling needs, in which case subsequent reductions in teacher numbers might simply reflect an efficient adjustment of local provision rather than a shortfall created by administrative frictions.

To test this hypothesis, we focus on municipalities that experienced large staffing losses during the relocation process – specifically, those in the bottom quartile of relocation balances, corresponding to declines of at least 40 percent of their pre-purge workforce. We then classify these municipalities into three groups according to the number of new hires they subsequently received. Our analysis compares the distribution of teacher-to-student ratios across these groups at three points in time: (i) immediately after the purge, before relocations; (ii) after relocations were completed; and (iii) in 1946, once all new hires had been assigned.

Figure B13 plots the cumulative distributions of teacher-to-student ratios for the

three groups. Panel (a) shows that, before relocations, these distributions were broadly comparable. Panel (b) illustrates that, after the relocation process, teacher-to-student ratios declined sharply across all groups. Although municipalities that would later receive more new hires became more understaffed at this stage, those receiving fewer hires were also far from the objective of 1 teacher per 50 students. Finally, Panel (c) shows that, by 1946 – once all hiring rounds were completed – municipalities receiving fewer hires remained substantially understaffed relative to their schooling needs: over three-quarters fell below the target, and their ratios were markedly worse than before relocations. Taken together, this evidence indicates that the decline in local staffing among these municipalities does not reflect an adjustment to lower local demand but rather frictions in the management of relocations and the allocation of new hires.

Punishing disloyal regions. Finally, we test an alternative explanation: that the regime deliberately limited teacher recovery in certain municipalities as a form of punishment for political disloyalty. Whether the regime would have pursued such a strategy is ambiguous. On one hand, education was viewed as a key instrument for securing ideological conformity and consolidating power, which could have led to prioritizing staffing even in disloyal areas. On the other hand, restricting educational provision could also have served as a mechanism of political control by reducing channels for civic mobilization in territories considered less reliable.

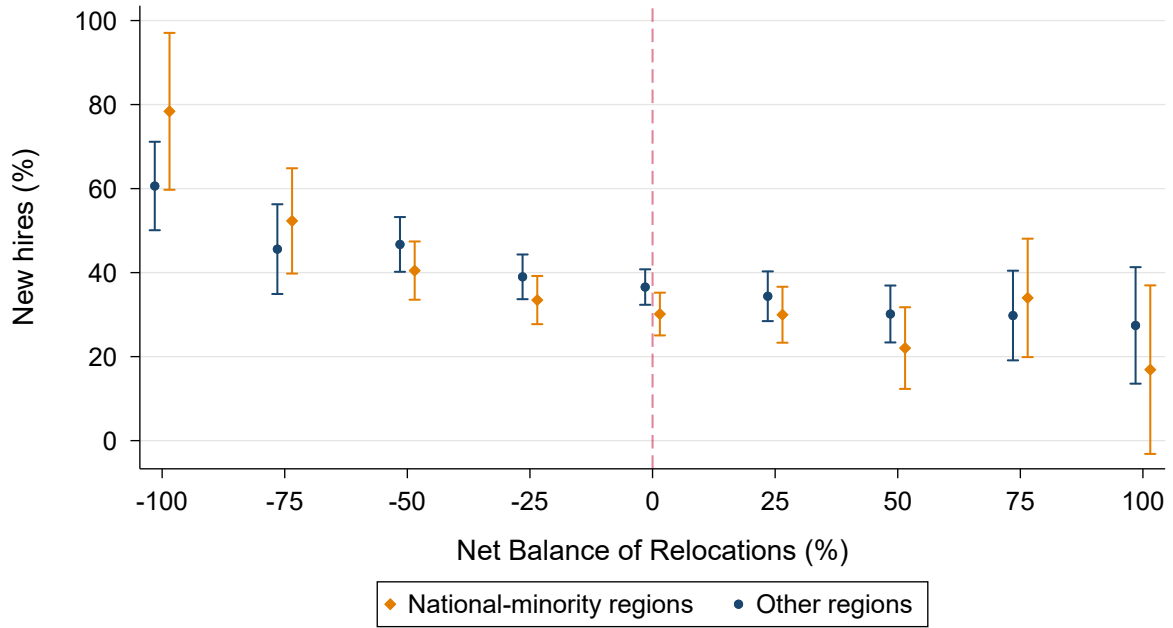
We assess this hypothesis by comparing the allocation of new hires across municipalities with different relocation balances in two groups of provinces: national-minority regions and other regions. This comparison is informative because national minority regions were more intensively repressed, making them plausible targets of punitive staffing policies. The first group includes Barcelona, Bizkaia, Girona, and Tarragona – the four national minority provinces in our sample.

Figure 9 presents the non-parametric relationship between municipal relocation balances and new hires for each group. The allocation of new hires does not differ significantly between national-minority and other provinces. If anything, municipalities with large negative relocation balances in national-minority regions received slightly more hires. Similar patterns are observed when we look at the allocation of early and late hires separately, as depicted in Figure B14. These findings rule out that the observed imbalances resulted from targeted punishment of disloyal or politically suspect regions.

Evaluating municipal allocation frictions

Our evidence shows that the regime implemented policies to compensate local staffing shortages, but that these efforts were uneven and characterized by coordination frictions. A natural next step is to assess the magnitude of these frictions in the local assignment of teachers. However, doing so is challenging, as it requires taking a stance on the optimal

Figure 9: Share of new hires across relocation balances in provinces with regional nationalist movements and other provinces



Note: The figure presents a non-parametric representation of the distribution of new hires across municipalities with different relocation balances, for two different subsamples. The orange set corresponds to provinces with strong regional nationalist movements (Barcelona, Bizkaia, Girona, Tarragona). The blue set corresponds to 11 provinces without strong regional nationalist movements. The dependent variable is the number of new hires appointed between 1940 and 1946 and assigned to the municipality by 1946, expressed as a percentage of the pre-purge teaching staff. The independent variable for binning is the net balance of relocations during the 1940-1946 period, expressed as a percentage of the pre-purge staff. It is constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels, and provinces. Each dot represents the mean number of new hires, expressed as a percentage of the pre-purge teaching staff, in the residualized data. Blue and orange lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

local allocation – a particularly difficult task in a context of severe staffing constraints. Instead, we perform a back-of-the-envelope exercise that quantifies aggregate municipal deviations from the policy target of having at least one teacher per 50 students. This approach simplifies the allocation problem and may overlook fine-grained adjustments – such as sharing of teachers across small, nearby municipalities – but it is informative regarding the relevant margin of misallocation in our context: local staffing deficits that could be solved by teacher relocation.

In-service teachers, aggregate needs, and overall staffing deficit. We begin by estimating the total number of in-service teachers, the aggregate local needs, and the resulting overall staffing deficit. First, we obtain the total number of in-service teachers by aggregating all teachers working in the municipalities of the 15 provinces in our sample.²¹ Second, we estimate the number of teachers required to meet local schooling needs. For this purpose, we approximate each municipality’s school-age population using total municipal population counts and the provincial share of individuals aged 6

²¹ Despite working with a subsample of provinces, our results remain valid insofar as local staffing patterns resemble those of the provinces not included.

to 12.²² Then, we compute the number of teachers needed to reach the policy target of at least one teacher per 50 students. Aggregating these municipal needs yields the total number of required teachers. Finally, we calculate the overall staffing deficit as the number of additional teachers that would need to be hired to meet aggregate local schooling demand, given by:

$$\text{Overall Staffing Deficit}_t = \sum_{m=1}^M \text{Required Teachers}_{m,t} - \sum_{m=1}^M \text{In-service Teachers}_{m,t} \quad (6)$$

where m denotes the municipality and t the year. $\text{Required Teachers}_{m,t}$ is the number of teachers needed in each municipality to achieve the policy target of at least one teacher per 50 students at time t . $\text{In-service Teachers}_{m,t}$ is the total number of teachers employed in each municipality at time t .

Observed local deficits and allocation frictions. In a context where employed teachers are low relative to total schooling demand, local misallocation arises when the number of teachers in some municipalities is above the minimum policy requirement, while others remain below. To quantify the extent of this imbalance, we first compute the aggregate local deficit of teachers, defined as:

$$\text{Aggregate Local Deficit}_t = \sum_{m=1}^M \text{Max} \left(0, \text{Required Teachers}_{m,t} - \text{In-service Teachers}_{m,t} \right) \quad (7)$$

where m denotes the municipality and t the year. While Equation 6 accounts for the deficit of teachers that must be covered through new hiring, Equation 7 also includes the deficit that could be addressed by relocating teachers. The difference between the two measures (7 - 6) captures the extent of allocation frictions relative to local schooling needs.

Allocation benchmarks. Evaluating allocation frictions after the purge by simply comparing the 1946 teacher distribution with the pre-purge period can be misleading, because overall staffing deficits differ markedly across years. To assess the extent of allocation frictions in 1946, we construct two additional benchmarks. In both cases,

²² We rely on this approximation because age-specific population counts are not available at the municipal level. First, we digitize census data for 1930, 1940, and 1950, which provide province-level population counts by age group, separately for the provincial capital and the rest of the province. Second, we interpolate across census years to estimate the school-age population for each year. Third, we use municipal-level total population counts in 1920, 1930, and 1940, and interpolate them to estimate total municipal population for each year. We finally estimate yearly school-age population for each municipality multiplying the corresponding provincial share of school-age population by the total municipal population, using the specific share for the case of provincial capitals.

we hold the overall staffing deficit fixed at its 1946 level. We treat the post-relocation local teacher distribution as given and vary only the assignment of new hires across municipalities.²³ As discussed earlier in this section, the overlap in the timing of relocations and hiring contributed to local imbalances. Accordingly, these benchmarks allow us to quantify how allocation frictions would differ under hiring processes carried out with more or less coordination with the relocation process.

First, in the *frictionless benchmark*, new hires are allocated exclusively to municipalities that fall below the target of one teacher per 50 students. This scenario delivers the largest possible reduction in local deficits given the existing teacher distribution and using only the available new hires. Second, in the *random benchmark*, the same number of new hires is distributed fully at random across municipalities, regardless of their staffing needs. This represents the opposite scenario, in which the allocation of new hires exhibits no coordination or targeting of shortages. Taken together, these benchmarks provide lower and upper bounds on allocation frictions to evaluate the observed teacher distribution in 1946.

Results. Table 4 shows our measures of in-service teachers, overall staffing deficit, and allocation frictions. We report them for the observed municipal distributions before the purge²⁴ and in 1946, and for the two 1946 benchmarks under alternative hiring allocations.

The number of in-service teachers before the purge and in 1946 was similar, and in both cases well below what was required to meet aggregate local schooling demand. However, because total teacher needs were higher before the purge, the resulting overall staffing deficit was about 17.5% larger than in 1946 (7,650 versus 6,512 additional teachers needed). Although differences in overall deficits may affect comparisons across periods, all measures suggest greater allocation frictions in 1946. Our preferred metric – measured relative to in-service teachers – was about 8% before the purge, compared with 10.1% in 1946. By relocating those teachers to understaffed municipalities, the regime would have reduced aggregate local deficits by 17.3%. Taken together, these figures suggest that allocation frictions may have been higher in 1946.

To assess the magnitude of allocation frictions in the 1946 local teacher distribution, we compare it with the two benchmarks constructed, both of which keep the number of in-service teachers fixed at its 1946 level. Measured relative to in-service teachers, allocation frictions in these benchmarks range from 2.7% in the *frictionless* case to 14.4% in the *random* one. Crucially, the magnitude of allocation frictions in the 1946 observed

²³ We take relocation frictions as given, because despite the regime did not restrict the origin municipalities from which teachers could apply (see Figure 6), it effectively targeted relocation vacancies toward municipalities that lost teachers during the purge (see Table 2).

²⁴ We use school-age population in 1938 as the reference for the pre-purge distribution, as most purge sanctions were issued between 1938 and 1941.

Table 4: In-service teachers, overall staffing deficit and allocation frictions before the purge and in 1946

		Pre-purge	1946		
		Observed	Observed	Frictionless	Random
	Total in-service teachers	13,721	13,562	13,562	13,562
	Total required teachers (1:50 rule)	21,371	20,074	20,074	20,074
(6)	Overall staffing deficit	7,650	6,512	6,512	6,512
Municipal allocation					
(7)	Aggregate local deficit	8,748	7,879	6,879	8,462
(7) - (6)	Allocation friction (teachers)	1,098	1,367	367	1,950
	Allocation friction (% in-service)	8.00	10.08	2.71	14.38
	Allocation friction (% staffing deficit)	14.35	20.99	5.64	29.94

Note: This table displays our estimates for in-service teachers, staffing needs and deficits, and aggregate local deficits. Our sample includes 2,477 municipalities in 15 provinces. Total in-service teachers is obtained by aggregating all teachers working in the municipalities in our sample. Total required teachers is the total number of teachers needed to reach the policy target of at least one teacher per 50 students in all municipalities. Overall staffing deficit is the number of additional teachers that would need to be hired to meet aggregate local schooling demand, and is computed using the following expression: $\text{Overall Staffing Deficit}_t = \sum_{m=1}^M \text{Required Teachers}_{m,t} - \sum_{m=1}^M \text{In-service Teachers}_{m,t}$. The aggregate local deficit is the sum of all municipal shortfalls relative to the number of teachers needed to reach the policy target of one teacher per 50 students, without considering the potential relocation of teachers across municipalities, and is given by: $\text{Aggregate Local Deficit}_t = \sum_{m=1}^M \text{Max}(0, \text{Required Teachers}_{m,t} - \text{In-service Teachers}_{m,t})$. The allocation friction refers to the aggregate local deficit that could be addressed by relocating in-service teachers across municipalities, and is given by the difference between the aggregate local deficit and the overall staffing deficit. Finally, we present the allocation friction as a percent of the number of in-service teachers and the overall staffing deficit. We provide those measures in 4 scenarios: the observed local distribution before the purge (that uses municipal-level school-age population measured in 1938); the observed local distribution in 1946; two additional scenarios, constructed using the total number of teachers in 1946. In the *frictionless benchmark*, new hires by the Francoist regime are allocated exclusively to municipalities that fall below the target of one teacher per 50 students after the relocation. In the *random benchmark*, new hires by the Francoist regime are distributed fully at random across municipalities, regardless of their staffing needs.

distribution lies much closer to the *random benchmark* than to the *frictionless* one. Had new hires been allocated without coordination frictions – as in the *frictionless benchmark* – the observed aggregate local deficit would have been reduced by about 12.7% (from 7,879 to 6,879 teachers). Overall, these results indicate sizable coordination frictions in the 1946 distribution, with the observed allocation far from what a perfectly coordinated process would deliver.

6 Conclusion

A central question in political economy is how states select the public workforce and how this decision shapes state capacity (Pepinsky, Pierskalla, and Sacks, 2017). Although modern civil-service systems aim to limit political discretion, ideology-based selection remains a common feature of many non-democratic regimes. Existing work has examined both the costs and potential benefits of discretionary appointments – including patronage (Xu, 2018) and mission alignment (Ashraf, Bandiera, and Jack, 2014) – yet we know comparatively little about the consequences of ideology-based

screening of frontline public workers. In particular, there is limited evidence on how ideology-based personnel policies affect the quality, allocation, and availability of civil servants who provide essential public services.

Our paper studies these questions in the context of the Francoist purge of primary-school teachers, a large-scale effort to enforce ideological conformity within a frontline occupation central to state provision. Using a newly digitized dataset on teacher careers, we show that ideology-based selection eroded the quality of the public workforce through the dismissal of more competent teachers and a decline in workforce experience. Moreover, we provide evidence that large-scale personnel reforms can create staffing and organizational constraints that weaken the local provision of essential services. Taken together, our results show how personnel policies used for autocratic consolidation can impose sizable costs on state capacity and weaken service delivery.

While our analysis sheds light on how ideology-based personnel policies reshape public workforce quality and the organization of local service provision, open questions remain. Building on work on the indoctrination origins of public education and state-building ([Paglayan, 2024](#)), future research could examine how ideology-based teacher selection affects human capital and civic values. Changes in teacher composition and disruptions in service delivery may have had lasting effects on educational attainment, labor market outcomes, and social mobility. Moreover, given the purge's indoctrination motive and the contrasting profiles of dismissed and newly appointed teachers, future work could explore its influence on political attitudes, gender roles, and social norms. These remain promising avenues for future work.

References

- Akhtari, Mitra, Diana Moreira, and Laura Trucco. 2022. "Political Turnover, Bureaucratic Turnover, and the Quality of Public Services." *American Economic Review* 112, no. 2 (February): 442–493. (Page 6).
- Albertus, Michael. 2023. "The Political Price of Authoritarian Control: Evidence from Francoist Land Settlements in Spain." *The Journal of Politics* 85, no. 4 (October): 1258–1274. (Page 6).
- Ashraf, Nava, Oriana Bandiera, Edward Davenport, and Scott S. Lee. 2020. "Losing Prosociality in the Quest for Talent? Sorting, Selection, and Productivity in the Delivery of Public Services." *American Economic Review* 110, no. 5 (May): 1355–1394. (Page 6).
- Ashraf, Nava, Oriana Bandiera, and B. Kelsey Jack. 2014. "No margin, no mission? A field experiment on incentives for public service delivery." *Journal of Public Economics* 120 (December): 1–17. (Page 42).
- Barro, Robert J., and Jong Wha Lee. 2013. "A new data set of educational attainment in the world, 1950–2010." *Journal of Development Economics* 104 (September): 184–198. (Page 7).
- Besley, Timothy, Robin Burgess, Adnan Khan, and Guo Xu. 2022. "Bureaucracy and Development." *Annual Review of Economics* 14:397–424. (Page 6).
- Besley, Timothy, and Torsten Persson. 2009. "The Origins of State Capacity: Property Rights, Taxation, and Politics." *American Economic Review* 99, no. 4 (August): 1218–1244. (Page 6).
- Brierley, Sarah. 2021. "Combining Patronage and Merit in Public Sector Recruitment." *The Journal of Politics* 83, no. 1 (January): 182–197. (Page 6).
- Caplan, Jane. 1988. *Government without administration: state and civil service in Weimar and Nazi Germany*. Oxford historical monographs. Oxford: Clarendon. (Page 2).
- Cattaneo, Matias D., Richard K. Crump, Max H. Farrell, and Yingjie Feng. 2024. "On Binscatter." *American Economic Review* 114, no. 5 (May): 1488–1514. (Pages 20, 31, 34, 37, 39, 54, 62, 68).
- Chambru, Cédric, Emeric Henry, and Benjamin Marx. 2024. "The Dynamic Consequences of State Building: Evidence from the French Revolution." *American Economic Review* 114, no. 11 (November): 3578–3622. (Page 6).

- Chen, Shuo, Xinyu Fan, L. Colin Xu, and Xun Yan. 2023. "Competence-loyalty tradeoff under dominant minority rule: The case of Manchu rule, 1650-1911." *Journal of Public Economics* 220 (April): 104843. (Page 6).
- Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso. 2020. "Patronage and Selection in Public Sector Organizations." *American Economic Review* 110, no. 10 (October): 3071–3099. (Page 6).
- Dal Bó, Ernesto, Frederico Finan, and Martín A. Rossi. 2013. "Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service." *The Quarterly Journal of Economics* 128, no. 3 (August): 1169–1218. (Page 6).
- Deserranno, Erika. 2019. "Financial Incentives as Signals: Experimental Evidence from the Recruitment of Village Promoters in Uganda." *American Economic Journal: Applied Economics* 11, no. 1 (January): 277–317. (Page 6).
- Dewey, John. 1899. *The school and society: Being three lectures*. University of Chicago Press. (Page 8).
- . 1916. *Democracy and Education: An Introduction to the Philosophy of Education*. New York: The Macmillan Company. (Page 8).
- Díaz Agulló, M. del Carmen, and Juan Manuel Fernández Soria. 1999. *Maestros valencianos bajo el franquismo: la depuración del magisterio: 1939-1944*. Institució Alfons el Magnànim. (Page 9).
- Egorov, Georgy, and Konstantin Sonin. 2011. "Dictators and their viziers: Endogenizing the loyalty–competence trade-off." *Journal of the European Economic Association* 9, no. 5 (October): 903–930. (Page 6).
- Finan, F., B.A. Olken, and R. Pande. 2017. "The Personnel Economics of the Developing State." In *Handbook of Economic Field Experiments*, 2:467–514. Elsevier. (Pages 2, 6).
- Fisman, Raymond, Jing Shi, Yongxiang Wang, and Weixing Wu. 2020. "Social Ties and the Selection of China's Political Elite." *American Economic Review* 110, no. 6 (June): 1752–1781. (Page 6).
- García Díaz, Jesús Manuel. 2018. "La historiografía sobre la depuración franquista del magisterio nacional de España." *Boletín Avriense* 48/49:409–448. (Page 9).
- Goerlich Gisbert, Francisco José, Francisco Ruiz González, Pilar Chorén Rodríguez, and Carlos Albert Pérez. 2015. *Cambios en la Estructura y Localización de la Población. Una visión de largo plazo (1842-2011)*. (Page 14).

- González Roldán, Gregorio. 2023. *Depuración republicana y franquista de los catedráticos de universidad. Antecedentes, contexto y legislación*. History of Universities 72. Carlos III University of Madrid. (Page 9).
- Jia, Ruixue, Masayuki Kudamatsu, and David Seim. 2015. "Political Selection In China: The Complementary Roles Of Connections And Performance." *Journal of the European Economic Association* 13, no. 4 (August): 631–668. (Page 6).
- La Parra-Pérez, Álvaro. 2020. "For a fistful of pesetas? The political economy of the army in a nonconsolidated democracy: the Second Spanish Republic and Civil War (1931–9)." *The Economic History Review* 73, no. 2 (May): 565–594. (Page 7).
- Landry, Pierre F, Xiaobo Lü, and Haiyan Duan. 2018. "Does Performance Matter? Evaluating Political Selection Along the Chinese Administrative Ladder." *Comparative Political Studies* 51 (8): 1074–1115. (Page 6).
- Liu, Kevin Zhengcheng, and Xiaoming Zhang. 2025. "Discretion, talent allocation, and governance performance: Evidence from China's imperial bureaucracy." *Journal of Development Economics* 172 (January): 103391. (Page 6).
- Lorenzo Vicente, Juan Antonio. 2003. *La enseñanza media en la España franquista(1936-1975)*. OCLC: 1443674982. Madrid: Editorial Complutense. (Page 9).
- Martinez-Marquina, Alejandro. 2024. *Indoctrination in the Classroom: Evidence from the Spanish Social Service*. (Page 6).
- Mastrorocco, Nicola, and Edoardo Teso. 2025. "State Capacity as an Organizational Problem." *Review of Economic Studies* (forthcoming). (Page 6).
- Meng, Anne, and Jack Paine. 2022. "Power Sharing and Authoritarian Stability: How Rebel Regimes Solve the Guardianship Dilemma." *American Political Science Review* 116, no. 4 (November): 1208–1225. (Page 6).
- Molero Pintado, Antonio, Clara-Eugenia Núñez, Manuel de De Fuelles Benítez, Paloma Flórez Plaza, Pedro Álvarez Lázaro, María José Subierla Bernat, Cristina Crespo Galindo, and María del Mar Pozo Andrés. 2001. *La educación en la España del siglo XX: primer centenario de la creación del Ministerio de Instrucción Pública y Bellas Artes: Biblioteca Nacional, septiembre-octubre 2001*. Madrid: Ministerio de Educación, Cultura y Deporte. (Page 8).
- Monés i Pujol-Busquets, Jordi. 1976. "Cuatro décadas de educación franquista. Aspectos ideológicos." *Cuadernos de Pedagogía* 21:10–16. (Page 2).

- Morente Valero, Francisco. 1997. *La escuela y el Estado Nuevo. La depuración del magisterio nacional (1936-1943)*. (Page 9).
- Muñoz-Blanco, Laura. 2024. *Reservoir-induced displacement and social participation: Evidence from the Spanish dictatorship*. (Page 6).
- Nord, Marina, Fabio Angiolillo, Ana Good God, David Altman, Tiago Fernandes, and Staffan I. Lindberg. 2025. *25 Years of Autocratization - Democracy Trumped? Democracy Report 2025*. V-Dem Institute. (Page 2).
- Oto-Peralías, Daniel. 2015. "The Long-term Effects of Political Violence on Political Attitudes: Evidence from the Spanish Civil War." *Kyklos* 68, no. 3 (August): 412–442. (Page 6).
- Paglayan, Agustina S. 2024. *Raised to obey: the rise and spread of mass education*. The Princeton economic history of the western world. Princeton, New Jersey: Princeton University Press. (Page 43).
- Pepinsky, Thomas B, Jan H Pierskalla, and Audrey Sacks. 2017. "Bureaucracy and Service Delivery." *Annual Review of Political Science* 20:249–68. (Page 42).
- Ransom, Michael R, and David P. Sims. 2010. "Estimating the Firm's Labor Supply Curve in a "New Monopsony" Framework: Schoolteachers in Missouri." *Journal of Labor Economics* 28, no. 2 (April): 331–355. (Page 21).
- Rodon, Toni. 2024. "The scars of violence and repression on founding elections: Evidence from Spain." *Conflict Management and Peace Science* 41, no. 1 (January): 47–71. (Page 6).
- Schubert, Gregor, Anna Stansbury, and Bledi Taska. 2022. *Employer Concentration and Outside Options*. Washington Center of Equitable Growth - Working Paper. (Page 21).
- Shih, Victor, Christopher Adolph, and Mingxing Liu. 2012. "Getting Ahead in the Communist Party: Explaining the Advancement of Central Committee Members in China." *American Political Science Review* 106, no. 1 (February): 166–187. (Page 6).
- Spenkuch, Jörg L., Edoardo Teso, and Guo Xu. 2023. "Ideology and Performance in Public Organizations." *Econometrica* 91 (4): 1171–1203. (Page 2).
- Tur-Prats, Ana, and Felipe Valencia Caicedo. 2025. *The Long Shadow of the Spanish Civil War*. (Page 6).

- Waldinger, Fabian. 2010. "Quality Matters: The Expulsion of Professors and the Consequences for PhD Student Outcomes in Nazi Germany." *Journal of Political Economy* 118, no. 4 (August): 787–831. (Page 9).
- Wang, Zhaoguang, Yang Yao, and Junni Zhang. 2022. "The competence-loyalty tradeoff in China's political selection." *Journal of Comparative Economics* 50, no. 2 (June): 369–393. (Page 6).
- Weaver, Jeffrey. 2021. "Jobs for Sale: Corruption and Misallocation in Hiring." *American Economic Review* 111, no. 10 (October): 3093–3122. (Page 6).
- Xu, Guo. 2018. "The Costs of Patronage: Evidence from the British Empire." *American Economic Review* 108, no. 11 (November): 3170–3198. (Pages 6, 42).
- Zakharov, Alexei V. 2016. "The Loyalty-Competence Trade-Off in Dictatorships and Outside Options for Subordinates." *The Journal of Politics* 78, no. 2 (April): 457–466. (Page 6).

Appendix

A Appendix: Data

Table A1: 1934 and 1946 teacher rosters descriptive statistics. ([back to text](#))

<i>Panel A: Full teacher sample</i>		
	1934 roster	1946 roster
Number of teachers	44217	55797
N° of teachers by hiring rounds		
1928	4506	3562
1931	4999	3989
1933	8944	7779
1935 + <i>Professional Plan</i>	-	5895
1939 + 1941 <i>Military officers</i>	-	1414
1941	-	5620
1944	-	9793
Female teachers (%)	48.7	52.2
Teachers working in province of birth (%)	51.5	53.1
Average experience (years)	10.9	14.0
Average age (years)	37.6	40.6
<i>Panel B: Teacher subsample linked to purge records</i>		
	1934 roster	1946 roster
Number of teachers	11034	8173
N° of teachers by hiring rounds		
1928	1184	976
1931	1182	993
1933	2105	1654
1935 + <i>Professional Plan</i>	-	-
1939 + 1941 <i>Military officers</i>	-	-
1941	-	-
1944	-	-
Female teachers (%)	48.3	52.4
Teachers working in province of birth (%)	47.5	49.3
Average experience (years)	10.8	21.2
Average age (years)	37.5	47.2

Note: This table reports descriptive statistics for the 1934 and 1946 teacher rosters. Panel A includes the full roster in each year. Panel B restricts the sample to teachers linked between the 1934 roster and purge records in fifteen provinces. In Panel B, the 1946 column reflects the subset of linked 1934 teachers who remained employed in 1946.

Table A2: Summary statistics of purge outcomes ([back to text](#))

	15 provinces	Linked teachers
Number of teachers	16923	10906
Any sanction (%)	27.3	30.9
Permanently dismissed (%)	9.7	10.7
Ban on promotion to managerial positions (%)	15.3	17.3
Temporary suspension (%)	7.5	8.5
Avg. temporary suspension duration (in months)	14.1	13.5
Forced relocation within province (%)	4.7	6.3
Avg. within-province relocation duration (in months)	29.2	29.2
Forced relocation outside province (%)	5.8	7.4
Avg. outside-province relocation duration (in months)	52.3	52.4

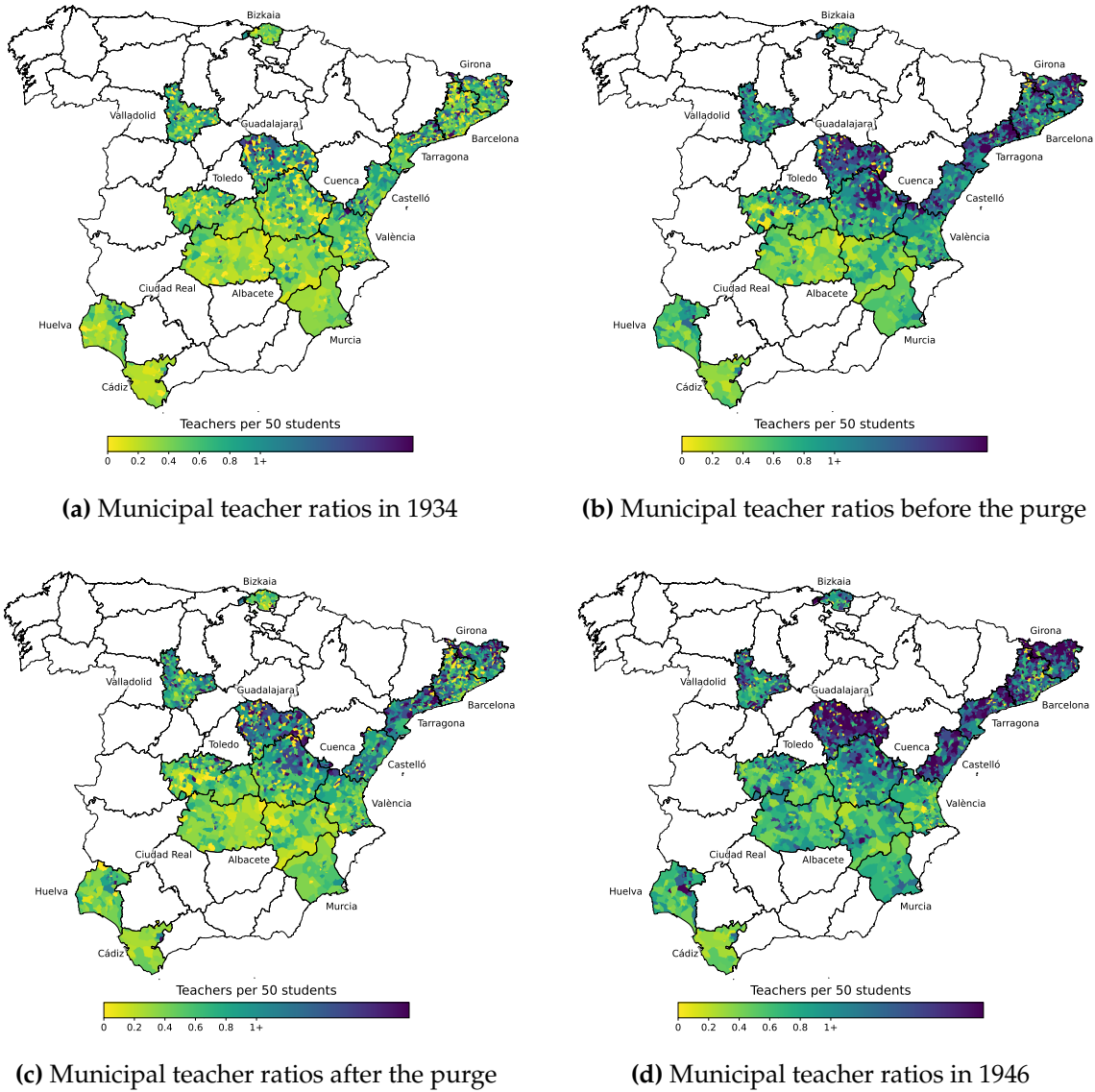
Note: This table reports summary statistics for purge outcomes. The 15 provinces column includes all teachers in the fifteen provinces for which purge records were digitized. The Linked teachers column includes teacher linked between the 1934 roster and the purge records with non-missing purge outcomes.

Table A3: Summary statistics in the municipal-level panel ([back to text](#))

	Mean	Std. Dev.	Min	Max
<i>Panel A: Teacher variables</i>				
Number of teachers in 1934	3.12	12.35	0	487
Number of teachers before the purge	5.55	20.73	0	745
Number of teachers in 1946	5.48	19.80	0	720
Share of purged teachers lost	23.63	29.05	0	100
Change number teachers (% pre-purge to 1946)	10.36	68.60	-100	800
New hires in the 1946 staff (% pre-purge staff)	38.37	51.24	0	400
Net balance of relocations (% pre-purge staff)	-4.38	58.96	-100	500
Teachers per 50 students in 1934	0.65	0.62	0	12
Teachers per 50 students before the purge	1.17	0.80	0	9
Teachers per 50 students in 1946	1.38	1.05	0	13
<i>Panel B: Population</i>				
Population	3,479.91	24,961.99	28	1,081,175
Population: Quintile 1	309.84	98.27	28	472
Population: Quintile 2	648.01	109.78	474	862
Population: Quintile 3	1,148.03	183.67	864	1,482
Population: Quintile 4	2,105.03	431.61	1,488	2,995
Population: Quintile 5	13,236.22	54,500.44	2,999	1,081,175
<i>Panel C: Provinces</i>				
Province: Albacete	0.03	0.18	0	1
Province: Barcelona	0.12	0.33	0	1
Province: Bizkaia	0.04	0.19	0	1
Province: Castelló	0.05	0.23	0	1
Province: Ciudad Real	0.04	0.19	0	1
Province: Cuenca	0.09	0.29	0	1
Province: Cádiz	0.02	0.13	0	1
Province: Girona	0.09	0.28	0	1
Province: Guadalajara	0.12	0.32	0	1
Province: Huelva	0.03	0.17	0	1
Province: Murcia	0.02	0.13	0	1
Province: Tarragona	0.07	0.26	0	1
Province: Toledo	0.08	0.28	0	1
Province: València	0.11	0.31	0	1
Province: Valladolid	0.09	0.29	0	1

Note: This table reports summary statistics for our municipal-level panel of teaching staffing, which includes 2,477 municipalities across 15 provinces for which purge records were digitized. Respectively, Columns (2) to (5) show the sample mean, the standard deviation, the sample minimum, and the sample maximum of the across municipalities distribution of each variable displayed in Column (1). Panel A displays teacher variables, Panel B population in 1940, and Panel C province indicators.

Figure A1: Evolution of municipal teacher-to-student ratios (teachers per 50 students) over time ([back to text](#))



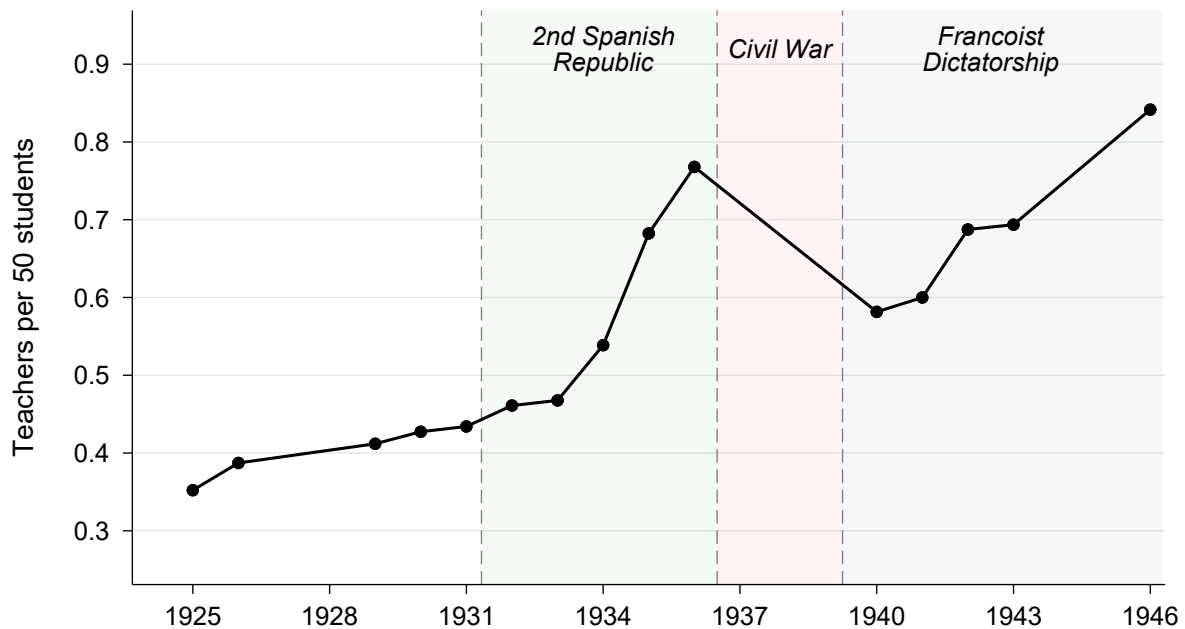
Note: The figure presents a map of mainland Spain showing municipalities in the 15 provinces for which individual teacher sanction records have been digitized: Albacete, Barcelona, Bizkaia, Cádiz, Castelló, Ciudad Real, Cuenca, Girona, Guadalajara, Huelva, Murcia, Tarragona, Toledo, València, and Valladolid. Shading indicates, at the municipal level, the number of teachers per 50 students in each municipality, at four points in time. Panel (a) corresponds to 1934, panel (b) to immediately before the purge, panel (c) to immediately after the purge, and panel (d) to 1946. The number of students in each municipality is estimated using two components: (i) total municipal population and (ii) the share of school-age population in the province. We estimate separate school-age shares for provincial capitals and for the rest of the province. The ratio of 50 students per teacher reflects the policy objectives of both the Second Spanish Republic and the Francoist regime (see Section 4.1). Note that the total number of teachers and the total school-age population vary across panels.

B Appendix: Additional results

Table B1: Evolution of the number of in-service teachers, school-age population and teacher-student ratios in the 1925-1946 period ([back to text](#))

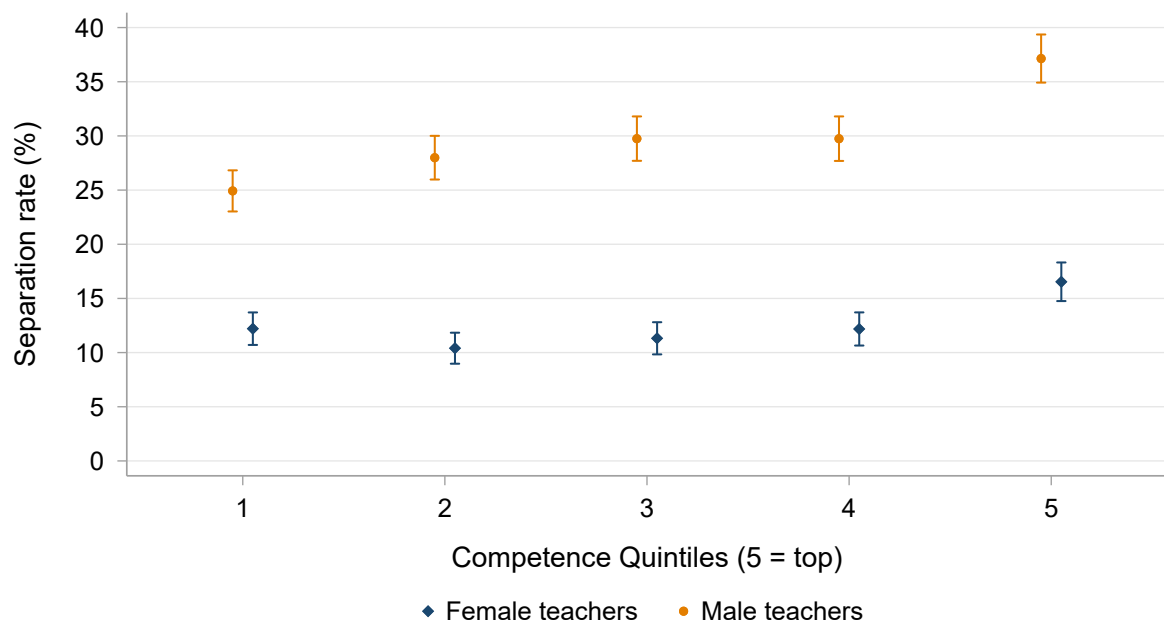
Year	Number of Teachers	School-age population	Teachers per 50 students
1925	20,451	2,903,250	0.352
1926	22,641	2,923,564	0.387
1929	24,587	2,984,509	0.412
1930	25,684	3,004,824	0.427
1931	26,368	3,036,607	0.434
1932	28,292	3,068,391	0.461
1933	28,996	3,100,174	0.468
1934	33,744	3,131,957	0.539
1935	43,182	3,163,741	0.682
1936	49,077	3,195,524	0.768
1940	38,644	3,322,657	0.582
1941	39,337	3,278,232	0.600
1942	44,464	3,233,807	0.687
1943	44,248	3,189,382	0.694
1946	51,441	3,056,108	0.842

Figure B1: Evolution of the teacher-student ratio in the 1925-1946 period ([back to text](#))



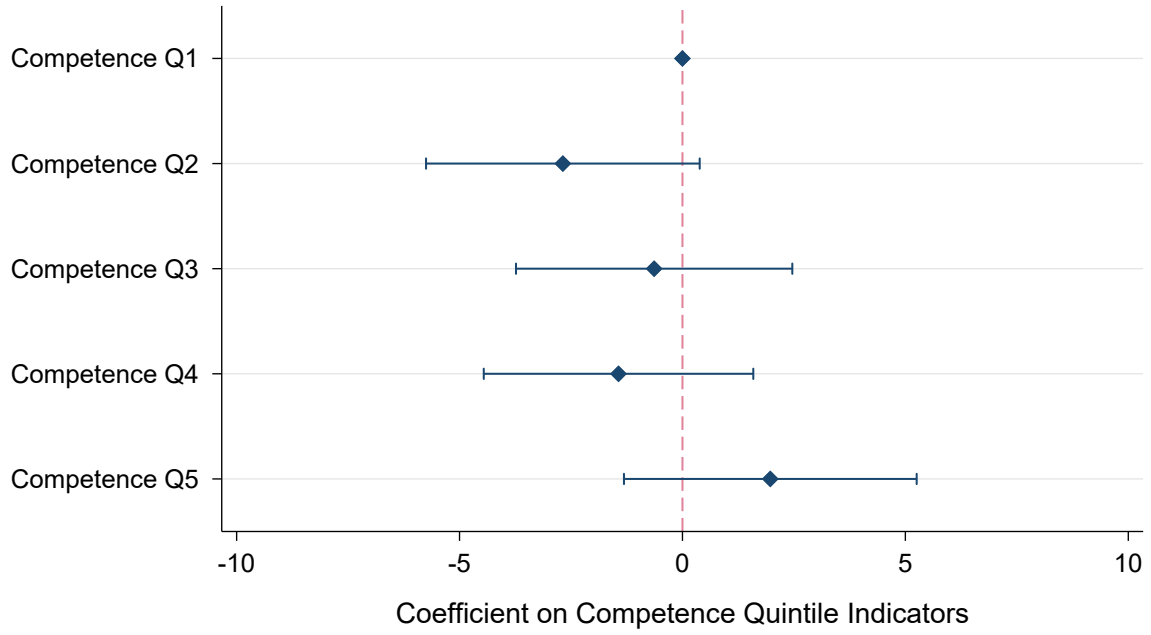
Note: This figure depicts the evolution of the teacher-student ratio for the 1925-1946 period. The method for reconstructing the series is described in Subsection 4.1, and uses data in our teacher panel and in historical census data, including population counts by age group. Table B1 details the evolution of the teacher-student ratio in our series.

Figure B2: Teacher separation rates across competence quintiles, by gender ([back to text](#))



Note: The figure presents a non-parametric representation of the results in Column (5) and (7) of Table 1. The orange set corresponds to the sample of male teachers, and mirrors results in Column (5). The blue set corresponds to the sample of female teachers, and mirrors results in Column (7). The dependent variable is an indicator variable taking value one for teachers who were separated from service following the purge. The independent variable for binning is a categorical variable for competence quintile. It is constructed by first partialling out gender by cohort fixed effects. Each dot represents the mean separation rate in the residualized data. Blue and orange lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

Figure B3: Teacher separation rates not directly explained to dismissal sanctions, across competence quintiles ([back to text](#))



Note: The figure depicts estimated coefficients for γ_c estimated using Equation 2. Our sample includes 4,336 teachers that entered the profession between 1928 and 1933 for which we have linked purge records; 2,229 male teachers and 2,107 female teachers. The complete specification estimated is $\text{Separation}_{i,c} = \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} + \sum_{q=2}^5 \gamma_q \left(\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} \times \text{Dismissal}_{i,c} \right) + \delta \text{Dismissal}_{i,c} + \psi_{a(i)} + \psi_{c,g(i)} + \varepsilon_{i,c}$, where i denotes the teacher and c the cohort. $\text{Separation}_{i,c}$ is an indicator variable for teachers who were separated from service by 1946. $\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\}$ are indicator variables for competence quintiles, with the bottom quintile omitted as the reference group. $\psi_{a(i)}$ and $\psi_{c,g(i)}$ denote age and cohort-by-gender fixed effects, respectively. $\text{Dismissal}_{i,c}$ is an indicator variable for teachers who were permanently dismissed in the purge. The coefficients γ_q capture differences in separation rates across competence quintiles not directly explained by dismissal sanctions. Blue lines denote 95% confidence intervals.

Table B2: OLS estimates: teacher separation, dismissal, and sanction rates across competence quintiles, in the subsample of teachers with linked purge records ([back to text](#))

	All teachers			Male teachers		Female teachers	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: Separations</i>							
Competence Q2	-2.167 (1.883)	-2.053 (1.881)	-1.906 (1.882)	1.322 (3.002)	2.056 (3.007)	-5.793 (2.161)**	-6.003 (2.159)**
Competence Q3	1.627 (1.885)	1.704 (1.882)	2.006 (1.885)	4.070 (2.942)	4.549 (2.942)	-0.950 (2.284)	-1.176 (2.287)
Competence Q4	0.892 (1.853)	0.982 (1.849)	1.208 (1.861)	4.295 (2.912)	4.941 (2.948)	-2.665 (2.206)	-2.972 (2.209)
Competence Q5	4.908 (1.958)*	5.008 (1.955)*	5.128 (1.968)**	8.799 (3.059)**	9.628 (3.077)**	0.897 (2.376)	0.543 (2.377)
Observations	4,336	4,336	4,336	2,229	2,229	2,107	2,107
R ²	0.05	0.05	0.06	0.01	0.03	0.01	0.01
Outcome average	18.27	18.27	18.27	25.93	25.93	10.16	10.16
<i>Panel B: Dismissals</i>							
Competence Q2	-0.819 (1.483)	-0.777 (1.482)	-0.582 (1.491)	-0.048 (2.493)	0.563 (2.508)	-1.619 (1.460)	-1.646 (1.464)
Competence Q3	3.202 (1.538)*	3.237 (1.536)*	3.591 (1.546)*	5.356 (2.576)*	5.916 (2.578)*	0.911 (1.572)	0.883 (1.583)
Competence Q4	2.334 (1.498)	2.369 (1.494)	2.762 (1.513)	4.972 (2.525)*	5.724 (2.564)*	-0.474 (1.479)	-0.513 (1.479)
Competence Q5	3.409 (1.554)*	3.448 (1.553)*	3.752 (1.574)*	6.916 (2.645)**	7.763 (2.667)**	-0.257 (1.528)	-0.301 (1.538)
Observations	4,336	4,336	4,336	2,229	2,229	2,107	2,107
R ²	0.05	0.05	0.05	0.01	0.02	0.00	0.00
Outcome average	11.28	11.28	11.28	17.72	17.72	4.46	4.46
<i>Panel C: Any sanction</i>							
Competence Q2	-0.684 (2.344)	-0.793 (2.348)	-0.788 (2.366)	3.608 (3.535)	3.726 (3.564)	-5.671 (3.017)	-5.652 (3.020)
Competence Q3	0.841 (2.285)	0.743 (2.288)	0.832 (2.302)	4.604 (3.434)	4.542 (3.460)	-3.525 (2.974)	-3.504 (2.984)
Competence Q4	2.725 (2.271)	2.632 (2.274)	2.643 (2.306)	6.054 (3.387)	6.156 (3.453)	-1.179 (2.982)	-1.151 (3.000)
Competence Q5	4.839 (2.323)*	4.735 (2.328)*	4.864 (2.361)*	10.228 (3.489)**	10.363 (3.542)**	-1.192 (3.035)	-1.159 (3.060)
Observations	4,336	4,336	4,336	2,229	2,229	2,107	2,107
R ²	0.07	0.07	0.07	0.00	0.02	0.00	0.00
Outcome average	32.80	32.80	32.80	44.46	44.46	20.46	20.46
Gender FE	Yes	No	No	No	No	No	No
Cohort FE	Yes	No	No	Yes	Yes	Yes	Yes
Gender × Cohort FE	No	Yes	Yes	No	No	No	No
Age FE	No	No	Yes	No	Yes	No	Yes

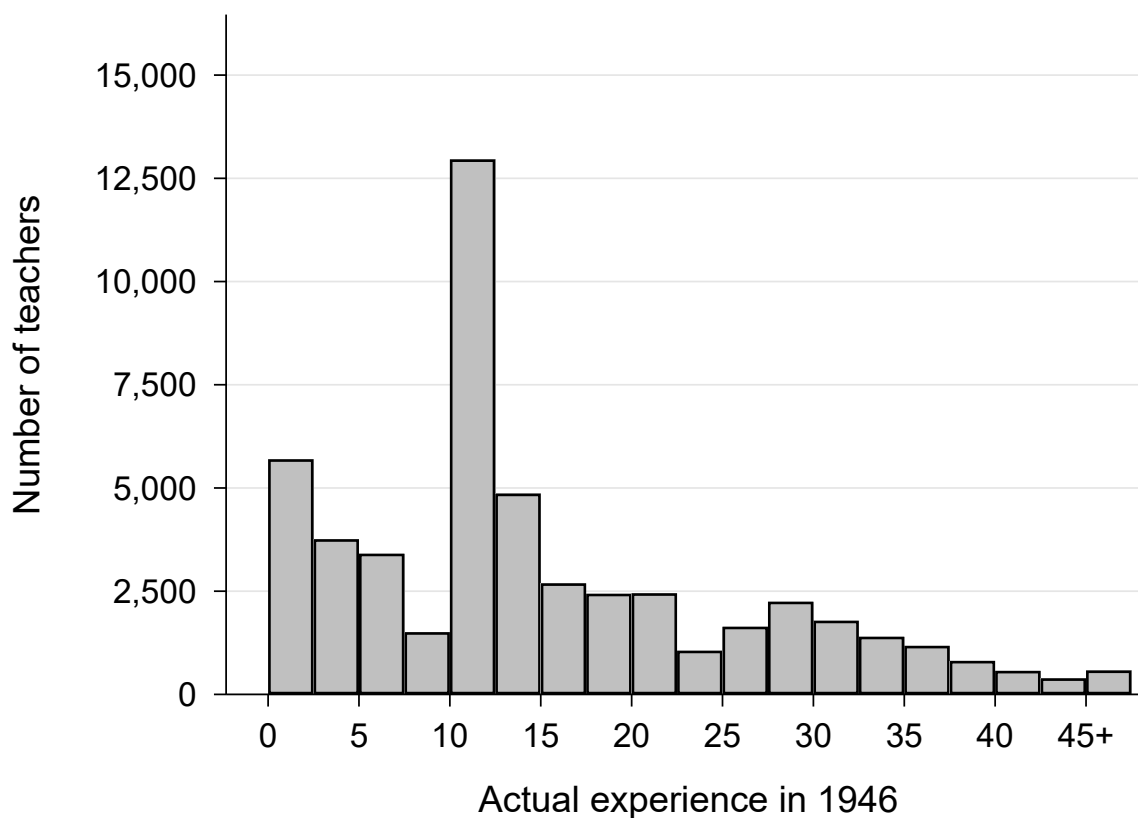
Note: This table displays the analogous estimates of the OLS regression described in Equation 1, for three outcome variables. Panel A reports estimates using as dependent variable and indicator taking value 1 if teachers were separated from service after the purge. Panel B reports estimates using as dependent variable and indicator taking value 1 if teachers faced a dismissal sanction. Panel C reports estimates using as dependent variable and indicator taking value 1 if teachers faced any type of sanction. Our sample includes 4,336 teachers that entered the profession between 1928 and 1933 for which we have linked purge records; 2,229 male teachers and 2,107 female teachers. Columns (1) to (3) show estimates for the sample of male and female teachers. Columns (4) and (5) show estimates for the subsample of male teachers, whereas columns (6) and (7) show them for the subsample of female teachers. The complete specification estimated in Column (3) is $Y_{i,c} = \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} + \psi_{a(i)} + \psi_{c,g(i)} + \varepsilon_{i,c}$, where i denotes the teacher and c the cohort. Separation $_{i,c}$ is an indicator variable taking value one for teachers who were separated from service following the purge. $\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\}$ are indicator variables for competence quintiles, with the bottom quintile omitted as the reference group. $\psi_{a(i)}$ and $\psi_{c,g(i)}$ denote age and cohort-by-gender fixed effects, respectively. We report heteroskedasticity-robust standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B3: OLS estimates: teacher separation rates across competence quintiles, by hiring cohort
(back to text)

	1928 cohort		1931 cohort		1933 cohort	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Male teachers</i>						
Competence Q2	2.172 (2.429)	2.061 (2.447)	0.569 (3.116)	0.151 (3.134)	5.122 (2.235)*	5.062 (2.230)*
Competence Q3	2.335 (2.429)	1.912 (2.453)	2.990 (3.114)	2.582 (3.126)	7.618 (2.237)***	7.298 (2.234)**
Competence Q4	4.820 (2.430)*	4.555 (2.453)	4.424 (3.116)	4.059 (3.136)	5.122 (2.235)*	4.424 (2.236)*
Competence Q5	9.190 (2.461)***	9.187 (2.518)***	6.990 (3.154)*	6.025 (3.185)	17.229 (2.265)***	16.230 (2.270)***
Observations	3,069	3,069	2,208	2,208	4,200	4,200
R ²	0.01	0.02	0.00	0.02	0.01	0.03
Outcome average	24.73	24.73	32.52	32.52	32.05	32.05
<i>Panel B: Female teachers</i>						
Competence Q2	-2.710 (2.697)	-1.867 (2.715)	-2.800 (1.838)	-2.691 (1.852)	-0.860 (1.574)	-1.198 (1.581)
Competence Q3	-1.116 (2.694)	-1.248 (2.711)	-1.918 (1.837)	-1.782 (1.854)	-0.133 (1.574)	-0.450 (1.584)
Competence Q4	-0.709 (2.694)	-0.564 (2.717)	-3.161 (1.838)	-3.012 (1.860)	1.995 (1.574)	1.680 (1.586)
Competence Q5	3.269 (2.733)	3.952 (2.762)	-0.063 (1.861)	0.059 (1.889)	7.201 (1.595)***	6.762 (1.616)***
Observations	1,228	1,228	2,778	2,778	4,731	4,731
R ²	0.00	0.03	0.00	0.01	0.01	0.01
Outcome average	10.18	10.18	10.76	10.76	14.06	14.06
Age FE	No	Yes	No	Yes	No	Yes

Note: This table displays the estimates of an OLS regression for a reduced version of Equation 1. Our sample includes teachers that entered the profession between 1928 and 1933. Panel A reports estimates for male teachers, whereas panel B does so for female teachers. Columns (1) and (2) show estimates for the 1928 cohort. Columns (3) and (4) show estimates for the 1931 cohort. Columns (5) and (6) show estimates for the 1933 cohort. The complete specification estimated in Column (3) is $\text{Separation}_{i,c} = \sum_{q=2}^5 \beta_q \mathbb{1}\{\text{Competence}_{i,c} \in Q_q\} + \psi_{a(i)} + \varepsilon_{i,c}$, where i denotes the teacher and c the cohort. $\text{Separation}_{i,c}$ is an indicator variable taking value one for teachers who were separated from service following the purge. $\mathbb{1}\{\text{Competence}_{i,c} \in Q_q\}$ are indicator variables for competence quintiles, with the bottom quintile omitted as the reference group. $\psi_{a(i)}$ denote age fixed effects. We report heteroskedasticity-robust standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure B4: Observed distribution of teacher experience in 1946 ([back to text](#))

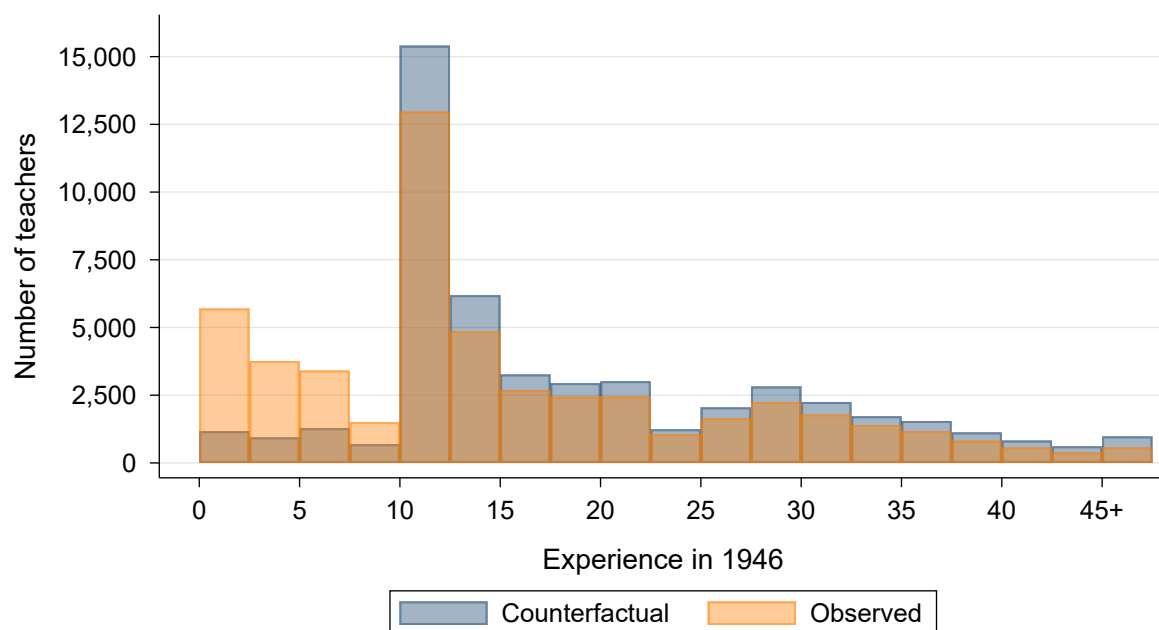


Note: The figure presents the observed distribution of teacher experience among in-service teachers in the 1946 teacher roster, in bins of 2.5 years.

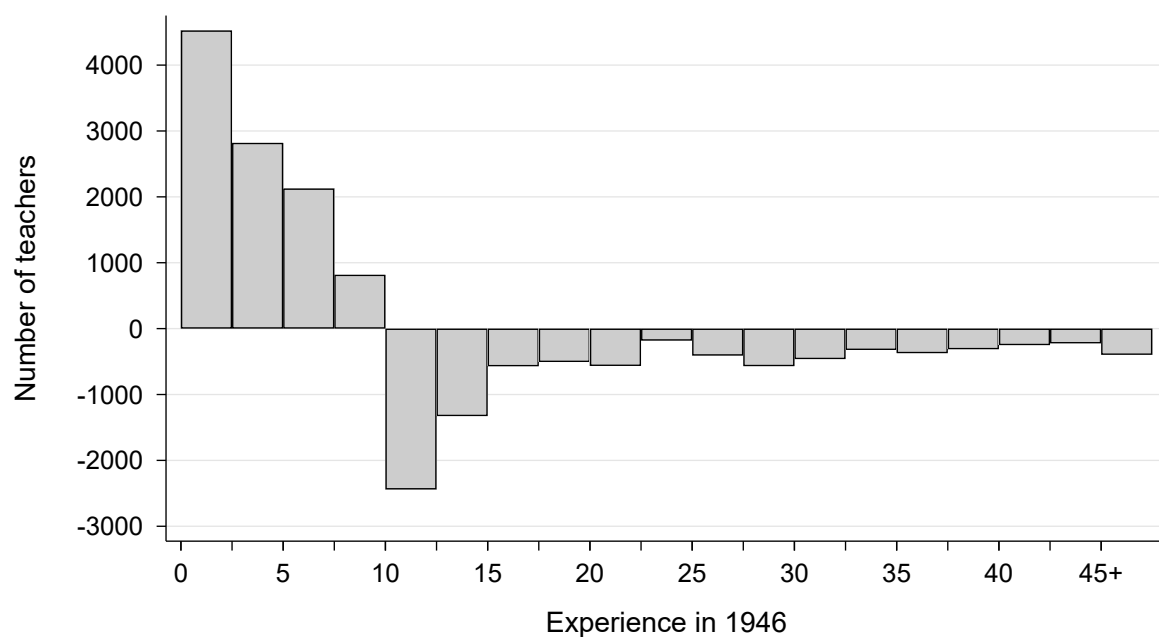
Table B4: Summary statistics of the observed and counterfactual distributions of teacher experience ([back to text](#))

	Obs.	Mean	Median	p10	p25	p75	P90
Real distribution	51,446	15.27	12.00	1.92	7.42	21.58	32.08
Counterf. - 10% attrition	50,076	17.83	13.00	6.00	10.33	25.25	34.25
Counterf. - 2.5% attrition	50,076	18.87	13.17	10.00	12.00	26.33	35.17
Counterf. - optimal teacher ratio	61,121	15.60	12.00	3.00	8.00	21.67	32.58

Figure B5: Observed and counterfactual distributions of teacher experience in 1946, under the alternative counterfactual scenario based on a 2.5% natural separation rate ([back to text](#))

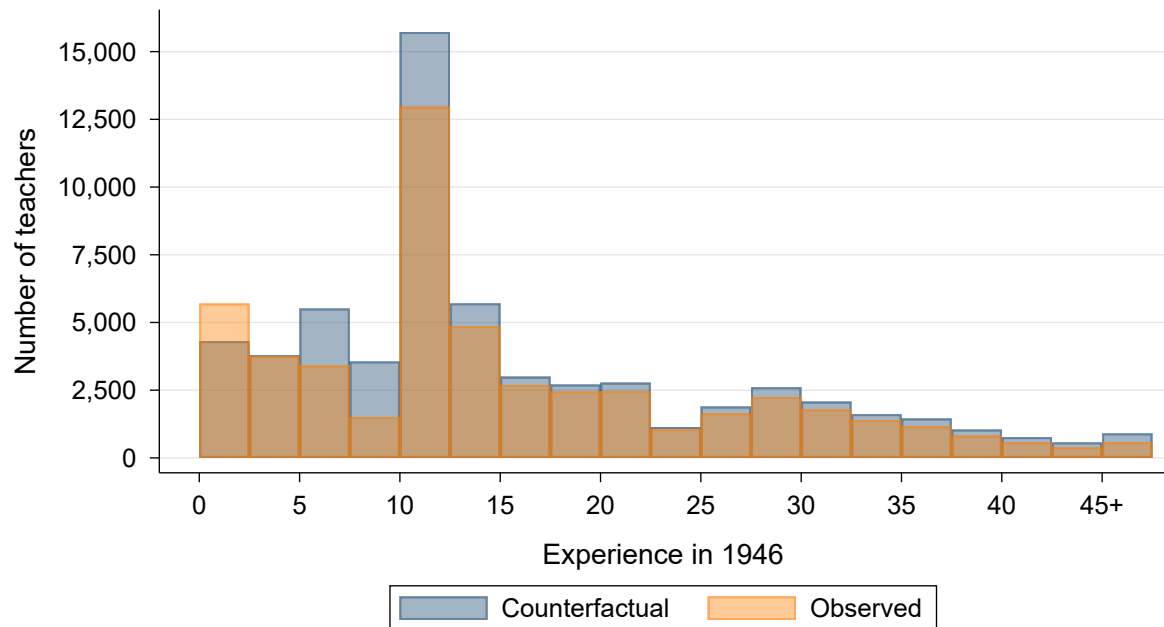


(a) Distribution of teacher experience in 1946: observed vs no-purge counterfactual

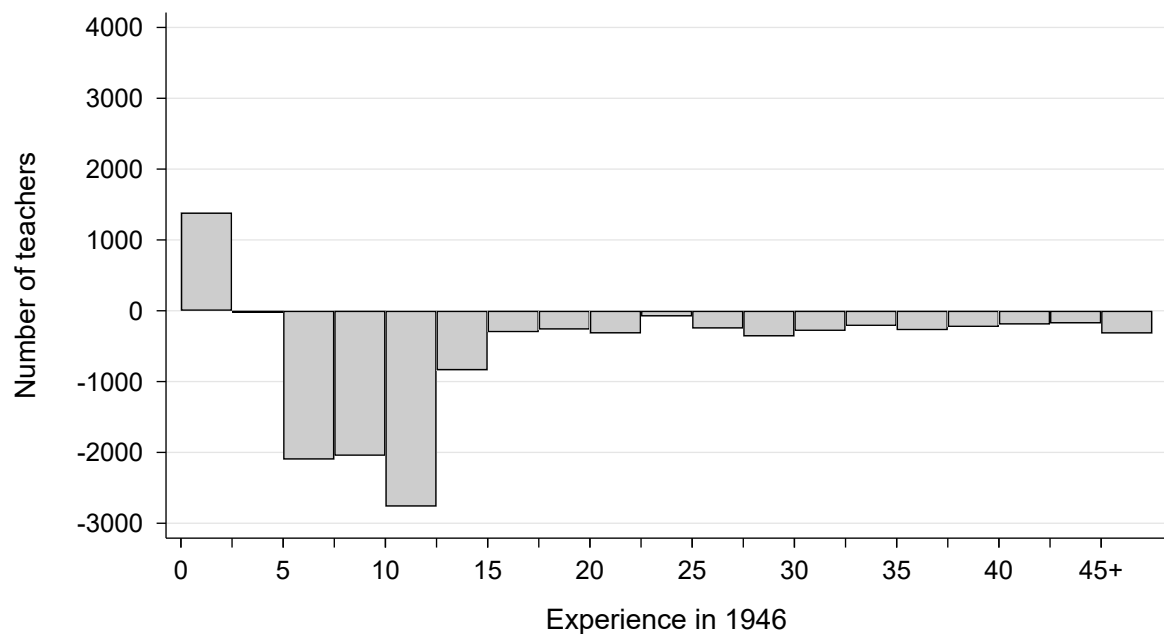


(b) Difference between actual and counterfactual distributions

Figure B6: Observed and counterfactual distributions of teacher experience in 1946, under the alternative counterfactual scenario based on additional hiring to reach the 1 teacher per 50 students ratio ([back to text](#))

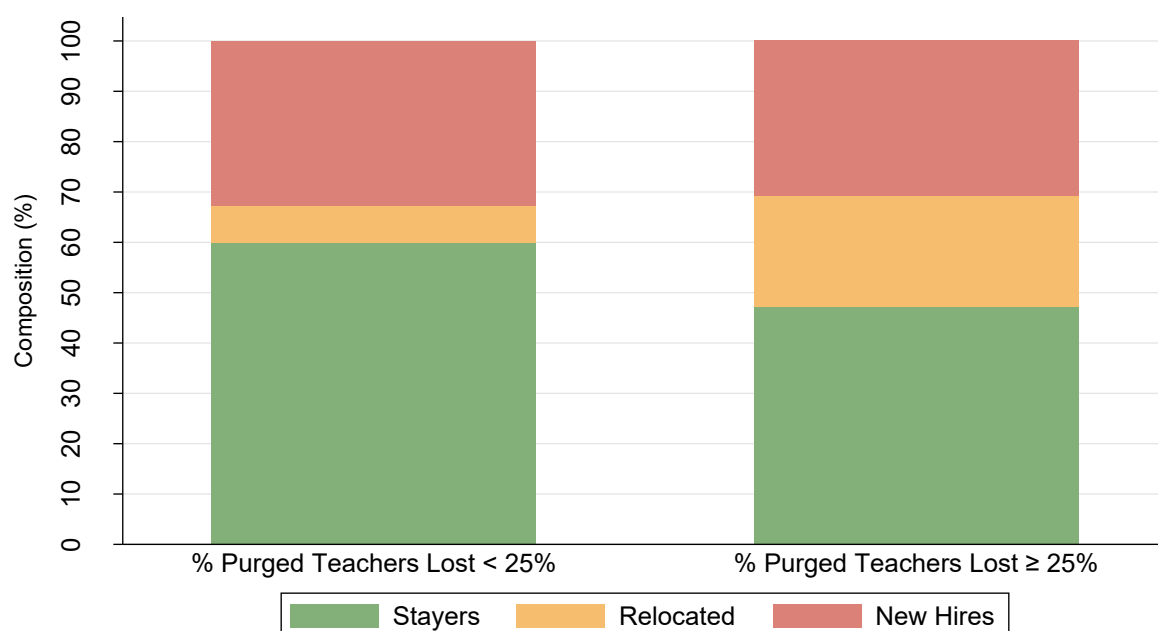


(a) Distribution of teacher experience in 1946: observed vs no-purge counterfactual



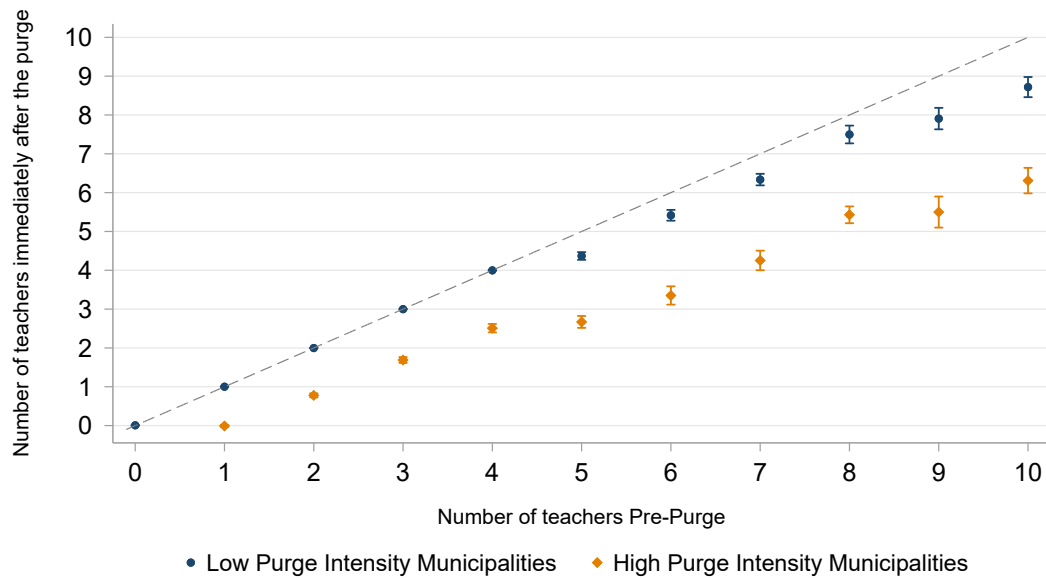
(b) Difference between actual and counterfactual distributions

Figure B7: Municipal composition of the 1946 teaching workforce (stayers, relocated, and newly hired teachers) by purge intensity ([back to text](#))

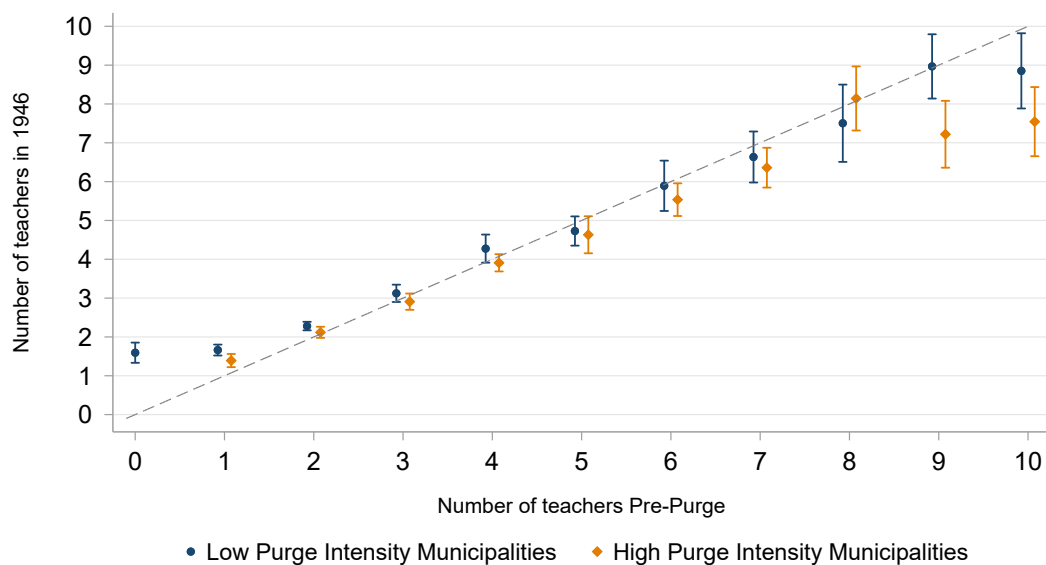


Note: The figure presents the average municipal composition of the local staff in 1946 for two groups of municipalities: those that lost less than a quarter of their staff in the purge process due to sanctions (low purge intensity), and those that lost a quarter or more of their staff (high purge intensity). The composition of the local staff is given by the share of teachers that were already working in the municipality before the purge, the share of incoming teachers that relocated to the municipality after the purge, and the share of new hires in the period 1940 to 1946.

Figure B8: Size of the local teaching staff immediately after the purge and in 1946, across pre-purge staff levels and by purge intensity (low- versus high-intensity municipalities). ([back to text](#))



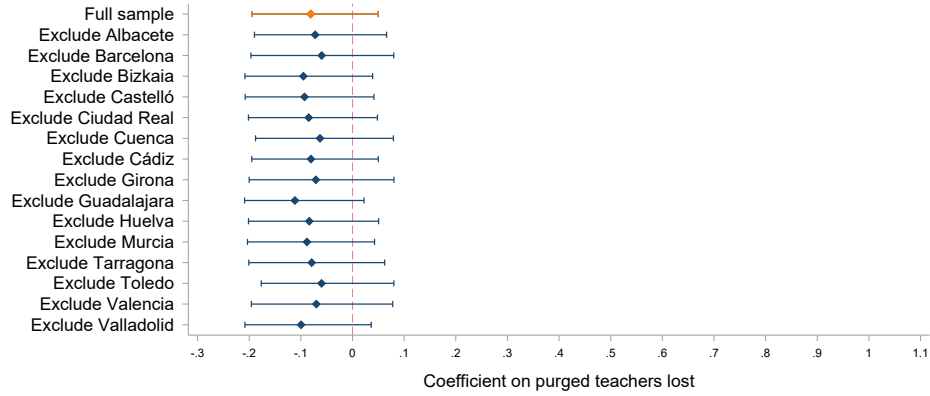
(a) Size of the local teaching staff immediately after the purge



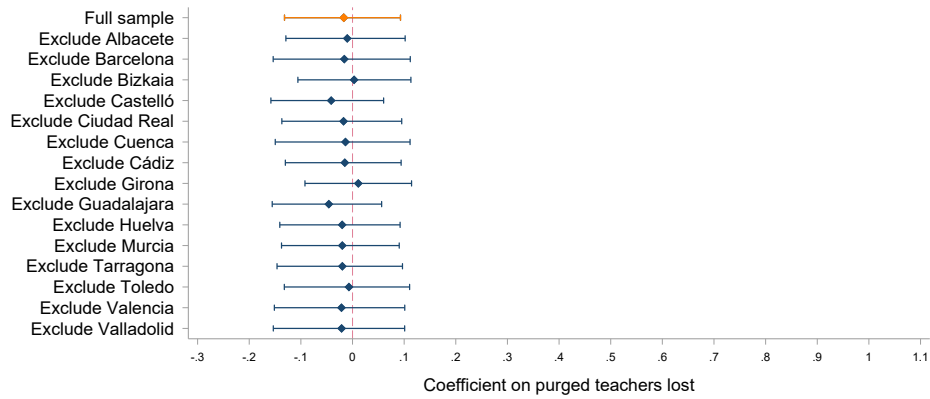
(b) Size of the local teaching staff in 1946

Note: The figure in panel (a) presents a non-parametric representation of the number of teachers immediately after the purge across municipalities with different number of teachers before the purge process. The figure in panel (b) presents a non-parametric representation of the number of teachers in 1946 across municipalities with different number of teachers before the purge process. The orange sets correspond to municipalities exposed to low purge intensity (lost less than a quarter of their staff in the purge process). The blue sets correspond to municipalities exposed to high purge intensity (lost more than a quarter of their staff in the purge process). Both figures are constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels, and provinces. Each dot represents the mean number of teachers working in the municipality, either immediately after the purge (a) or in 1946 (b), in the residualized data. Blue and orange lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

Figure B9: Robustness to Province Exclusions: municipal purge intensity, size and composition of the 1946 teaching staff. ([back to text](#))



(a) Outcome: percent change in the number of teachers (pre-purge to 1946)



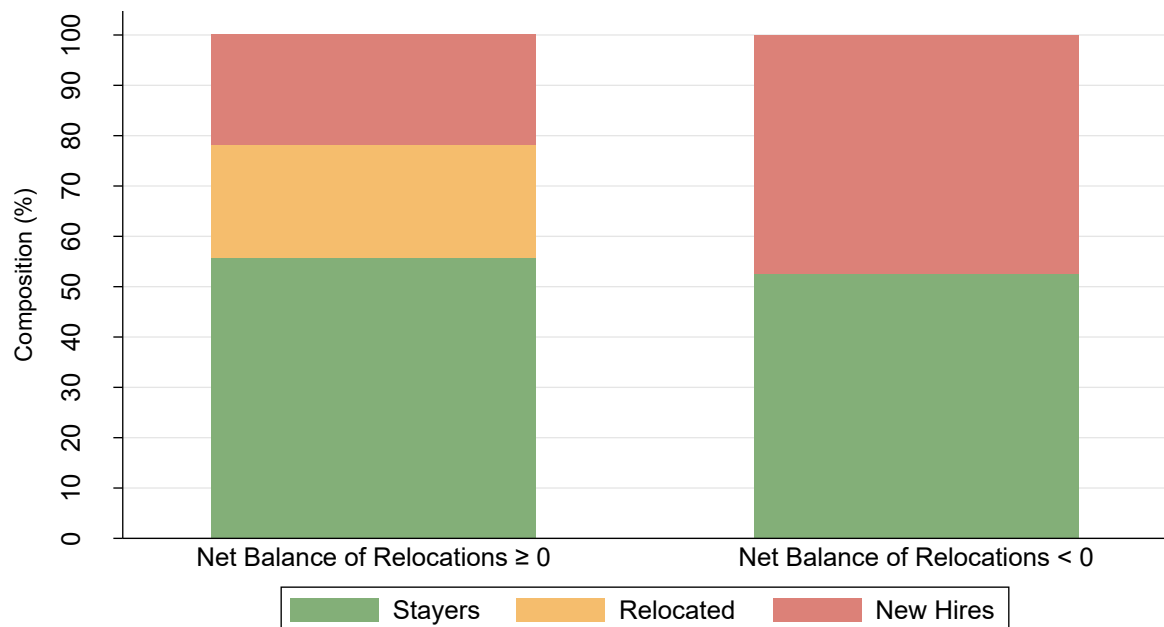
(b) Outcome: New hires (1940–1946) assigned to the municipality in 1946, as a percentage of pre-purge staff.



(c) Outcome: Net balance of relocations in the 1940 to 1946 period, as a percentage of pre-purge staff

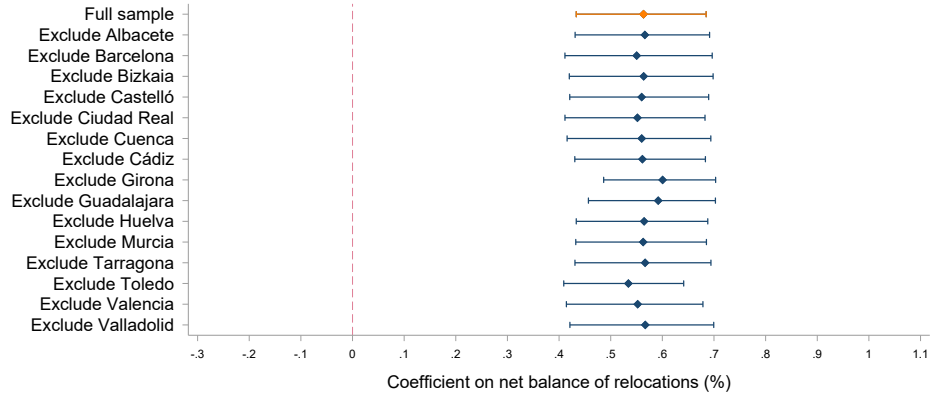
Note: This figure displays robustness check estimates of the OLS regression described in Equation 4 obtained by sequentially excluding one province at a time. Our full sample includes 2,388 municipalities in 15 provinces (89 municipalities are excluded as they did not have any teacher assigned at the time of the purge). The specification we estimate is $Y_m = \beta \text{Purged Teachers Lost}_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m$ where m denotes the municipality and $p(m)$ the province. Panel (a) shows estimates using as an outcome the percent change in the number of teachers between the pre-purge level and 1946. Panel (b) uses the number of new hires appointed between 1940 and 1946 and assigned to the municipality in 1946. Panel (c) uses the net balance of relocations during the period, expressed as a percentage of the pre-purge staff. All outcomes are expressed relative to the initial (pre-purge) number of teachers, for comparability. Purged Teachers Lost _{m} is the share of teachers removed in the purge due to sanctions, including dismissals, temporary suspensions, or forced relocations. In all cases, we include fixed effects for population quintiles, pre-purge staffing levels and province ($\psi_{q(m)}$, $\psi_{n(m)}$, $\psi_{p(m)}$). The lines denote 95% confidence intervals estimated using wild-cluster bootstrap by province, using Webb weights and 100,000 replications, to account for the low number of clusters.

Figure B10: Municipal composition of the 1946 teaching workforce (stayers, relocated, and newly hired teachers) by relocation balance ([back to text](#))

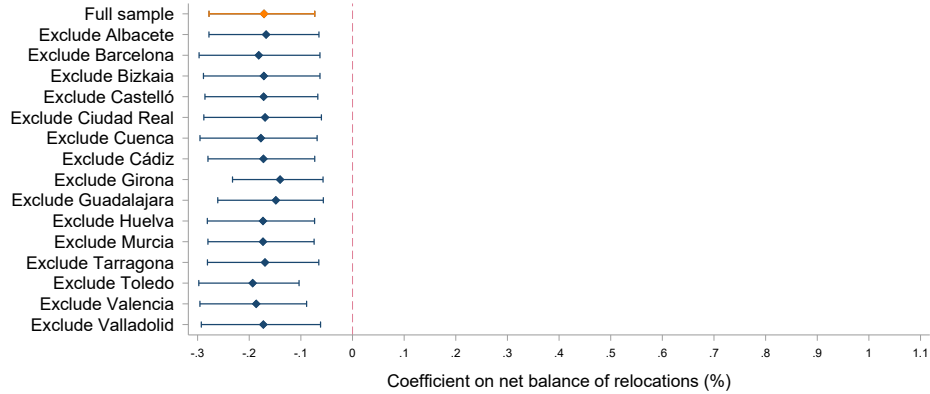


Note: The figure presents the average municipal composition of the local staff in 1946 for two groups of municipalities: those that experienced neutral or positive relocation balances between 1940 and 1946, and those that experienced negative relocation balances in the same period. The composition of the local staff is given by the share of teachers that were already working in the municipality before the purge, the share of incoming teachers that relocated to the municipality after the purge, and the share of new hires in the period 1940 to 1946.

Figure B11: Robustness to Province Exclusions: relocation balances, size of the 1946 teaching staff and share of new hires. ([back to text](#))



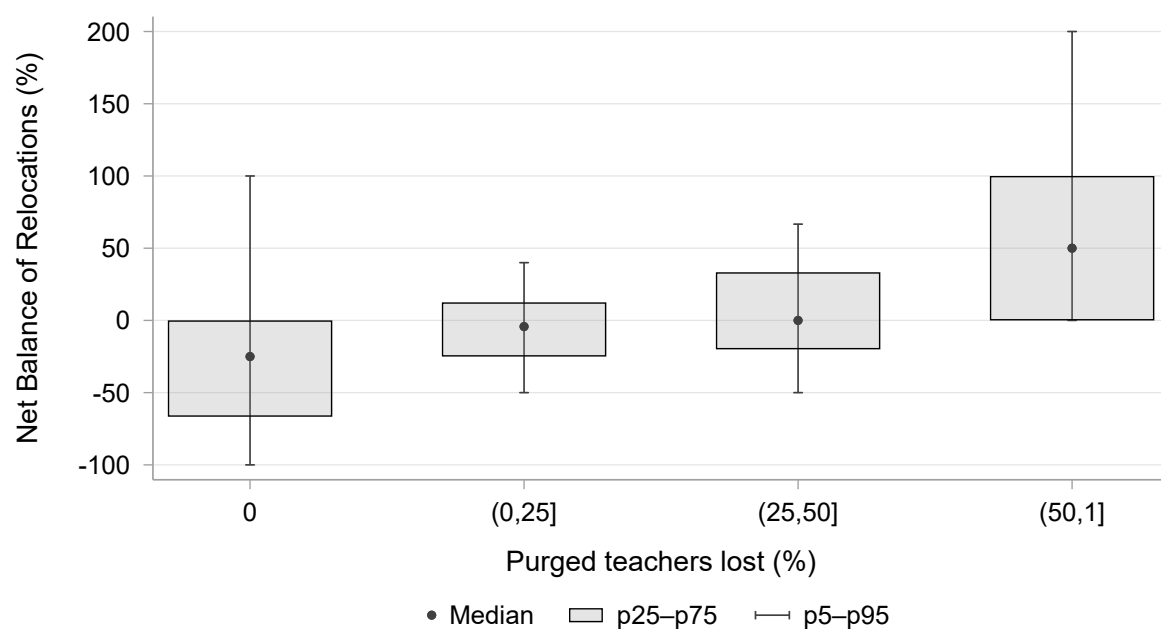
(a) Outcome: percent change in the number of teachers (pre-purge to 1946)



(b) Outcome: New hires (1940–1946) assigned to the municipality in 1946, as a percentage of pre-purge staff.

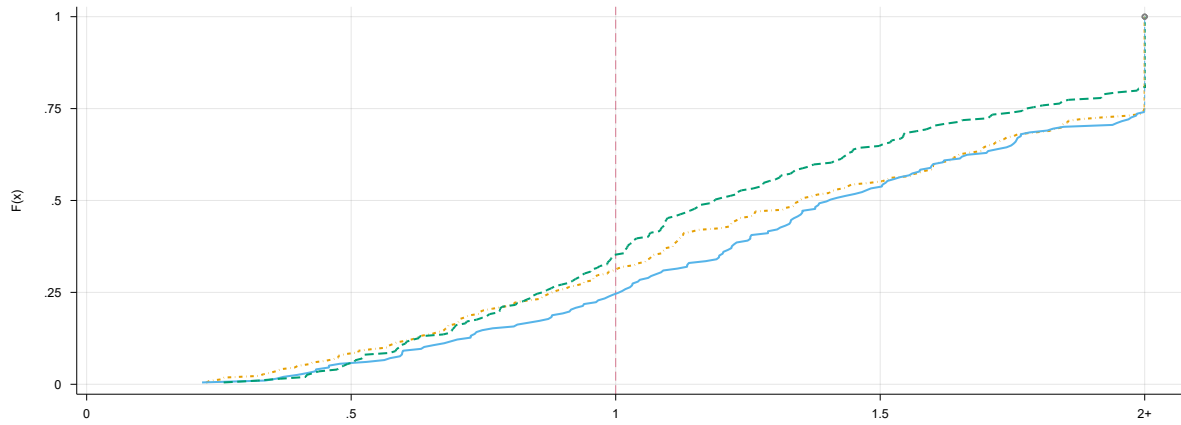
Note: This figure displays robustness check estimates of the OLS regression described in Equation 5 obtained by sequentially excluding one province at a time. Our full sample includes 2,388 municipalities in 15 provinces (89 municipalities are excluded as they did not have any teacher assigned at the time of the purge). The specification we estimate is $Y_m = \beta \Delta \text{Relocations}_m + \psi_{q(m)} + \psi_{n(m)} + \psi_{p(m)} + \varepsilon_m$ where m denotes the municipality and $p(m)$ the province. Panel (a) shows estimates using as an outcome the percent change in the number of teachers between the pre-purge level and 1946. Panel (b) uses the number of new hires appointed between 1940 and 1946 and assigned to the municipality in 1946. All outcomes are expressed relative to the initial (pre-purge) number of teachers, for comparability. $\Delta \text{Relocations}_m$ is the net balance of relocations during the period, expressed as a percentage of the pre-purge staff. In all cases, we include fixed effects for population quintiles, pre-purge staffing levels and province ($\psi_{q(m)}$, $\psi_{n(m)}$, $\psi_{p(m)}$). The lines denote 95% confidence intervals estimated using wild-cluster bootstrap by province, using Webb weights and 100,000 replications, to account for the low number of clusters.

Figure B12: Boxplot with 5th–95th percentile whiskers of relocation balances across levels of municipal purge intensity ([back to text](#))

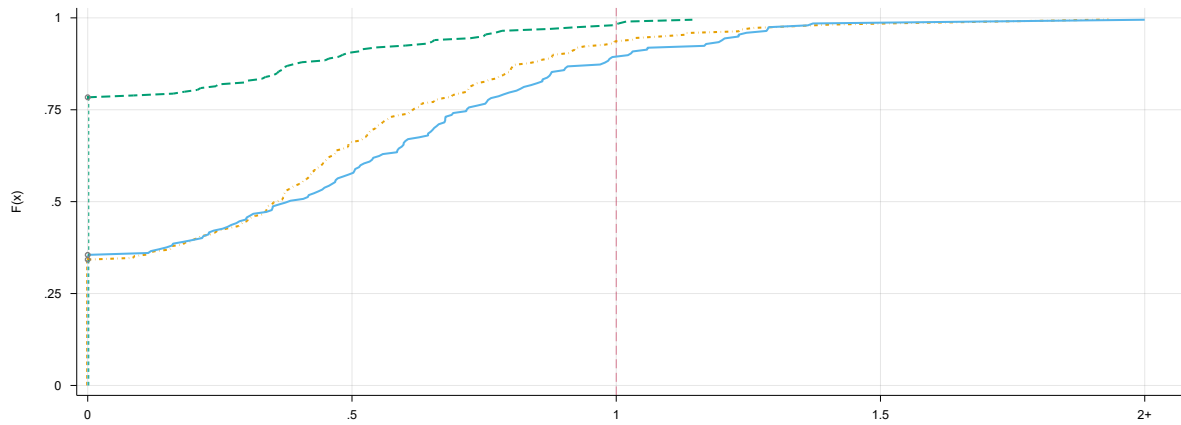


Note: The figure presents boxplots with 5th–95th percentile whiskers of the net balance of relocations between 1940 and 1946 for municipalities with different levels of purge-related teacher losses. From left to right, the 4 groups are: municipalities that lost no teachers, municipalities that lost teachers but not more than a quarter of their staff, municipalities that lost between a quarter and a half of their staff, and municipalities that lost more than half of their staff.

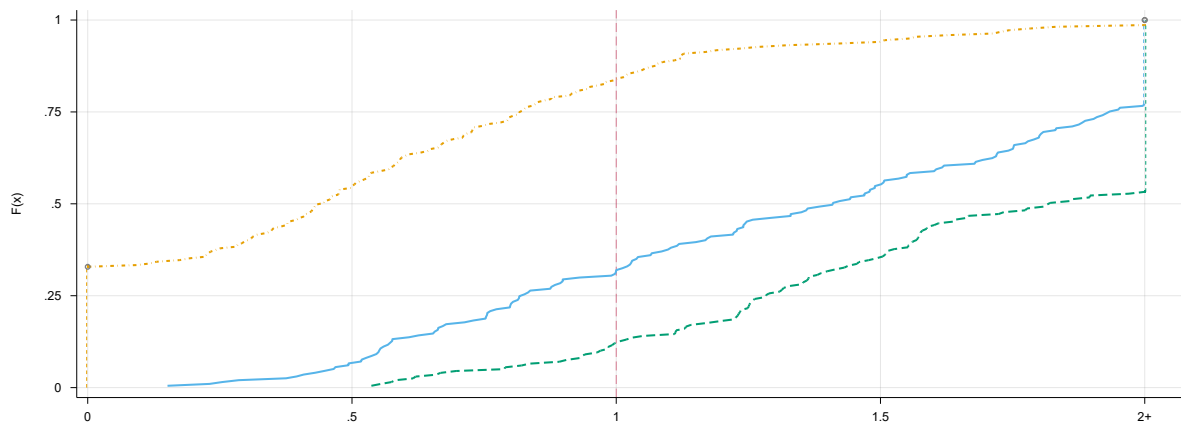
Figure B13: Cumulative distribution of teacher-to-student ratios for municipalities in the bottom quartile of relocation balances, by terciles of new hires. ([back to text](#))



(a) Immediately after the purge



(b) After the purge and relocation

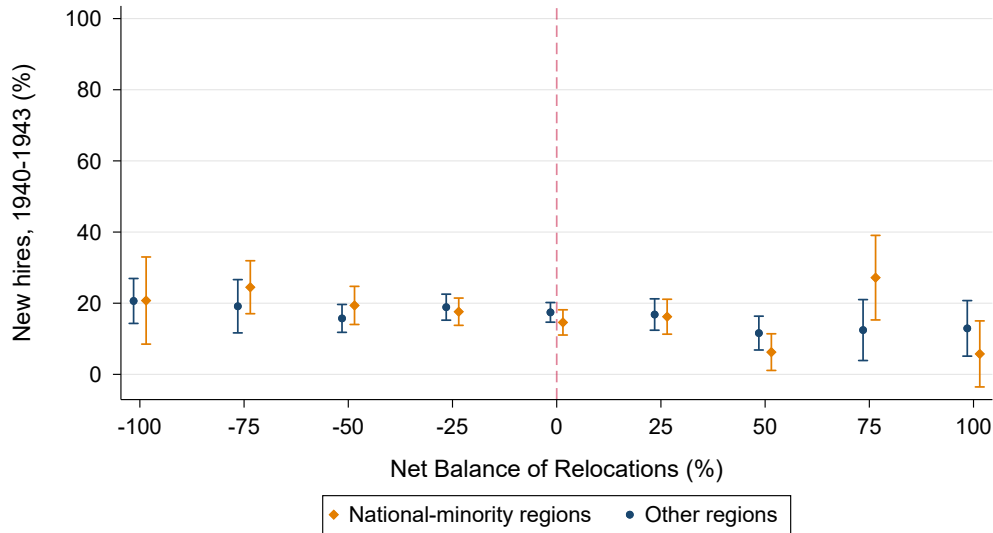


(c) In 1946, after the purge, relocation, and allocation of new hires

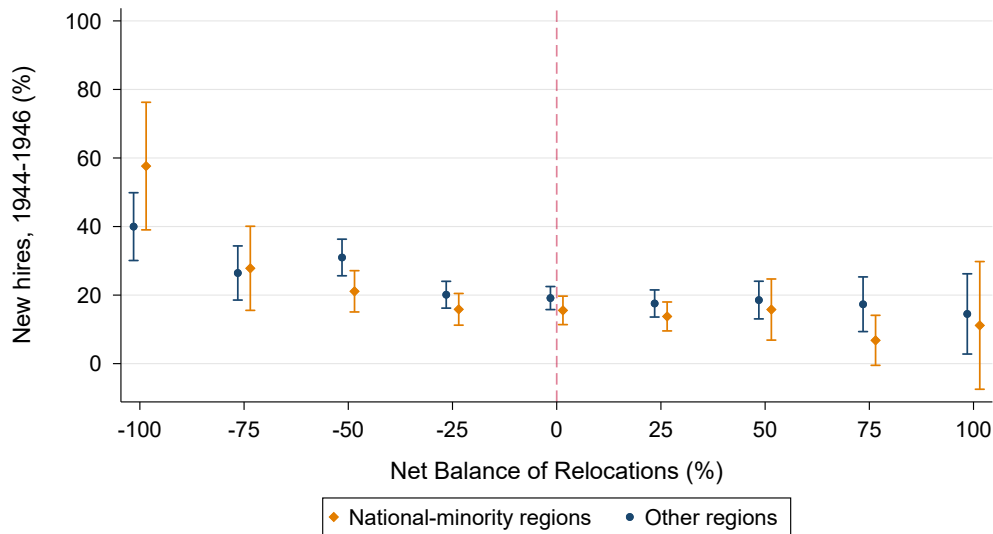
--- Low hires (T1) — Medium Hires (T2) - - High hires (T3)

Note: The figure presents the cumulative distribution of teacher-to student ratios for municipalities in the bottom quartile of relocation balances (this includes municipalities with Net Balance of Relocations $\leq -40\%$, measured as a percentage of pre-purge staff). We classify these municipalities in three terciles, according to the new hires they receive by 1946: Panel (a) depicts the cumulative distribution of teacher-to-student ratios immediately after the purge; Panel (b) does so after the relocation; finally, Panel (c) depicts them in 1946.

Figure B14: Share of new hires across relocation balances in provinces with strong regional nationalist movements and other regions, by recruitment rounds ([back to text](#))



(a) Allocation of early hires (1940-1943)



(b) Allocation of late hires (1944-1946)

Note: The figures present a non-parametric representation of the distribution of new hires across municipalities with different relocation balances. The orange set corresponds to provinces with strong regional nationalist movements (Barcelona, Bizkaia, Girona, Tarragona). The blue set corresponds to 11 provinces without strong regional nationalist movements. In panel (a), the dependent variable is the number of new hires appointed between 1940 and 1943 and assigned to the municipality by 1946, expressed as a percentage of the pre-purge teaching staff. In panel (b), the dependent variable is analogous, but for the 1944 to 1946 hires. The independent variable for binning is the net balance of relocations during the 1940-1946 period, expressed as a percentage of the pre-purge staff. It is constructed by first partialling out fixed effects for population quintiles, pre-purge staffing levels and provinces. Each dot represents the mean number of new hires, expressed as a percentage of the pre-purge teaching staff, in the residualized data. Blue and orange lines denote 95% confidence intervals. The graph is obtained following robust inference procedures (Cattaneo et al., 2024).

C Appendix: Original data records

Figure C1: Sample page of a purge record opening file. ([back to text](#))

Núm.

EXPEDIENTE

instruido al Maestro Nacional
de Plan Profesional
Don José Morales Malva

por la Comisión depuradora del Magisterio de la provincia de Valencia, en virtud del
Decreto de 8 de Noviembre de 1936 ("Boletín Oficial" n.º 27), Orden de 18 de Marzo
de 1939 y nombramientos de 10 de Julio último.



FECHA DE INCOACIÓN

1 marzo 1940

COMISIÓN DEPURADORA

Presidente.....	D. Francisco Morote Greus
Vocal.....	D. Vicente Garrido Pastor, Obro.
"	B. Francisco Marco Marañiano
"	D. Alfredo Soriano Hernandez
Secretaria.....	D ^a Mariana Ruiz Vallecillo

Vocales adjuntos de F.E.T. y de las J.O.N.S.

D. Vicente Garcia Flaócer
D. Justo de Avila San Pascual

Figure C2: Sample page of charges issued in a purge record. ([back to text](#))

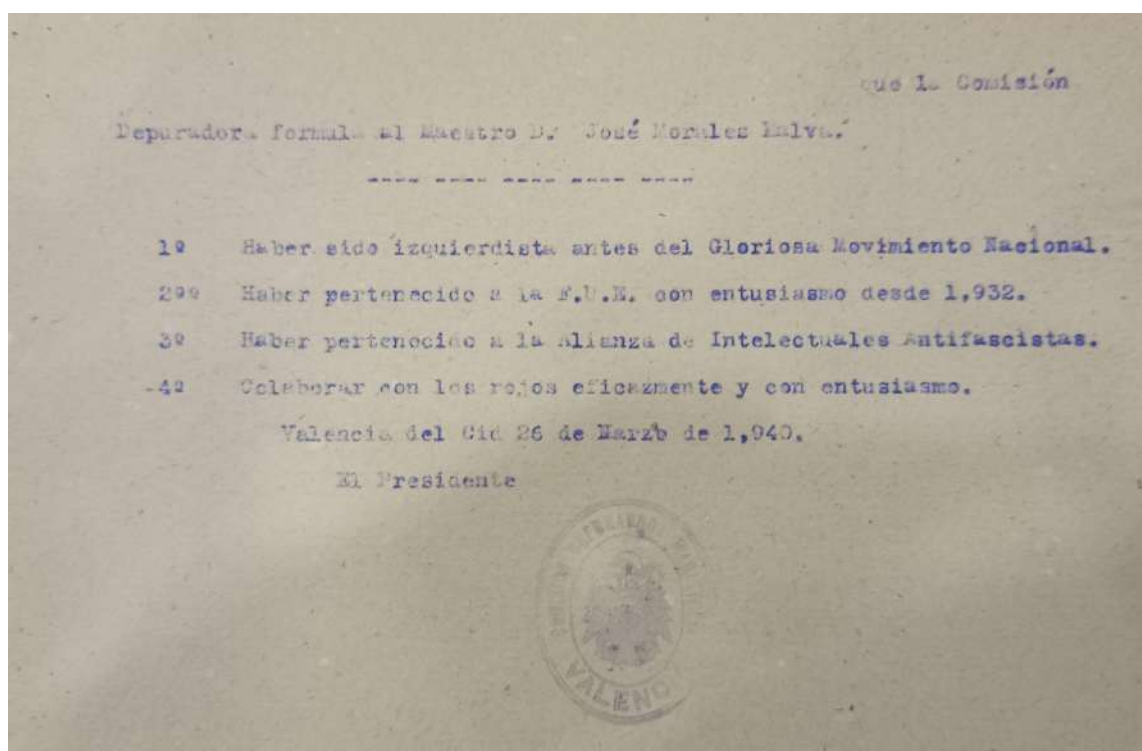


Figure C3: Sample page of a final sanction issued by the central commission. ([back to text](#))

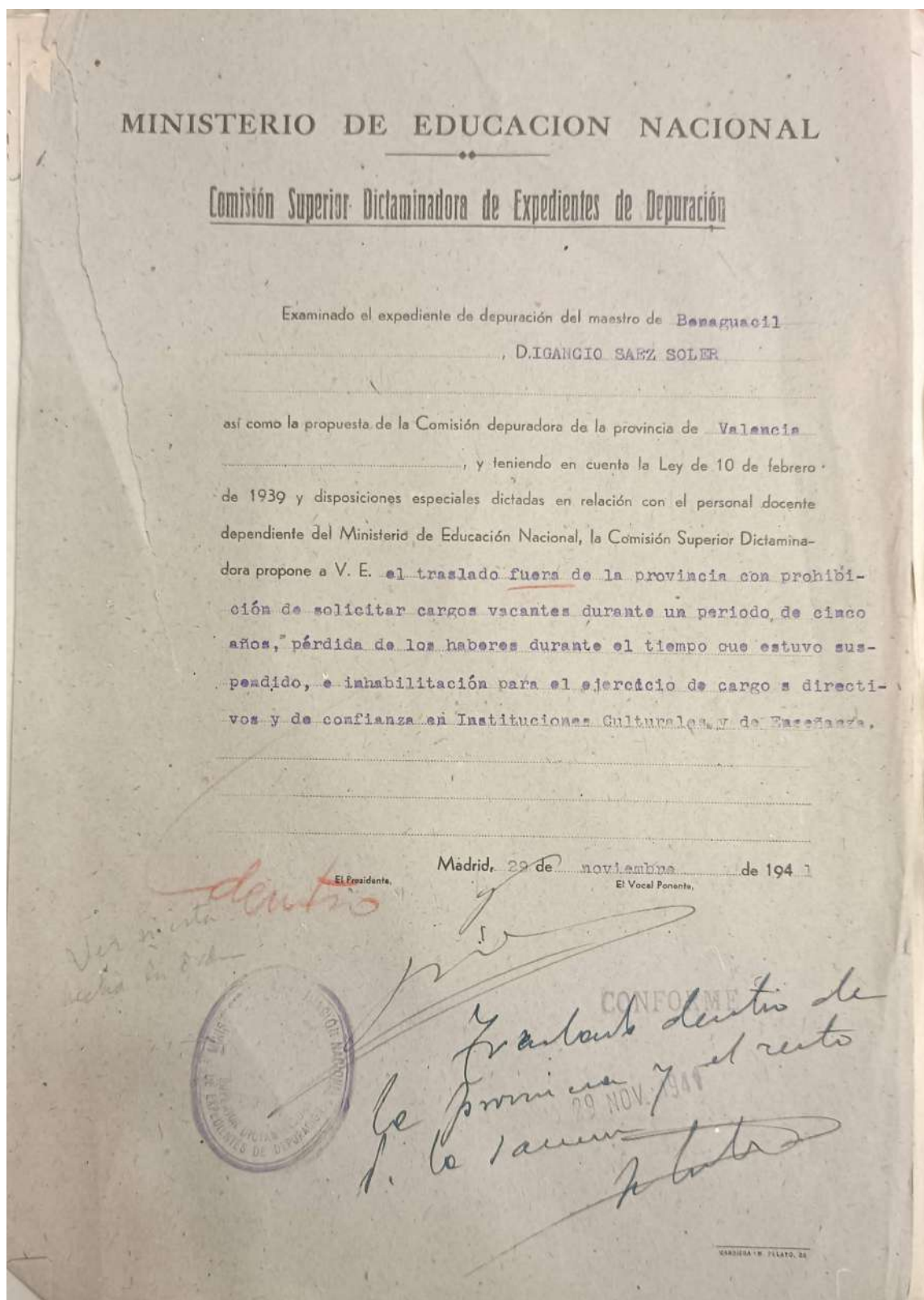


Figure C4: Teacher rosters sample pages. (back to text)

— 20 —
— 21 —

CATEGORIA TERCERA: 7.000 pesetas -:- 375 plazas

Número del Escalafón general	Número en la categoría	NOMBRES Y APELLIDOS	TÍTULO Y NOTA DEL MISMO	NACIMIENTO		SERVICIOS						PUNTO DONDE PRESTA SUS SERVICIOS		Número del Escalafón general	OBSERVACIONES			
				FECHA	PROVINCIA	En la categoría		En propiedad		Interinos		PUEBLO	PROVINCIA					
						Días	Años	Años	MeSES	Días	Años					MeSES	Días	
291	1	D. Luis Alabart Bailester...	S. St.	14	9 1888	Barcelona	2	4	20	9	3	3	Barcelona	Barcelona	291			
292	2	Tomás Masías García...	M. y B.	29	8 1888	Guadalajara	2	4	17	6	3	3	Madrid	Madrid	292			
293	3	José Vilaplana Ebel...	S.	25	4 1875	Castellón	2	4	27	7	21	2	1	Vinaros	Castellón	293		
294	4	Luis Conejo Ramos...	S.	25	10 1886	León	2	4	20	9	5	2	10	21	Madrid	Madrid	294	
295	5	Andrés Santamaría Chavarria...	N.	10	11 1889	Burgos	2	4	20	9	14	3	4	14	Madrid	Madrid	295	
296	6	Luis Moreno Torres...	S.	19	2 1881	Huesca	2	4	31	2	14	3	3	3	Salv	Cerona	296	
297	7	Antonio Magariños Pastoriza...	S. St.	7	12 1880	Ceruñá	2	4	34	1	7	1	15	Cambados	Pontevedra	297		
298	8	Rafael González Cuadrado...	S. St.	1	4 1882	Cuenca	2	4	29	9	3	3	29	Madrid	Madrid	298		
299	9	Lisardo Fernández Fernández...	E.	27	2 1883	Logroño	2	4	17	7	3	3	3	Madrid	Madrid	299		
300	10	Martin Serra Molins...	S.	7	9 1889	Barcelona	2	4	17	9	25	3	5	Barcelona	Barcelona	300		
301	11	Claudio Franco Angé...	S.	30	10 1876	Navarra	2	4	35	2	5	4	4	Madrid	Madrid	301		
302	12	Pedro Mejías Rodríguez...	S. St.	31	5 1885	Huelva	2	4	25	7	20	3	5	Madrid	Madrid	302		
303	13	Damián Ricart Lafont...	S.	1	1 1879	Lérida	2	4	31	6	20	3	5	Barcelona	Barcelona	303		
304	14	Francisco Fernández del Castillo...	S. St.	4	10 1881	Alava	2	4	18	4	10	3	3	Vitoria	Alava	304		
305	15	José Sánchez Asensi...	S.	16	2 1883	Valencia	2	4	27	7	16	5	15	Vinaros	Castellón	305		
306	16	Ambrosio Martín Sáez y Lázaro...	N. y B. St.	7	12 1873	Logroño	2	4	30	9	12	3	3	San Sebastián	Valencia	306		
307	17	José Licerías Aguilera...	S.	20	10 1874	Málaga	2	4	31	11	16	3	3	Valencia	Gulpiá	307		
308	18	José Hoyos García...	S.	20	2 1885	Málaga	2	4	48	22	3	3	3	Málaga	Málaga	308		
309	19	Celestino Ojeda Angulo...	S.	21	2 1876	Logroño	2	4	46	7	17	3	3	Bermeo	Vizcaya	309		

Plan 1901.

(a) 1934 teacher roster.

CATEGORIA QUINTA

— 7 —

CON 3.400 PESETAS

Número del Escalafón general	Número en la categoría	NOMBRES Y APELLIDOS	TÍTULO Y NOTA DEL MISMO	NACIMIENTO		SERVICIOS						PUNTO DONDE PRESTA SUS SERVICIOS		Número del Escalafón general	OBSERVACIONES			
				FECHA	PROVINCIA	En la categoría		En propiedad		Interinos		PUEBLO	PROVINCIA					
						Días	Años	Años	MeSES	Días	Años					MeSES	Días	
2925	188	D.ª Crispina C. Sánchez Morante Eiroa...	M.	8	12 1888	Cuenca	1	3	27	1	19	3	S. Pedro Palmiches	Cuenca	2925			
2926	189	Elvira Alvarez Díez...	E.	7	4 1889	León	1	3	27	1	19	3	Astorga	León	2926			
2927	190	Encarnación Aláez Cervera...	M. St.	2	7 1897	León	1	3	27	1	17	3	La Seca	Valladolid	2927			
2928	191	María V. Martínez-Unda Morguez...	E.	10	11 1894	Cuenca	1	3	27	1	16	1	2	4	Barajas de Melo	Cuenca	2928	
2929	192	Emérita Valcarlos Gutiérrez...	N.	12	5 1890	León	1	3	24	9	3	3	3	3	Puertollano	Ciudad Real	2929	R.º serv. E. E.
2930	193	Brigida Silva Antón...	S. St.	23	7 1897	Zamora	1	3	27	1	16	3	3	3	Montamarta	Zamora	2930	
2931	194	María del Carmen Alvarez Peláez...	S.	3	8 1886	Zamora	1	3	23	8	14	3	3	3	Monclal del Pan	Zamora	2931	
2932	195	María Concepción Ovelheiro Marcos...	S.	14	2 1891	Valladolid	1	3	27	1	16	3	3	3	León	León	2932	
2933	196	Teresa Crespo de la Fuente...	S.	17	6 1895	Zamora	1	3	27	1	16	3	3	3	Malpartida Cáceres	Cáceres	2933	
2934	197	María Clotilde Casares García...	M.	27	3 1894	Zamora	1	3	27	1	16	3	3	3	Infantes	Ciudad Real	2934	
2935	198	Teresa Martín Araujo...	M.	4	1 1895	Salamanca	1	3	27	1	16	3	3	3	Vilvestre	Salamanca	2935	
2936	199	Josefa Rodríguez García...	S. St.	23	7 1894	Zamora	1	3	27	1	15	1	2	25	Andavías	Zamora	2936	
2937	200	Josefa González Fernández...	S. B.	10	12 1892	León	1	3	27	1	15	3	3	3	León	León	2937	
2938	201	María Herminia de Paula Pardal...	S.	17	8 1896	Zamora	1	3	27	1	15	3	3	3	Cereceda	Salamanca	2938	
2939	202	Daniela Serrano Díaz...	M. St.	10	4 1895	Almería	1	3	27	1	15	3	3	3	Cuenca	Cuenca	2939	
2940	203	Petronila Méndez Peñín...	M.	23	3 1897	Zamora	1	3	27	1	15	3	3	3	Granja de Morenuela	Zamora	2940	
2941	204	Encarnación Villaverde Sánchez...	S. St.	4	3 1896	Granada	1	3	27	1	13	3	3	3	Padul	Granada	2941	
2942	205	Avelina Soté Valenzuela...	M.	13	11 1897	Pontevedra	1	3	27	1	13	3	3	3	Vigo	Pontevedra	2942	
2943	206	Manuela Suárez Díaz...	M.	11	5 1896	León	1	3	27	1	8	3	3	3	Carballino	Orense	2943	
2944	207	María Alen Solla...	S.	13	9 1895	Pontevedra	1	3	27	1	5	3	3	3	Pontevedra	Pontevedra	2944	
2945	208	Juana María Marín Onrubia...	S. St.	16	5 1894	Albacete	1	3	27	1	2	3	3	3	Tobarra	Albacete	2945	
2946	209	Paz Paula García Priego...	E.	7	6 1894	Cuenca	1	3	27	1	3	3	3	3	Villarrobledo	Albacete	2946	
2947	210	Josefa del Valle Rodríguez...	E.	25	5 1889	León	1	3	27	3	11	11	3	3	Villamañá	León	2947	
2948	211	María Josefa Moreno Margallo...	S.	21	4 1885	Badajoz	1	3	27	3	26	3	3	3	Helechal (Benquerén)	Badajoz	2948	
2949	212	María C. García Sáenz Lacuesta...	S.	26	2 1887	Alava	1	3	27	3	26	3	3	3	Mallén	Zaragoza	2949	
2950	213	Teresa Oliván Rafael...	N.	16	10 1897	Huesca	1	3	19	11	14	3	3	3	Angüés	Huesca	2950	

(b) 1946 teacher roster.

Table C1: List of the most frequent charges brought against sanctioned teachers. ([back to text](#))

Political charges

1. Leader in a pro-Republic party
2. Member of a pro-Republic party
3. Collaboration with pro-Republic parties
4. Sympathizer of pro-Republic parties
5. Member/collaborator of Red Aid or other republican organizations
6. Disaffection toward the Movement

Union-related charges

1. Member of the Teachers' Union
2. Member of other labor organizations

Moral charges

1. Immorality in private conduct
2. Immorality in public conduct

Religious charges

1. Religious indifference
2. Atheism
3. Anti-religiosity or anti-clericalism

Professional charges

1. Applying republican educational principles
2. Neglect of professional duties

Other charges

1. Membership in Freemasonry
2. Collaboration in republican cultural, union, or political events
3. Collaboration with republicans in the press

Figure C5: Sample pages documenting post-purge teacher hiring. ([back to text](#))

Núm. 194	BOLETIN OFICIAL DEL ESTADO	Página 4829
ORDEN de 6 de julio de 1940 por la que se nombran para los destinos que se indican a los Oficiales del Ejército que han sido admitidos en el Concurso al Magisterio Nacional. Tmo. Sr.: Vista la clasificación verificada por el Ministerio del Ejército de los Oficiales que han solicitado tomar parte en el concurso de		
ingreso en el Magisterio Nacional, así como los antecedentes facilitados por la Oficina de Depuración de este Departamento y los documentos que figuran en los expedientes de cada peticionario. Este Ministerio, en cumplimiento de lo dispuesto en el artículo sexto de la Ley de 26 de enero último (B. O. del 7 de febrero), ha resuelto:		
Primero. Nombrar Maestros propietarios provisionales, con el sueldo anual de 4.000 pesetas, de las Escuelas que se indican y para realizar las prácticas y pruebas a que se alude en dicho precepto, a los Oficiales que se relacionan a continuación, los cuales causarán baja en el Ejército, pasando a las Escuelas que les corresponda, al tomar posesión de su nuevo cargo.		
Abad Benito (Fidel) Abrido del Rey (Marino) Abrio Pulido (José) Achelegui Rodríguez (Benito) Adiego Martínez (Ignacio) Alfonso Padrón (Eugenio) Aguayo Sáenz (Domitilo) Agüero Porset (Miguel) Aguila Sánchez (Juan) Aguilar Coya (Fernando) Aguilar Hermano (Manuel) Aguilar Manzano (Francisco) Aguilo Forteza (Gaspar) Aguirre Cárdenas (Jorge) Alamo Coca (Francisco) Alava Concor (Argeo) Alva Garrido (Antonio) Alva Navas (Manuel) Alvala Tejerina (Alberto)	Almendrales (Badajoz). G.ª «Suárez Somonte». Salamanca. G.ª «Santa Teresa». Jerez de la Frontera (Cádiz). G.ª «Carmen Benítez». Calahorra (Logroño). G.ª «Quintillano». Lérida. G.ª «Caballeros». G.ª Tarifa (Cádiz). G.ª Baltanal (Palencia). Bilbao (Vizcaya). G.ª «Zorroza». Madrid. G.ª «José Calvo Sotelo». Barcelona (capital). Sevilla (capital). G.ª «Cervantes». Sevilla (capital). G.ª «Hospicio». G.ª La Puebla (Balears). Albacete. G.ª «P. Manjón». Avila (capital). G.ª «Cervantes». Cuenca (capital). G.ª «Ramón y Cajal». G.ª Noalejo (Jaén). Sevilla (capital). G.ª A. a Normal. Palencia (capital). G.ª «Jorge Manriques».	

(a) 1940 resolution listing locality assignments for the 1939 military opening.

Núm. 233

BOLETIN OFICIAL DEL ESTADO

Página 6301

MINISTERIO DE EDUCACION NACIONAL

Continuación a la lista general de los Maestros que obtuvieron plaza en las oposiciones a ingreso en el Magisterio Nacional, convocadas por Orden ministerial de 19 de mayo de 1941. (Publicadas la primera, segunda y tercera relación según Orden de 27 de julio de 1944, BOLETIN OFICIAL DEL ESTADO números 230, 231 y 232, de 17, 18 y 19 del mes corriente)

Número de orden	NOMBRES Y APELLIDOS	Tribunal en que actuó	Número obtenido	Puntuación media	Servicios internos	Otros méritos
1.729	D. Salustiano Fernández Asensio	Cáceres	39	7,08		
1.730	D. Alejandro Fernández Santamaría	Burgos	39	7,03		
1.731	D. Bernardo Pulido Santiago	Jaén	39	6,90		
1.732	D. Gonzalo Frean Barreira	Lugo	39	6,89		
1.733	D. Antonio Vidal Juan	Baleares	39	6,80		
1.734	D. Fernando Jordán Gras	Alicante	39	6,63		
1.735	D. Antonio Megia Salmerón	Almería	39	6,62		
1.736	D. Félix Garrido Escobar	Zamora	39	6,04		
1.737	D. Gabriel García Boquer	B.ª de Oña	39	6,00		
1.738	D. Mariano Rodríguez Martín	Valladolid	39	6,94		
1.739	D. José Conesa Hernández	Murcia	39	5,80		
1.740	D. Manuel Gordillo Pentinel	Badajoz	39	5,33		
1.741	D. Felipe Paradela Muñoz	Avila	39	5,30		
1.742	D. Amador García Gómez	Salamanca	40	11,93		
1.743	D. Francisco Valverde Alvarez	León	40	11,90		
1.744	D. José Moreno Urbano	Córdoba	40	9,83		
1.745	D. Jesús Dacal Vázquez	Pontevedra	40	9,82		
1.746	D. Julián Estornes Anaut	Navarra	40	9,38		
1.747	D. Pascual Pablo Canales	Valencia	40	9,10		
1.748	D. Antonio Martínez Luque	Granada	40	9,03		

(b) 1944 resolution reporting teacher scores and rank ordering for the 1941 hiring round.

Figure C6: Sample page from the resolution documenting teachers' locality origin and assigned destinations in the 1934 relocation contest. ([back to text](#))

Gaceta de Madrid.—Núm. 224	12 Agosto 1934	1447
PROVINCIA DE SANTANDER	"Ramón Pelayo" (Santander); serie B.	tialay (Soria-Alconava). Menos de 500 habitantes.
MAESTROS	Número 7.405, excedente, grupo D. Doña María de la Concepción Aguirre González, de Castañedo (Oviedo), con 5-0-24; para Santander, Grupo "Ramón Pelayo"; serie D.	Número 17.021, grupo D.—D. Basilio Pastor Vallejo, de La Losilla (Soria), con 8-9-0; para Tozalmoro-Alarcón (Soria). Menos de 500 habitantes.
Número 238, grupo A.—D. Guillermo García López, de Badalona (Barcelona), con 19-0-21; para Santander, Sección graduada grupo Ramón Pelayo; A.	Número 910, grupo de 1928, grupo D.—Doña Cipriana Pereda Herrera, de Alconchel de la Estrella (Cuenca), con 2-3-23; para Barcelona de Pie de Concha (Santander); serie A.	Número 17.226, grupo D.—D. Crispín Gonzalo de la Orden, de Matute de la Sierra (Soria), con 0-8-0; para Rollamienta (Soria). Menos de 500 habitantes.
Número 2.430, grupo B.—D. Marcelino Rodríguez del Pozo, de Santillana del Mar (Santander), con 17-5-0; para Santander, Sección graduada Ramón Pelayo.	Número 8.884, excedente, grupo D. Doña Florentina Cimiano Galván, de Entrambasestas (Santander), con 1-0-28; para Pontejos-Marina de Cudeyo (Santander); serie A.	Número 17.532, grupo D.—D. Esteban Crespo Heras, de Estepa de San Juan (Soria), con 7-10-0; para Narrod (Soria). Menos de 500 habitantes.
Número 2.871, grupo B.—D. Nemesio Calderón Cortés, de Novales (Santander), con 13-2-0; para Santander, Sección graduada, barriada de San Román; A.	Número 86, 1923, grupo D.—Doña Anlana Fernández Collantes, de Fontecha (Santander), con 1-0-0; para Elechas-Marina de Cudeyo (Santander); serie A.	Número 17.704, grupo D.—D. Juan Manrique Sanz, de Villanueva de Gormaz (Soria), con 10-7-16; para la de Valdemalque (Soria). Menos de 500 habitantes.
Número 3.549, grupo C.—D. José Torres Fernández, de Bacia, en Luarca (Oviedo), con 8-4-13; para Laredo, unitaria A (Santander).	Número 2.296, 1931, grupo D.—Doña Sebastiana Acha González, de Molledo (Burgos), con 0-0-0; para Silió-Molledo (Santander); serie A.	Número 18.042, grupo D.—D. Casimiro Carro Ortega, de Casarejos (Soria), con 10-7-0; para Gormaz (Soria).
Número 10.667, grupo C.—D. Maximiliano Montero Romero, de Salarzón (Santander), con 6-9-5; para Requejada y Mar, en Polanco (Santander), mixta, A.	Número 15.431, grupo D.—Doña María Mercedes Ruiz Ochoa, de Cabarceo (Santander), con 0-8-0; para Vargas-Puenteviego (Santander); serie A.	Número 912-1931, grupo D.—Don David Ranz Lafuente, de Araoz (Guipúzcoa), con 0-0-0; para Muñoz-Viana de Duero (Soria). Menos de 500 habitantes.
Número 241, grupo C.—D. Clemente Morante de Alles, de Vega de Villazufre (Santander), con 3-7-10; para San Mamés de Polaciones, en Polaciones (Santander).	Número 15.920, grupo D.—Doña Victoria Feijóo Santos, de Valmeo (Santander), 0-8-0; para Barrio Arriba-Rio-tuerto (Santander); serie A.	Número 1.540-1931, grupo D.—Don Toribio Crespo Jiménez, de Pozuelo (Soria), con 0-0-0; para Casillas de Berlanga-Caltójar (Soria). Menos de 500 habitantes.
Número 409 C, grupo D.—D. José Manuel García García, de San Martín de Valdelomar (Santander), con 3-4-13; para Roiz, en Valdáliga (Santander); A.	Número 3.157, segundo Escalafón.—Doña Francisca M. García Berganzo, de Barrio de Arriba (Santander), con 8-2-28; para Rubalcaba-Liérganes (Santander). Menos de 500 habitantes.	Número 2.146-1931, grupo D.—Don Enrique Gordón Jiménez, de Tamagarda (Santa Cruz de Tenerife), con 0-0-0; para Useño de la Sierra (Soria). Menos de 500 habitantes.
Número 811 C, grupo D.—D. Saturnino Gutiérrez Díez, de Peones de Amaya (Burgos), con 2-8-11; para Viernoles, en Torrelavega (Santander); A.	Número 17.001, grupo D.—D. Vicente Lombraña Rodríguez, de Cosgalla-Camaleño, con 5-1-16; para Armaño y San Sebastián, en Castro-Cillorigo (Santander).	Número 1.683, segundo Escalafón. D. Fermín Martínez Ballano, de Zayas de Vascoles (Soria), con 5-2-28; para Villaciervitos (Soria). Menos de 500 habitantes.
Número 17.001, grupo D.—D. Vicente Lombraña Rodríguez, de Cosgalla-Camaleño, con 5-1-16; para Armaño y San Sebastián, en Castro-Cillorigo (Santander).	Número 4.364, segundo Escalafón.—Doña Leonor Sierra Puig, de Bascuñana (Burgos), con 3-4-6; para Herrera de Ibio-Mazcuerras (Santander). Menos de 500 habitantes.	Número 1.411-1931, grupo D.—Doña Petra Ibáñez Morales, de Valtueñas (Soria), con 0-0-0; para Castilruiz (Soria); serie A.
Número 17.842, grupo D.—D. José Burgos Ortega, de Berzosilla (Palencia), con 4-2-21; para Vispieres, en Santillana (Santander).	Número, alta, excedente, segundo Escalafón.—Doña Felicidad González Zabala, de Izara (Santander), con 0-2-29; para Ubiarco-Santillana (Santander). Menos de 500 habitantes.	Segundo Escalafón.—Doña Margarita Heliadora Marijuán Manso, de Carazuelo (Soria), con 19-6-24; para Saldueiro (Soria). Menos de 500 habitantes.
Número 17.875, grupo D.—D. Francisco Peláez Barreda, de Nuñorodero (Santander), con 3-5-15; para Pechón, en Val de San Vicente (Santander).	PROVINCIA DE SORIA	Segundo Escalafón.—Doña Josefa Afienza Campos, de Marazovel (Soria), con 16-10-0; para Tajuelo (Soria). Menos de 500 habitantes.
Número 18.077, grupo D.—D. Teófilo Villalba González, de Ogarrio, con 5-0-0; para Angustina, en Voto (Santander).	MAESTROS	PROVINCIA DE LOGROÑO
Número 1.566 H, grupo D.—D. Antonio Villarreal Pellón, de Río Castillo (Oviedo), con 0-0-0; para Tezanos, en Villacarriedo (Santander); A.	Número 2.970, grupo B.—D. Vicente del Río García, de Osma (Soria), con 14-0-17; para Soria, Sección graduada mixta; serie B.	MAESTROS
Número 1.670 H, grupo D.—D. Ildefonso Pastor Hernández, de Sobrepenilla (Santander), con 0-0-0; para Bedoya, en Castro-Cillorigo (Santander).	Número 3.740, grupo C.—D. Vicente Garcés García, de Lomparedes (Soria), con 5-2-4; para Canos-Aldehuela de Perlañez (Soria). Menos de 500 habitantes.	Número 2.969, grupo B.—D. Daniel Monja Agustín, de Arenys de Mar (Barcelona), con 4-1-0; para Haro, Sección graduada (Logroño); serie B.
Número 1.928 H, grupo D.—D. Pablo Muñoz Rodríguez, de Hoz de Abiaga, con 0-0-0; para Requejo, en Enmedio (Santander).	Número 5.441, grupo C.—D. Florencio del Amo Marina, de Alfaro (Logroño), con 5-1-9; para la de Revilla de Calatañazor (Soria). Menos de 500 habitantes.	Número 5.271, grupo C.—D. Aureliano Lezana Fernández, de Bañares (Logroño), con 16-0-5; para Santo Domingo de la Calzada, Sección graduada (Logroño); serie A.
Número 2032, segundo Escalafón.—D. Juan Gómez Huidobro, excedente de Llano de Valdearroyo (Santander); con 5-7-28; para Mogro de Miengo (Santander).	Número 178 A, grupo C.—D. Mariano Labadía Aramburu, de Morales (Soria), con 3-7-20; para Agreda, Sección graduada (Soria); serie C.	Número 7.039, grupo C.—D. Miguel Esteban Montalvo, de Siete Iglesias de Cabaneros (Valladolid), con 5-11-26; para Casalarreina, unitaria número 1 (Logroño); serie A.
Número 2.443, segundo Escalafón.—D. Eustasio Villalba González, de Rascon (Santander), con 5-0-0; para Bernales, en Ampuero (Santander).	Número 80 F, grupo D.—D. Doro-teo Molina Martínez, de Garabelos-Balar (Orense), con 1-1-24; para Valavellano de Tera (Soria); serie A.	Número 898 H, grupo D.—D. Aurelio de Diego Sánchez, de San Andrés de Cameros (Logroño), con 0-0-0; para Pajares, mixta (Logroño).
MAESTRAS	Número 16.970, grupo D.—D. Hilarión Enciso Guerrero, de Tardajos de Duero (Soria), con 22-4-29; para Mar-	Número 1.573 H, grupo D.—D. José Martínez Lázaro, de Dombellas (So-
Número 2.628, grupo B.—Doña Angela Mazorra González, de Ramales (Santander), con 18-3-0, para el Grupo		

Figure C7: Original legal provision establishing a maximum class size of 50 students per teacher. ([back to text](#))

22 Septiembre 1934
Gaceta de Madrid.—Núm. 265

Las disposiciones vigentes sobre organización de las Escuelas graduadas señalan la cifra máxima de 50 alumnos para cada una de las clases. En algunas Escuelas, atendiendo al ritmo de la enseñanza graduada, el número de niños matriculados, especialmente en los grados superiores, es muy inferior al que fija la Ley, y esta conducta en la Dirección de las graduadas es muy laudable cuando todos los niños tengan Centros primarios donde recibir la enseñanza, pero no puede autorizarse si las Escuelas creadas en la localidad son insuficientes para recoger la población escolar infantil. Y aunque pierda algo la unidad de trabajo y se aumente éste en las graduadas, los Directores de las Escuelas y los Inspectores de Primera enseñanza, siempre que haya niños que reclamen la asistencia a las Escuelas nacionales, completarán la matrícula de alumnos en cada grado seleccionándolos cuidadosamente para que dentro de estas normas, se conserve el espíritu y la eficacia de la enseñanza graduada.

Lo digo a VV. SS. para su conocimiento y demás efectos. Madrid, 2 de Septiembre de 1934.—El Director general, Victoriano Lucas.

Señores Inspectores Jefes de Primera enseñanza y Directores de Escuelas graduadas.

- (a) Directorate General of Primary Education, "Circular," *Gaceta de Madrid* (Official Gazette), no. 265 (22 September 1934): 2542.

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18 julio 1945

B. O. del E.—Núm. 199

Artículo cincuenta y uno.—Se considera edificio público escolar, a los efectos de este artículo, el que albergue servicios docentes de enseñanza primaria nacional.

Todo edificio escolar habrá de estar emplazado, en lo posible, en el centro geográfico de mayor densidad de alumnos, en lugar sano, sin peligro de accidentes y con vecindad salubre y moral. Ha de comprender el aula o aulas capaces, según la matrícula que arroje el censo, dentro de los límites fijados en el artículo diecisiete, siempre que dicha matrícula no exceda por aula de cincuenta alumnos; los servicios higiénicos y complementarios proporcionales, asimismo, a la matrícula total de la Escuela o grupo escolar y los campos de juego y deportes.

- (b) *Law on Primary Education* (17 July 1945), *Boletín Oficial del Estado* (Official State Gazette), no. 199 (18 July 1945), art. 51

Figure C8: Sample page from original hiring lists reporting teacher scores and rank ordering.
(back to text)

Anexo Único.—Página 306		19 Octubre 1930		Gaceta de Madrid.—Núm. 232				
Número de orden	NOMBRES Y APELLIDOS	PUNTUACIONES			TOTAL	PUNTUACIONES		
		EJERCICIOS PROVINCIALES				COMISIONES GENERALES		
		Primer ejercicio	Segundo ejercicio	Ejercicio práctico		Primer ejercicio	Segundo ejercicio	tercer ejercicio
62	D. Arturo Rodríguez García.....	36	60	89	175	23	25	24
63	Julian Hernández Olivares.....	36	49	100	185	29	30	24
64	Juan García Madueño.....	47	42	88	177	20	25	24
65	José Peinado Altable.....	28	42	98	168	23	30	20
66	Eduardo Candilejo Adame.....	43	44	90	177	25	20	23
67	Rafael Rodríguez Díaz.....	25	17,5	68	110,5	15	25	17
68	Julian Vázquez Santos.....	41,5	46,5	100	188	25	20	20
69	Manuel Prada Tablatax.....	47	47,5	90	184,5	20	30	22
70	Fernando García Calero.....	46	48	100	194	28	30	15
71	Juan Díaz de la Torre.....	45	41	86	172	20	20	25
72	Francisco Rueda Pardo.....	50	36	96	181	30	15	24
73	Antonio Chavarino Gómez.....	46,5	35	97,5	179	25	20	20
74	Natalio Fernández Llamero.....	50	49,5	85	184,5	20	25	24
75	A. Eloy Martín Carmona.....	44	48	95	187	30	15	20
76	Adolfo Villaiba Carralero.....	50	50	95	195	20	25	24
77	Alfonso Ortega Fernández.....	33	41	92	166	19	15	24
78	Juan Avilés Alia.....	44	46	88	172	17	20	27
79	José Varela Otero.....	40	50	80	170	20	20	23
80	Francisco Valverde Martínez.....	43	45	90	178	23	20	26
81	Juan Fraile Bernal.....	36,5	45	65	147,5	20	20	23
82	Adrián González López.....	45,5	48	71	164,5	25	20	20

(a) 1930 resolution reporting teacher scores and rank ordering for the 1928 hiring round

2328		15 Septiembre 1934		Gaceta de Madrid.—Núm. 258	
sillista 1931); Almería; 12-5-1912; 198,500 puntos.		consume plaza); Córdoba; 30-6-1914; 178,100.		84-1.—Doña Asunción Nicoláu Senent (hija de Maestro); Alicante; 23-3-1910; 150,200 puntos.	
8-1.—Doña Marina López López; Cáceres; 18-6-1902; 197,600.		46-1.—Doña María Concepción Rodríguez Matías; Huesca; 18-2-1911; 176,750.		85-1.—Doña Carmen Guerra de la Mota; Teruel; 6-2-1913; 150,200.	
9-1.—Doña María del Pilar Arroyo González; Madrid; 2-2-1893; 197,500.		47-1.—Doña María del Rosario Gasco Pascual; Valencia; 1-8-1909; 176.		86-1.—Doña María del Carmen García Salinero; Segovia; 7-6-1913; 150,010.	
10-1.—Doña Pilar de la Torre Rodríguez; Madrid; 27-12-1913; 197.		48-1.—Doña Rita Vallespi Torrent; Barcelona; 5-1-1911; 176.		87-1.—Doña Dolores Leira Luaces; Coruña; 24-5-1914; 149,750.	
11-1.—Doña María Llenas Tremols; Barcelona; 15-6-1912; 196.		49-1.—Doña Simona Rambal Piquer; Zaragoza; 28-10-1911; 174,750.		88-1.—Doña Libertad Serra Bartrina; Gerona; 2-1-1913; 149,740.	
12-1.—Doña Gerarda Romero Ramos; Burgos; 11-7-1913; 196.		50-1.—Doña Rosalina Calderón García; Palencia; 24-1-1909; 174,590.		89-1.—Doña Eloísa Galisteo Gualart; Tarragona; 1-12-1904; 148,500.	
13-1.—Doña Anselma Hernández Seiquer; Soria; 12-6-1911; 194,500.		51-1.—Doña María R. Jaureguizar Isasi; Santiago; 30-9-1913; 174,500.		90-1.—Doña Iluminada Marcos Pérez; Salamanca; 19-6-1913; 148,060.	
14-1.—Doña Remedios Bardina Soronellas; Barcelona; 6-9-1907; 193.		52-1.—Doña Juliana Esparza Ugaide; Navarra; 19-2-1908; 173,080.		91-1.—Doña Jesusa Fernández Monjardín; Oviedo; 18-2-1910; 146,300.	
15-1.—Doña Raquel Moura Rivas; Orense; 24-4-1913; 193.		53-1.—Doña María T. Llovet Serra; Lérida; 18-9-1909; 171,500.		92-1.—Doña Francisca Cabezas Cerro; Badajoz; 12-8-1914; 146,200.	
16-1.—Doña Luciana Visitación Martínez y Martínez; Albacete; 26-10-1908; 192,900 puntos.		54-1.—Doña Pilar Beranga Berges; Huesca; 12-10-1912; 171,500.		93-1.—Doña María Consuelo Celia Carid Vázquez; Orense; 5-5-1913; 145.	
17-1.—Doña Juliana María del Pilar García Subero (no consume plaza); Zaragoza; 20-3-1910; 192,650.		55-1.—Doña Presentación Hernández Martín; Granada; 17-4-1913; 170.		94-1.—Doña María López Cabanillas; Lugo; 6-2-1904 144,720.	
18-1.—Doña María del Pilar Liñán Larrucea; Madrid; 11-6-1905; 192.		56-1.—Doña Pilar Rodríguez Domínguez; Pontevedra; 22-11-1912; 169,200.		95-1.—Doña Concepción Fernández Cabal (hija de Maestro); León; 21-5-1911; 144,300.	
19-1.—Doña María de los Dolores Molina y Luna; Ciudad Real; 13-2-1913; 192.		57-1.—Doña Encarnación María Lili Galdames; Bilbao; 25-4-1904; 169.		96-1.—Doña María Manuela López Sánchez; Avila; 17-6-1909; 144,250.	
20-1.—Doña Dolores Sánchez Bolea; Murcia; 20-2-1909; 191,200.		58-1.—Doña Carmen Peña Gil; Pontevedra; 24-4-1895; 168.		97-1.—Doña Josefa Gómez Cacho (hija de Maestro); Avila; 31-12-1914; 144,200.	
		59-1.—Doña Inés Moreno Bejarano; Cáceres; 17-12-1902; 166,900.			

(b) 1934 resolution reporting teacher scores and rank ordering for the 1933 hiring round

Figure C9: Original text listing vacancies to be filled in the 1941 relocation process ([back to text](#))

BOLETIN OFICIAL DEL ESTADO
4 abril 1941

ORDEN de 2 de abril de 1941 por la que se convoca concurso general de traslados entre el personal del Magisterio Nacional Primario.

Elmo. Sr.: En cumplimiento de lo dispuesto en el Decreto de 25 de noviembre último.

Este Ministerio dispone:

1.º Convocar concurso general de traslados para proveer, por los turnos que establece el artículo segundo del citado Decreto, todas las vacantes definitivas de Escuelas Nacionales que se hubieran producido hasta el 31 de diciembre de 1940 por excedencias, jubilaciones, fallecimientos, expedientes gubernativos o de depuración, en Escuelas mixtas, unitarias y Secciones de graduadas, aunque se hallen servidas provisional o interinamente, excepto las de anejas a Normales y aquellas otras cuya provisión esté reservada a procedimiento especial por la vigente legislación.

Las escuelas vacantes definitivas en 31 de diciembre de 1940, provistas en propiedad por Maestros trasladados a consecuencia de depuración, con posterioridad a dicha fecha, se considerarán eliminadas. Serán incluidas las escuelas que hayan quedado vacantes por idénticos traslados producidas hasta la fecha de publicación de esta Orden.